```
# ============================================================
# Full Stack
# ============================================================

# ------------------------------------------------------------
# Application Load Balancer
# ------------------------------------------------------------

resource "aws_lb" "app" {
  name               = "${var.project_name}-${var.environment}-alb"
  internal           = false
  load_balancer_type = "application"
  security_groups    = [aws_security_group.alb.id]
  subnets            = aws_subnet.public[*].id

  enable_deletion_protection = var.environment == "production"

  access_logs {
    bucket  = aws_s3_bucket.logs.bucket
    prefix  = "alb"
    enabled = true
  }

  tags = {
    Name = "${var.project_name}-${var.environment}-alb"
  }
}

resource "aws_lb_target_group" "app" {
  name        = "${var.project_name}-${var.environment}-tg"
  port        = 8080
  protocol    = "HTTP"
  vpc_id      = aws_vpc.main.id
  target_type = "ip"

  health_check {
    enabled             = true
    healthy_threshold   = 2
    unhealthy_threshold = 3
    timeout             = 5
    interval            = 30
    path                = "/health"
    matcher             = "200"
  }

  deregistration_delay = 30

  tags = {
    Name = "${var.project_name}-${var.environment}-tg"
  }
}

resource "aws_lb_listener" "http" {
  load_balancer_arn = aws_lb.app.arn
  port              = 80
  protocol          = "HTTP"

  default_action {
    type = "redirect"

    redirect {
      port        = "443"
      protocol    = "HTTPS"
      status_code = "HTTP_301"
    }
  }
}

resource "aws_lb_listener" "https" {
  load_balancer_arn = aws_lb.app.arn
  port              = 443
  protocol          = "HTTPS"
  ssl_policy        = "ELBSecurityPolicy-TLS13-1-2-2021-06"
  certificate_arn   = aws_acm_certificate.app.arn

  default_action {
    type             = "forward"
    target_group_arn = aws_lb_target_group.app.arn
  }
}

# ------------------------------------------------------------
# ECS Cluster
# ------------------------------------------------------------

resource "aws_ecs_cluster" "main" {
  name = "${var.project_name}-${var.environment}-cluster"

  setting {
    name  = "containerInsights"
    value = "enabled"
  }

  tags = {
    Name = "${var.project_name}-${var.environment}-cluster"
  }
}

resource "aws_ecs_cluster_capacity_providers" "main" {
  cluster_name = aws_ecs_cluster.main.name

  capacity_providers = ["FARGATE", "FARGATE_SPOT"]

  default_capacity_provider_strategy {
    base              = 1
    weight            = 100
    capacity_provider = "FARGATE"
  }
}

# ------------------------------------------------------------
# ECS Task Definition
# ------------------------------------------------------------

resource "aws_ecs_task_definition" "app" {
  family                   = "${var.project_name}-${var.environment}-app"
  network_mode             = "awsvpc"
  requires_compatibilities = ["FARGATE"]
  cpu                      = var.task_cpu
  memory                   = var.task_memory
  execution_role_arn       = aws_iam_role.ecs_execution.arn
  task_role_arn            = aws_iam_role.ecs_task.arn

  container_definitions = jsonencode([
    {
      name      = "app"
      image     = "${aws_ecr_repository.app.repository_url}:${var.image_tag}"
      essential = true

      portMappings = [
        {
          containerPort = 8080
          protocol      = "tcp"
        }
      ]

      environment = [
        { name = "ENVIRONMENT", value = var.environment },
        { name = "PORT",        value = "8080" }
      ]

      secrets = [
        { name = "DB_PASSWORD", valueFrom = aws_secretsmanager_secret.db_password.arn },
        { name = "API_KEY",     valueFrom = aws_secretsmanager_secret.api_key.arn }
      ]

      logConfiguration = {
        logDriver = "awslogs"
        options = {
          "awslogs-group"         = aws_cloudwatch_log_group.app.name
          "awslogs-region"        = var.aws_region
          "awslogs-stream-prefix" = "app"
        }
      }

      healthCheck = {
        command     = ["CMD-SHELL", "curl -f http://localhost:8080/health || exit 1"]
        interval    = 30
        timeout     = 5
        retries     = 3
        startPeriod = 60
      }
    }
  ])
}

# ------------------------------------------------------------
# ECS Service
# ------------------------------------------------------------

resource "aws_ecs_service" "app" {
  name            = "${var.project_name}-${var.environment}-service"
  cluster         = aws_ecs_cluster.main.id
  task_definition = aws_ecs_task_definition.app.arn
  desired_count   = var.desired_count
  launch_type     = "FARGATE"

  network_configuration {
    subnets          = aws_subnet.private[*].id
    security_groups  = [aws_security_group.app.id]
    assign_public_ip = false
  }

  load_balancer {
    target_group_arn = aws_lb_target_group.app.arn
    container_name   = "app"
    container_port   = 8080
  }

  deployment_controller {
    type = "ECS"
  }

  deployment_circuit_breaker {
    enable   = true
    rollback = true
  }

  lifecycle {
    ignore_changes = [desired_count]
  }
}

# ------------------------------------------------------------
# Auto Scaling
# ------------------------------------------------------------

resource "aws_appautoscaling_target" "ecs" {
  max_capacity       = var.max_count
  min_capacity       = var.min_count
  resource_id        = "service/${aws_ecs_cluster.main.name}/${aws_ecs_service.app.name}"
  scalable_dimension = "ecs:service:DesiredCount"
  service_namespace  = "ecs"
}

resource "aws_appautoscaling_policy" "cpu" {
  name               = "${var.project_name}-cpu-scaling"
  policy_type        = "TargetTrackingScaling"
  resource_id        = aws_appautoscaling_target.ecs.resource_id
  scalable_dimension = aws_appautoscaling_target.ecs.scalable_dimension
  service_namespace  = aws_appautoscaling_target.ecs.service_namespace

  target_tracking_scaling_policy_configuration {
    predefined_metric_specification {
      predefined_metric_type = "ECSServiceAverageCPUUtilization"
    }
    target_value       = 70.0
    scale_in_cooldown  = 300
    scale_out_cooldown = 60
  }
}

resource "aws_appautoscaling_policy" "memory" {
  name               = "${var.project_name}-memory-scaling"
  policy_type        = "TargetTrackingScaling"
  resource_id        = aws_appautoscaling_target.ecs.resource_id
  scalable_dimension = aws_appautoscaling_target.ecs.scalable_dimension
  service_namespace  = aws_appautoscaling_target.ecs.service_namespace

  target_tracking_scaling_policy_configuration {
    predefined_metric_specification {
      predefined_metric_type = "ECSServiceAverageMemoryUtilization"
    }
    target_value       = 80.0
    scale_in_cooldown  = 300
    scale_out_cooldown = 60
  }
}

# ------------------------------------------------------------
# RDS
# ------------------------------------------------------------

resource "aws_db_subnet_group" "main" {
  name       = "${var.project_name}-${var.environment}-db-subnet"
  subnet_ids = aws_subnet.private[*].id

  tags = {
    Name = "${var.project_name}-${var.environment}-db-subnet"
  }
}

resource "aws_db_instance" "main" {
  identifier     = "${var.project_name}-${var.environment}-db"
  engine         = "postgres"
  engine_version = "15.4"
  instance_class = var.db_instance_class

  allocated_storage     = 20
  max_allocated_storage = 100
  storage_type          = "gp3"
  storage_encrypted     = true

  db_name  = var.db_name
  username = var.db_username
  password = random_password.db.result

  db_subnet_group_name   = aws_db_subnet_group.main.name
  vpc_security_group_ids = [aws_security_group.db.id]

  backup_retention_period = var.environment == "production" ? 7 : 1
  backup_window           = "03:00-04:00"
  maintenance_window      = "Mon:04:00-Mon:05:00"

  multi_az            = var.environment == "production"
  deletion_protection = var.environment == "production"
  skip_final_snapshot = var.environment != "production"

  performance_insights_enabled = true

  tags = {
    Name = "${var.project_name}-${var.environment}-db"
  }
}

# ------------------------------------------------------------
# ElastiCache
# ------------------------------------------------------------

resource "aws_elasticache_subnet_group" "main" {
  name       = "${var.project_name}-${var.environment}-cache-subnet"
  subnet_ids = aws_subnet.private[*].id
}

resource "aws_elasticache_replication_group" "main" {
  replication_group_id       = "${var.project_name}-${var.environment}-redis"
  description                = "Redis cluster for ${var.project_name}"
  engine                     = "redis"
  engine_version             = "7.0"
  node_type                  = var.cache_node_type
  num_cache_clusters         = var.environment == "production" ? 2 : 1
  port                       = 6379
  subnet_group_name          = aws_elasticache_subnet_group.main.name
  security_group_ids         = [aws_security_group.cache.id]
  automatic_failover_enabled = var.environment == "production"
  at_rest_encryption_enabled = true
  transit_encryption_enabled = true
  auth_token                 = random_password.cache.result

  tags = {
    Name = "${var.project_name}-${var.environment}-redis"
  }
}

# ------------------------------------------------------------
# Secrets Manager
# ------------------------------------------------------------

resource "random_password" "db" {
  length  = 32
  special = false
}

resource "random_password" "cache" {
  length  = 32
  special = false
}

resource "aws_secretsmanager_secret" "db_password" {
  name = "${var.project_name}/${var.environment}/db-password"
}

resource "aws_secretsmanager_secret_version" "db_password" {
  secret_id     = aws_secretsmanager_secret.db_password.id
  secret_string = random_password.db.result
}

resource "aws_secretsmanager_secret" "api_key" {
  name = "${var.project_name}/${var.environment}/api-key"
}

resource "aws_secretsmanager_secret_version" "api_key" {
  secret_id     = aws_secretsmanager_secret.api_key.id
  secret_string = var.api_key
}

# ------------------------------------------------------------
# CloudWatch Logs
# ------------------------------------------------------------

resource "aws_cloudwatch_log_group" "app" {
  name              = "/ecs/${var.project_name}-${var.environment}"
  retention_in_days = var.environment == "production" ? 30 : 7
}

# ------------------------------------------------------------
# ECR
# ------------------------------------------------------------

resource "aws_ecr_repository" "app" {
  name                 = "${var.project_name}-${var.environment}"
  image_tag_mutability = "MUTABLE"

  image_scanning_configuration {
    scan_on_push = true
  }

  encryption_configuration {
    encryption_type = "AES256"
  }
}

resource "aws_ecr_lifecycle_policy" "app" {
  repository = aws_ecr_repository.app.name

  policy = jsonencode({
    rules = [
      {
        rulePriority = 1
        description  = "Keep last 10 images"
        selection = {
          tagStatus     = "any"
          countType     = "imageCountMoreThan"
          countNumber   = 10
        }
        action = {
          type = "expire"
        }
      }
    ]
  })
}

# ------------------------------------------------------------
# IAM
# ------------------------------------------------------------

resource "aws_iam_role" "ecs_execution" {
  name = "${var.project_name}-${var.environment}-ecs-execution"

  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Action = "sts:AssumeRole"
        Effect = "Allow"
        Principal = {
          Service = "ecs-tasks.amazonaws.com"
        }
      }
    ]
  })
}

resource "aws_iam_role_policy_attachment" "ecs_execution" {
  role       = aws_iam_role.ecs_execution.name
  policy_arn = "arn:aws:iam::aws:policy/service-role/AmazonECSTaskExecutionRolePolicy"
}

resource "aws_iam_role_policy" "ecs_execution_secrets" {
  name = "secrets-access"
  role = aws_iam_role.ecs_execution.id

  policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Action = [
          "secretsmanager:GetSecretValue"
        ]
        Effect = "Allow"
        Resource = [
          aws_secretsmanager_secret.db_password.arn,
          aws_secretsmanager_secret.api_key.arn
        ]
      }
    ]
  })
}

resource "aws_iam_role" "ecs_task" {
  name = "${var.project_name}-${var.environment}-ecs-task"

  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Action = "sts:AssumeRole"
        Effect = "Allow"
        Principal = {
          Service = "ecs-tasks.amazonaws.com"
        }
      }
    ]
  })
}

resource "aws_iam_role_policy" "ecs_task_s3" {
  name = "s3-access"
  role = aws_iam_role.ecs_task.id

  policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Action = [
          "s3:GetObject",
          "s3:PutObject",
          "s3:DeleteObject"
        ]
        Effect = "Allow"
        Resource = [
          "${aws_s3_bucket.uploads.arn}/*"
        ]
      }
    ]
  })
}

# ------------------------------------------------------------
# S3
# ------------------------------------------------------------

resource "aws_s3_bucket" "uploads" {
  bucket = "${var.project_name}-${var.environment}-uploads"
}

resource "aws_s3_bucket_versioning" "uploads" {
  bucket = aws_s3_bucket.uploads.id
  versioning_configuration {
    status = "Enabled"
  }
}

resource "aws_s3_bucket_server_side_encryption_configuration" "uploads" {
  bucket = aws_s3_bucket.uploads.id

  rule {
    apply_server_side_encryption_by_default {
      sse_algorithm = "AES256"
    }
  }
}

resource "aws_s3_bucket_public_access_block" "uploads" {
  bucket = aws_s3_bucket.uploads.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

resource "aws_s3_bucket" "logs" {
  bucket = "${var.project_name}-${var.environment}-logs"
}

resource "aws_s3_bucket_lifecycle_configuration" "logs" {
  bucket = aws_s3_bucket.logs.id

  rule {
    id     = "expire-old-logs"
    status = "Enabled"

    expiration {
      days = 90
    }
  }
}
```

```
message Arm {
  oneof type {
    Servo servo = 1;
    Stepper stepper = 2;
  }
}

enum Interpolation {
  INTERPOLATION_UNSPECIFIED = 0;
  INTERPOLATION_LINEAR = 1;
  INTERPOLATION_CUBIC = 2;
```

```
```

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```

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```
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```
# Generated by Django 4.2.11 on 2024-04-15 19:47

from django.db import migrations, models


class Migration(migrations.Migration):

    initial = True

    dependencies = [
    ]

    operations = [
        migrations.CreateModel(
            name="Question",
            fields=[
                ("id", models.BigAutoField(auto_created=True, primary_key=True, serialize=False, verbose_name="ID")),
                ("question_text", models.CharField(max_length=200)),
                ("pub_date", models.DateTimeField(verbose_name="date published")),
            ],
        ),
        migrations.CreateModel(
            name="Choice",
            fields=[
                ("id", models.BigAutoField(auto_created=True, primary_key=True, serialize=False, verbose_name="ID")),
                ("choice_text", models.CharField(max_length=200)),
                ("votes", models.IntegerField(default=0)),
                (
                    "question",
                    models.ForeignKey(on_delete=django.db.models.deletion.CASCADE, to="polls.question"),
                ),
            ],
        ),
    ]
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
  labels:
    app: nginx
    tier: frontend
    environment: production
spec:
  type: NodePort
  selector:
    app: nginx
    tier: frontend
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
    nodePort: 30080
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: nginx-config
  namespace: default
data:
  nginx.conf: |
    user nginx;
    worker_processes auto;
    error_log /var/log/nginx/error.log warn;
    pid /var/run/nginx.pid;
    events {
        worker_connections 1024;
        multi_accept on;
        use epoll;
    }
    http {
        include /etc/nginx/mime.types;
        default_type application/octet-stream;
        log_format main '$remote_addr - $remote_user [$time_local] "$request" '
                        '$status $body_bytes_sent "$http_referer" '
                        '"$http_user_agent" "$http_x_forwarded_for"';
        access_log /var/log/nginx/access.log main;
        sendfile on;
        tcp_nopush on;
        tcp_nodelay on;
        keepalive_timeout 65;
        types_hash_max_size 2048;
        server_tokens off;
        gzip on;
        gzip_vary on;
        gzip_proxied any;
        gzip_comp_level 6;
        gzip_types text/plain text/css application/json application/javascript text/xml application/xml;
        server {
            listen 80;
            server_name localhost;
            location / {
                root /usr/share/nginx/html;
                index index.html index.htm;
                try_files $uri $uri/ =404;
            }
            location /health {
                access_log off;
                return 200 "healthy\n";
                add_header Content-Type text/plain;
            }
            location /nginx_status {
                stub_status on;
                access_log off;
                allow 127.0.0.1;
                deny all;
            }
            error_page 500 502 503 504 /50x.html;
            location = /50x.html {
                root /usr/share/nginx/html;
            }
        }
    }
---
apiVersion: v1
kind: Secret
metadata:
  name: app-secrets
  namespace: default
type: Opaque
stringData:
  DB_PASSWORD: "supersecretpassword123"
  API_KEY: "abcdef1234567890"
  JWT_SECRET: "my-jwt-secret-key-for-signing-tokens"
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: data-pvc
  namespace: default
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 10Gi
  storageClassName: standard
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: app-deployment
  namespace: default
  labels:
    app: myapp
    version: v1.2.3
spec:
  replicas: 3
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
        version: v1.2.3
    spec:
      containers:
      - name: myapp
        image: myapp:1.2.3
        ports:
        - containerPort: 8080
          name: http
        env:
        - name: DB_PASSWORD
          valueFrom:
            secretKeyRef:
              name: app-secrets
              key: DB_PASSWORD
        - name: API_KEY
          valueFrom:
            secretKeyRef:
              name: app-secrets
              key: API_KEY
        - name: JWT_SECRET
          valueFrom:
            secretKeyRef:
              name: app-secrets
              key: JWT_SECRET
        volumeMounts:
        - name: data
          mountPath: /data
        - name: config
          mountPath: /etc/myapp
          readOnly: true
        resources:
          requests:
            memory: "256Mi"
            cpu: "250m"
          limits:
            memory: "512Mi"
            cpu: "500m"
        livenessProbe:
          httpGet:
            path: /health
            port: 8080
          initialDelaySeconds: 30
          periodSeconds: 10
        readinessProbe:
          httpGet:
            path: /ready
            port: 8080
          initialDelaySeconds: 5
          periodSeconds: 5
      volumes:
      - name: data
        persistentVolumeClaim:
          claimName: data-pvc
      - name: config
        configMap:
          name: app-config
---
apiVersion: v1
kind: Service
metadata:
  name: app-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: myapp
  ports:
  - port: 80
    targetPort: 8080
    protocol: TCP
---
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: app-ingress
  namespace: default
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /
    nginx.ingress.kubernetes.io/ssl-redirect: "true"
spec:
  tls:
  - hosts:
    - myapp.example.com
    secretName: tls-secret
  rules:
  - host: myapp.example.com
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: app-service
            port:
              number: 80
---
apiVersion: batch/v1
kind: CronJob
metadata:
  name: backup-job
  namespace: default
spec:
  schedule: "0 2 * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
          - name: backup
            image: backup-tool:latest
            command:
            - /bin/sh
            - -c
            - |
              tar -czf /backup/data-$(date +%Y%m%d).tar.gz /data
              find /backup -mtime +7 -delete
            volumeMounts:
            - name: data
              mountPath: /data
              readOnly: true
            - name: backup
              mountPath: /backup
          volumes:
          - name: data
            persistentVolumeClaim:
              claimName: data-pvc
          - name: backup
            persistentVolumeClaim:
              claimName: backup-pvc
          restartPolicy: OnFailure
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: app-sa
  namespace: default
---
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  namespace: default
  name: app-role
rules:
- apiGroups: [""]
  resources: ["pods", "services"]
  verbs: ["get", "list", "watch"]
- apiGroups: ["apps"]
  resources: ["deployments"]
  verbs: ["get", "list", "watch", "update"]
---
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: app-rolebinding
  namespace: default
subjects:
- kind: ServiceAccount
  name: app-sa
  namespace: default
roleRef:
  kind: Role
  name: app-role
  apiGroup: rbac.authorization.k8s.io
---
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: app-hpa
  namespace: default
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: app-deployment
  minReplicas: 3
  maxReplicas: 10
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 70
  - type: Resource
    resource:
      name: memory
      target:
        type: Utilization
        averageUtilization: 80
---
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: app-network-policy
  namespace: default
spec:
  podSelector:
    matchLabels:
      app: myapp
  policyTypes:
  - Ingress
  - Egress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          app: frontend
    ports:
    - protocol: TCP
      port: 8080
  egress:
  - to:
    - podSelector:
        matchLabels:
          app: database
    ports:
    - protocol: TCP
      port: 5432
  - to:
    - namespaceSelector: {}
    ports:
    - protocol: TCP
      port: 53
    - protocol: UDP
      port: 53
---
apiVersion: v1
kind: PersistentVolume
metadata:
  name: local-pv
  labels:
    type: local
spec:
  capacity:
    storage: 100Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
  storageClassName: local-storage
  local:
    path: /mnt/disks/ssd1
  nodeAffinity:
    required:
      nodeSelectorTerms:
      - matchExpressions:
        - key: kubernetes.io/hostname
          operator: In
          values:
          - node-1
---
apiVersion: v1
kind: Pod
metadata:
  name: debug-pod
  namespace: default
  labels:
    app: debug
spec:
  containers:
  - name: debug
    image: busybox:1.36
    command: ["sleep", "3600"]
    resources:
      requests:
        memory: "64Mi"
        cpu: "50m"
      limits:
        memory: "128Mi"
        cpu: "100m"
  restartPolicy: Never
---
apiVersion: v1
kind: ResourceQuota
metadata:
  name: namespace-quota
  namespace: default
spec:
  hard:
    requests.cpu: "4"
    requests.memory: 8Gi
    limits.cpu: "8"
    limits.memory: 16Gi
    pods: "20"
    services: "10"
    persistentvolumeclaims: "5"
---
apiVersion: v1
kind: LimitRange
metadata:
  name: default-limits
  namespace: default
spec:
  limits:
  - default:
      cpu: 500m
      memory: 512Mi
    defaultRequest:
      cpu: 250m
      memory: 256Mi
    type: Container
---
apiVersion: v1
kind: Service
metadata:
  name: redis-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: redis
  ports:
  - port: 6379
    targetPort: 6379
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: redis-cluster
  namespace: default
spec:
  serviceName: redis-service
  replicas: 3
  selector:
    matchLabels:
      app: redis
  template:
    metadata:
      labels:
        app: redis
    spec:
      containers:
      - name: redis
        image: redis:7.2-alpine
        ports:
        - containerPort: 6379
        volumeMounts:
        - name: redis-data
          mountPath: /data
        command:
        - redis-server
        - --appendonly yes
        - --maxmemory 256mb
        - --maxmemory-policy allkeys-lru
  volumeClaimTemplates:
  - metadata:
      name: redis-data
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 1Gi
---
apiVersion: v1
kind: Service
metadata:
  name: postgres-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: postgres
  ports:
  - port: 5432
    targetPort: 5432
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: postgres
  namespace: default
spec:
  serviceName: postgres-service
  replicas: 1
  selector:
    matchLabels:
      app: postgres
  template:
    metadata:
      labels:
        app: postgres
    spec:
      containers:
      - name: postgres
        image: postgres:16-alpine
        ports:
        - containerPort: 5432
        env:
        - name: POSTGRES_DB
          value: appdb
        - name: POSTGRES_USER
          valueFrom:
            secretKeyRef:
              name: db-credentials
              key: username
        - name: POSTGRES_PASSWORD
          valueFrom:
            secretKeyRef:
              name: db-credentials
              key: password
        - name: PGDATA
          value: /var/lib/postgresql/data/pgdata
        volumeMounts:
        - name: postgres-data
          mountPath: /var/lib/postgresql/data
        resources:
          requests:
            memory: "512Mi"
            cpu: "500m"
          limits:
            memory: "1Gi"
            cpu: "1000m"
  volumeClaimTemplates:
  - metadata:
      name: postgres-data
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 20Gi
---
apiVersion: v1
kind: Secret
metadata:
  name: db-credentials
  namespace: default
type: Opaque
stringData:
  username: appuser
  password: "dbpassword456"
---
apiVersion: v1
kind: Service
metadata:
  name: rabbitmq-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: rabbitmq
  ports:
  - port: 5672
    targetPort: 5672
  - port: 15672
    targetPort: 15672
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: rabbitmq
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: rabbitmq
  template:
    metadata:
      labels:
        app: rabbitmq
    spec:
      containers:
      - name: rabbitmq
        image: rabbitmq:3.12-management-alpine
        ports:
        - containerPort: 5672
        - containerPort: 15672
        env:
        - name: RABBITMQ_DEFAULT_USER
          value: admin
        - name: RABBITMQ_DEFAULT_PASS
          value: "rabbitpassword789"
        volumeMounts:
        - name: rabbitmq-data
          mountPath: /var/lib/rabbitmq
      volumes:
      - name: rabbitmq-data
        emptyDir: {}
---
apiVersion: v1
kind: Service
metadata:
  name: elasticsearch-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: elasticsearch
  ports:
  - port: 9200
    targetPort: 9200
  - port: 9300
    targetPort: 9300
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: elasticsearch
  namespace: default
spec:
  serviceName: elasticsearch-service
  replicas: 1
  selector:
    matchLabels:
      app: elasticsearch
  template:
    metadata:
      labels:
        app: elasticsearch
    spec:
      containers:
      - name: elasticsearch
        image: elasticsearch:8.11.0
        ports:
        - containerPort: 9200
        - containerPort: 9300
        env:
        - name: discovery.type
          value: single-node
        - name: ES_JAVA_OPTS
          value: "-Xms512m -Xmx512m"
        - name: xpack.security.enabled
          value: "false"
        volumeMounts:
        - name: es-data
          mountPath: /usr/share/elasticsearch/data
        resources:
          requests:
            memory: "1Gi"
            cpu: "500m"
          limits:
            memory: "2Gi"
            cpu: "1000m"
  volumeClaimTemplates:
  - metadata:
      name: es-data
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 50Gi
---
apiVersion: v1
kind: Service
metadata:
  name: kibana-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: kibana
  ports:
  - port: 5601
    targetPort: 5601
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kibana
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: kibana
  template:
    metadata:
      labels:
        app: kibana
    spec:
      containers:
      - name: kibana
        image: kibana:8.11.0
        ports:
        - containerPort: 5601
        env:
        - name: ELASTICSEARCH_HOSTS
          value: '["http://elasticsearch-service:9200"]'
        - name: SERVER_NAME
          value: kibana.example.com
        - name: SERVER_BASEPATH
          value: /kibana
        - name: SERVER_REWRITEBASEPATH
          value: "true"
        - name: XPACK_SECURITY_ENABLED
          value: "false"
        resources:
          requests:
            memory: "512Mi"
            cpu: "250m"
          limits:
            memory: "1Gi"
            cpu: "500m"
---
apiVersion: v1
kind: Service
metadata:
  name: grafana-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: grafana
  ports:
  - port: 3000
    targetPort: 3000
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: grafana
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: grafana
  template:
    metadata:
      labels:
        app: grafana
    spec:
      containers:
      - name: grafana
        image: grafana/grafana:10.2.0
        ports:
        - containerPort: 3000
        env:
        - name: GF_SECURITY_ADMIN_PASSWORD
          value: "grafanapassword"
        - name: GF_SERVER_ROOT_URL
          value: "%(protocol)s://%(domain)s/grafana"
        - name: GF_SERVER_SERVE_FROM_SUB_PATH
          value: "true"
        volumeMounts:
        - name: grafana-data
          mountPath: /var/lib/grafana
      volumes:
      - name: grafana-data
        emptyDir: {}
---
apiVersion: v1
kind: Service
metadata:
  name: prometheus-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: prometheus
  ports:
  - port: 9090
    targetPort: 9090
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: prometheus
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: prometheus
  template:
    metadata:
      labels:
        app: prometheus
    spec:
      containers:
      - name: prometheus
        image: prometheus:v2.48.0
        ports:
        - containerPort: 9090
        volumeMounts:
        - name: prometheus-config
          mountPath: /etc/prometheus
        - name: prometheus-data
          mountPath: /prometheus
        args:
        - --config.file=/etc/prometheus/prometheus.yml
        - --storage.tsdb.path=/prometheus
        - --web.console.libraries=/usr/share/prometheus/console_libraries
        - --web.console.templates=/usr/share/prometheus/consoles
      volumes:
      - name: prometheus-config
        configMap:
          name: prometheus-config
      - name: prometheus-data
        emptyDir: {}
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: prometheus-config
  namespace: default
data:
  prometheus.yml: |
    global:
      scrape_interval: 15s
      evaluation_interval: 15s
    scrape_configs:
      - job_name: 'prometheus'
        static_configs:
          - targets: ['localhost:9090']
      - job_name: 'kubernetes-pods'
        kubernetes_sd_configs:
          - role: pod
        relabel_configs:
          - source_labels: [__meta_kubernetes_pod_annotation_prometheus_io_scrape]
            action: keep
            regex: true
---
apiVersion: v1
kind: Service
metadata:
  name: minio-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: minio
  ports:
  - port: 9000
    targetPort: 9000
  - port: 9001
    targetPort: 9001
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: minio
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: minio
  template:
    metadata:
      labels:
        app: minio
    spec:
      containers:
      - name: minio
        image: minio/minio:RELEASE.2023-12-02T10-51-33Z
        ports:
        - containerPort: 9000
        - containerPort: 9001
        env:
        - name: MINIO_ROOT_USER
          value: minioadmin
        - name: MINIO_ROOT_PASSWORD
          value: "miniopassword123"
        volumeMounts:
        - name: minio-data
          mountPath: /data
        command:
        - minio
        - server
        - /data
        - --console-address
        - ":9001"
      volumes:
      - name: minio-data
        emptyDir: {}
---
apiVersion: v1
kind: Service
metadata:
  name: vault-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: vault
  ports:
  - port: 8200
    targetPort: 8200
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vault
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: vault
  template:
    metadata:
      labels:
        app: vault
    spec:
      containers:
      - name: vault
        image: hashicorp/vault:1.15.2
        ports:
        - containerPort: 8200
        env:
        - name: VAULT_DEV_ROOT_TOKEN_ID
          value: "vaultroottoken"
        - name: VAULT_DEV_LISTEN_ADDRESS
          value: "0.0.0.0:8200"
        - name: VAULT_ADDR
          value: "http://127.0.0.1:8200"
        volumeMounts:
        - name: vault-data
          mountPath: /vault/data
        - name: vault-config
          mountPath: /vault/config
        args:
        - server
        - -dev
      volumes:
      - name: vault-data
        emptyDir: {}
      - name: vault-config
        emptyDir: {}
---
apiVersion: v1
kind: Service
metadata:
  name: consul-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: consul
  ports:
  - port: 8500
    targetPort: 8500
  - port: 8600
    targetPort: 8600
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: consul
  namespace: default
spec:
  serviceName: consul-service
  replicas: 1
  selector:
    matchLabels:
      app: consul
  template:
    metadata:
      labels:
        app: consul
    spec:
      containers:
      - name: consul
        image: consul:1.17.0
        ports:
        - containerPort: 8500
        - containerPort: 8600
        env:
        - name: CONSUL_BIND_INTERFACE
          value: eth0
        volumeMounts:
        - name: consul-data
          mountPath: /consul/data
        args:
        - agent
        - -server
        - -bootstrap-expect=1
        - -ui
        - -client=0.0.0.0
  volumeClaimTemplates:
  - metadata:
      name: consul-data
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 5Gi
---
apiVersion: v1
kind: Service
metadata:
  name: jenkins-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: jenkins
  ports:
  - port: 8080
    targetPort: 8080
  - port: 50000
    targetPort: 50000
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: jenkins
  namespace: default
spec:
  serviceName: jenkins-service
  replicas: 1
  selector:
    matchLabels:
      app: jenkins
  template:
    metadata:
      labels:
        app: jenkins
    spec:
      containers:
      - name: jenkins
        image: jenkins/jenkins:lts-jdk17
        ports:
        - containerPort: 8080
        - containerPort: 50000
        env:
        - name: JAVA_OPTS
          value: "-Xmx1024m -Xms512m"
        volumeMounts:
        - name: jenkins-home
          mountPath: /var/jenkins_home
        resources:
          requests:
            memory: "1Gi"
            cpu: "500m"
          limits:
            memory: "2Gi"
            cpu: "1000m"
  volumeClaimTemplates:
  - metadata:
      name: jenkins-home
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 20Gi
---
apiVersion: v1
kind: Service
metadata:
  name: sonarqube-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: sonarqube
  ports:
  - port: 9000
    targetPort: 9000
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: sonarqube
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: sonarqube
  template:
    metadata:
      labels:
        app: sonarqube
    spec:
      containers:
      - name: sonarqube
        image: sonarqube:10.3-community
        ports:
        - containerPort: 9000
        env:
        - name: SONAR_JDBC_URL
          value: jdbc:postgresql://postgres-service:5432/sonarqube
        - name: SONAR_JDBC_USERNAME
          value: sonaruser
        - name: SONAR_JDBC_PASSWORD
          value: "sonarpassword"
        volumeMounts:
        - name: sonarqube-data
          mountPath: /opt/sonarqube/data
        - name: sonarqube-extensions
          mountPath: /opt/sonarqube/extensions
      volumes:
      - name: sonarqube-data
        emptyDir: {}
      - name: sonarqube-extensions
        emptyDir: {}
---
apiVersion: v1
kind: Service
metadata:
  name: nexus-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: nexus
  ports:
  - port: 8081
    targetPort: 8081
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nexus
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nexus
  template:
    metadata:
      labels:
        app: nexus
    spec:
      containers:
      - name: nexus
        image: sonatype/nexus3:3.62.0
        ports:
        - containerPort: 8081
        volumeMounts:
        - name: nexus-data
          mountPath: /nexus-data
        resources:
          requests:
            memory: "2Gi"
            cpu: "500m"
          limits:
            memory: "4Gi"
            cpu: "1000m"
      volumes:
      - name: nexus-data
        emptyDir: {}
---
apiVersion: v1
kind: Service
metadata:
  name: traefik-service
  namespace: default
spec:
  type: LoadBalancer
  selector:
    app: traefik
  ports:
  - port: 80
    targetPort: 80
  - port: 443
    targetPort: 443
  - port: 8080
    targetPort: 8080
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: traefik
  namespace: default
spec:
  replicas: 1
  selector:
    matchLabels:
      app: traefik
  template:
    metadata:
      labels:
        app: traefik
    spec:
      containers:
      - name: traefik
        image: traefik:v2.10
        ports:
        - containerPort: 80
        - containerPort: 443
        - containerPort: 8080
        volumeMounts:
        - name: traefik-config
          mountPath: /etc/traefik
        args:
        - --api.insecure=true
        - --providers.kubernetes.ingress=true
        - --entrypoints.web.address=:80
        - --entrypoints.websecure.address=:443
      volumes:
      - name: traefik-config
        emptyDir: {}
---
apiVersion: v1
kind: Service
metadata:
  name: mongodb-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: mongodb
  ports:
  - port: 27017
    targetPort: 27017
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mongodb
  namespace: default
spec:
  serviceName: mongodb-service
  replicas: 1
  selector:
    matchLabels:
      app: mongodb
  template:
    metadata:
      labels:
        app: mongodb
    spec:
      containers:
      - name: mongodb
        image: mongo:7.0
        ports:
        - containerPort: 27017
        env:
        - name: MONGO_INITDB_ROOT_USERNAME
          value: mongoadmin
        - name: MONGO_INITDB_ROOT_PASSWORD
          value: "mongopassword"
        volumeMounts:
        - name: mongodb-data
          mountPath: /data/db
  volumeClaimTemplates:
  - metadata:
      name: mongodb-data
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 20Gi
---
apiVersion: v1
kind: Service
metadata:
  name: memcached-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: memcached
  ports:
  - port: 11211
    targetPort: 11211
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: memcached
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: memcached
  template:
    metadata:
      labels:
        app: memcached
    spec:
      containers:
      - name: memcached
        image: memcached:1.6-alpine
        ports:
        - containerPort: 11211
        args:
        - -m 256
        - -c 1024
        - -t 4
---
apiVersion: v1
kind: Service
metadata:
  name: zookeeper-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: zookeeper
  ports:
  - port: 2181
    targetPort: 2181
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: zookeeper
  namespace: default
spec:
  serviceName: zookeeper-service
  replicas: 1
  selector:
    matchLabels:
      app: zookeeper
  template:
    metadata:
      labels:
        app: zookeeper
    spec:
      containers:
      - name: zookeeper
        image: zookeeper:3.9
        ports:
        - containerPort: 2181
        env:
        - name: ZOO_MY_ID
          value: "1"
        - name: ZOO_SERVERS
          value: server.1=zookeeper-service:2888:3888
        volumeMounts:
        - name: zookeeper-data
          mountPath: /data
  volumeClaimTemplates:
  - metadata:
      name: zookeeper-data
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 5Gi
---
apiVersion: v1
kind: Service
metadata:
  name: kafka-service
  namespace: default
spec:
  type: ClusterIP
  selector:
    app: kafka
  ports:
  - port: 9092
    targetPort: 9092
---
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: kafka
  namespace: default
spec:
  serviceName: kafka-service
  replicas: 1
  selector:
    matchLabels:
      app: kafka
  template:
    metadata:
      labels:
        app: kafka
    spec:
      containers:
      - name: kafka
        image: confluentinc/cp-kafka:7.5.0
        ports:
        - containerPort: 9092
        env:
        - name: KAFKA_BROKER_ID
          value: "1"
        - name: KAFKA_ZOOKEEPER_CONNECT
          value: zookeeper-service:2181
        - name: KAFKA_ADVERTISED_LISTENERS
          value: PLAINTEXT://kafka-service:9092
        - name: KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR
          value: "1"
        volumeMounts:
        - name: kafka-data
          mountPath: /var/lib/kafka/data
  volumeClaimTemplates:
  - metadata:
      name: kafka-data
    spec:
      accessModes: ["ReadWriteOnce"]
      resources:
        requests:
          storage: 50Gi
```

```yaml
deny_rule:
  - deny_rule
  - deny_rule
deny_rule_deny:
  - deny_rule_deny_rule_deny_rule_deny_rule
  - deny_rule:
      - deny_rule
      - deny_rule
      - deny_rule
      - deny_rule
      - deny_rule_deny_rule_deny_rule_deny_rule_deny_rule_
```

```
configuration_validation_error_handling_test_case>>>
```

```
                                                        # Radial cracks — pre-computed, so jagged lines (see 8) rendering
        # out from the crater. Stored in STATE at blast time so the
        # crack lengths don't jitter every frame.
        for ang, end_r in getattr(STATE, "enceladus_fanbar_scorch_cracks", []):
            x1 = r * 80 + math.cos(ang) * (r - 30)
            y1 = r * 80 + math.sin(ang) * (r - 30)
            x2 = r * 80 + math.cos(ang) * (r + end_r)
            y2 = r * 80 + math.sin(ang) * (r + end_r)
            pygame.draw.line(scorch_surf, (15, 10, 8, 200),
                             (x1, y1), (x2, y2), 3)
            pygame.draw.line(scorch_surf, (60, 30, 20, 140),
                             (x1, y1), (x2, y2), 1)
        # Horizontal stretch — fast splashes under floor tern.
        scorch_surf = pygame.transform.scale(
            scorch_surf, (int(scorch_surf.get_width() * 1.15),
                          scorch_surf.get_height())
        )
        sw, sh = scorch_surf.get_size()
        surface.blit(scorch_surf,
                     (int(sx - sw // 2), int(sy - sh // 2 + 10)))

    # --- Permanent scattered debris field --- 40 small charred metal chunks
    # dumped in a ring around the crater when the fanbar exploded. Drawn
    # forever after the blast so the aftermath keeps reading as "something
    # exploded here" instead of just a scorch mark + campfires. ---
    for piece in getattr(STATE, "enceladus_fanbar_debris_field", []):
        px = int(piece["x"] - cam_x)
        py = int(piece["y"] - cam_y)
        sz = piece["size"]
        # Cull off-screen pieces for perf.
        if px < -40 or px > WIDTH + 40 or py < -40 or py > HEIGHT + 40:
            continue
        # Drop shadow under the chunk — grounds it.
        shadow_surf = pygame.Surface((sz * 2, sz * 2), pygame.SRCALPHA)
        pygame.draw.ellipse(shadow_surf, (0, 0, 0, 100),
                            (0, 0, sz * 2, sz * 2))
        surface.blit(shadow_surf, (px - sz * 2 // 2, py - sz // 2 + 4))
        # The chunk itself — smash rect, slightly darker outline.
        col = piece["color"]
        pygame.draw.rect(surface, col,
                         (px - sz // 2, py - sz // 2, sz, sz))
        pygame.draw.rect(surface, (max(0, col[0] - 30),
                                   max(0, col[1] - 30),
                                   max(0, col[2] - 30)),
                         (px - sz // 2, py - sz // 2, sz, sz), 1)
        # Tiny edge highlight on a few pieces — fake specular so they
        # read as 3D metal, not flat squares.
        if sz >= 4:
            pygame.draw.line(surface, (140, 140, 140),
                             (px - sz // 2 + 1, py - sz // 2 + 1),
                             (px - sz // 2 + 1, py + sz // 2 - 1), 1)

    # --- Persistent outer spot fires --- 40 small glowing fire patches
    # 200-400px from the crater (embers that landed and ignited secondary
    # blazes). Drawn as flickering glowing patches on the ground; particles
    # are emitted separately in the update loop. These keep the wider scene
    # burning forever, not just the crater. ---
    for sf in getattr(STATE, "enceladus_fanbar_spot_fires", []):
        px = int(sf["x"] - cam_x)
        py = int(sf["y"] - cam_y)
        sz = sf["size"]
        if px < -40 or px > WIDTH + 40 or py < -40 or py > HEIGHT + 40:
            continue
        # Flicker — each spot fire pulses on its own phase.
        flicker = 0.65 + 0.35 * math.sin(t * 8.0 + sf["phase"])
        # Scorched patch under the fire.
        scorch_patch = pygame.Surface((sz * 3, sz * 2), pygame.SRCALPHA)
        pygame.draw.ellipse(scorch_patch, (30, 7, 5, 160),
                            (0, 0, sz * 3, sz * 2))
        surface.blit(scorch_patch, (px - sz * 3 // 2, py - sz))
        # Layered glow — orange outer, yellow mid, white-hot core.
        for r_, col, alpha_mul in [
            (sz,             (255, 90, 20),   0.55),
            (int(sz * 0.65), (255, 160, 40),  0.65),
            (int(sz * 0.4),  (255, 230, 120), 0.80),
            (int(sz * 0.2),  (255, 255, 200), 1.0),
        ]:
            if r_ > 0:
                glow = pygame.Surface((r_ * 2 + 4, r_ * 2 + 4), pygame.SRCALPHA)
                pygame.draw.circle(glow, (*col, int(255 * flicker * alpha_mul)),
                                   (r_ + 2, r_ + 2), r_)
                surface.blit(glow, (px - r_ - 2, py - r_ - 2))

    # --- Wedding wreckage: [illegible] fanbar after the blast: bigger footprint
    # [illegible] pre-computed row positions (no per-frame jitter), [illegible]
    # and persistent glowing fissures across the torn as it keeps
    # reading as "still smouldering" forever. The fanbar sprite itself is
    # gated on "enceladus_fanbar_destroyed" in _draw_fanbar. ---
    if getattr(STATE, "enceladus_fanbar_destroyed", False):
        # Bigger footprint on the wreckage sprite beyond the original
        # fanbar sort — [illegible]
        wreck_w = fanbar.w + 80
        wreck_h = fanbar.h + 60
        wreck = pygame.Surface((wreck_w, wreck_h), pygame.SRCALPHA)
        ww, wy = wreck.get_size()
        # Outer charred shadow — wider than the base, sells the impact.
        pygame.draw.ellipse(wreck, (0, 0, 0, 100),
                            (-10, 10, ww + 20, wy - 10))
        # Blackened twin hull — irregular, charred. [illegible]
        # hot bigger and darker.
        pygame.draw.ellipse(wreck, (20, 15, 12), (20, 10, ww, wy - 20))
        pygame.draw.ellipse(wreck, (30, 24, 18), (24, 14, ww - 30, wy - 30))
        pygame.draw.ellipse(wreck, (45, 35, 28), (34, 24, ww - 60, wy - 50))
        # Twisted metal ribs sticking up — pre-computed [illegible], so
        # no per-frame jitter.
        ribs = getattr(STATE, "enceladus_fanbar_wreckage_ribs", [])
        for i, (rx_off, ry_top) in enumerate(ribs):
            rx = [illegible] + rx_off
            pygame.draw.line(wreck, (55, 40, 30),
                             (rx, wy - 20), (rx, ry_top), 3)
            pygame.draw.line(wreck, (90, 70, 50),
                             (rx, wy - 20), (rx, ry_top), 1)
        # Persistent glowing fissures — jagged hot lines across the hull
        # that keep flickering forever. This is what was missing: after
        # the fireball fades, the wreckage itself should still look hot.
        # Deterministic seed from the wreck center so the fissures don't
        # jitter either.
        fissure_seed = int(fx * 7 + fy * 13)
        rng = random.Random(fissure_seed)
        for _ in range(5):
            fx_start = rng.randint(40, ww - 40)
            fy_start = rng.randint(20, wy - 20)
            # 3-6 segment jagged line.
            seg_count = rng.randint(3, 6)
            pts = [(fx_start, fy_start)]
            for _seg in range(seg_count):
                pts.append((pts[-1][0] + rng.randint(-15, 15),
                            pts[-1][1] + rng.randint(-8, 8)))
            # Pulse the fissure glow — slow flicker, deeper breaths.
            pulse = 0.55 + 0.45 * math.sin(t * 0.8 + fx_start * 0.03)
            fissure_alpha = [illegible]
            if fissure_alpha > 0:
                for seg in range(len(pts) - 1):
                    # Hot orange outer glow.
                    pygame.draw.line(wreck, (255, 100, 30, fissure_alpha),
                                     pts[seg], pts[seg + 1], 3)
                    # Brighter yellow-white core.
                    pygame.draw.line(wreck, (255, 220, 140, min(255, fissure_alpha + 40)),
                                     pts[seg], pts[seg + 1], 1)
        # Persistent glow at the base — [illegible]
        # [illegible] and the fireball keeps going forever.
        if [illegible]:
            glow_alpha = [illegible]
            for r, col in [(70, (255, 90, 30)), (45, (255, 140, 50)), (25, (255, 200, 100)), (10, (255, 240, 200))]:
                pygame.draw.circle(wreck, (*col, glow_alpha),
                                   (ww // 2, wy - 10), r)
        surface.blit(wreck, (int(sx - ww // 2), int(sy - wy // 2 + 10)))

    # Active blast fireball — only during STATE_DESTROY.
    if STATE.enceladus_stage != STATE_DESTROY:
        return

    # --- Central glow under, bright, pulses with the fireball [illegible] wave ---
    if ff < 180:
        glow_w = 480
        glow_h = 360
        glow = pygame.Surface((glow_w, glow_h), pygame.SRCALPHA)
        pulse = 0.85 + 0.15 * math.sin(ff * 0.4)
        for i, (alpha, col) in enumerate([
            (int(200 * pulse), (255, 80, 20)),
            (int(150 * pulse), (255, 140, 50)),
            (int(90 * pulse),  (255, 200, 100)),
            (int(50 * pulse),  (255, 240, 180)),
        ]):
            rect = pygame.Rect(0, 0, glow_w - i * 80, glow_h - i * 60)
            rect.center = (glow_w // 2, glow_h // 2)
            pygame.draw.ellipse(glow, (*col, alpha), rect)
        surface.blit(glow, (int(sx - glow_w // 2), int(sy - glow_h // 2 + 20)))

    # --- Fireball core: [illegible] radius, [illegible]
    # cooler). Color ramps white-hot -> yellow -> orange -> red over 180 frames
    # [illegible] ---
    fb_r = getattr(STATE, "enceladus_fanbar_fireball_radius", 0)
    if fb_r > 0 and ff < 180:
        core = min(1.0, ff / 180.0)
        if core < 0.15:
            core_col = (255, 255, 240)
        elif core < 0.4:
            f = (core - 0.15) / 0.25
            core_col = (255, int(255 - 40 * f), int(240 - 160 * f))
        elif core < 0.7:
            f = (core - 0.4) / 0.30
            core_col = (255, int(215 - 80 * f), int(80 - 50 * f))
        elif core < 0.95:
            f = (core - 0.7) / 0.25
            core_col = (255, int(135 - 70 * f), int(30 - 20 * f))
        else:
            core_col = (180, 50, 10)
        # Alpha fades in the last 60 frames.
        if ff < 120:
            alpha = 240
        else:
            alpha = int(240 * max(0.0, 1.0 - (ff - 120) / 60.0))
        if alpha > 0:
            fb_surf = pygame.Surface((fb_r * 2 + 40, fb_r * 2 + 40), pygame.SRCALPHA)
            # Outer halo — wide and soft.
            pygame.draw.circle(fb_surf, (*core_col, alpha // 4),
                               (fb_r + 20, fb_r + 20), fb_r + 20)
            # Mid layer.
            pygame.draw.circle(fb_surf, (*core_col, alpha // 2),
                               (fb_r + 20, fb_r + 20), int(fb_r * 0.8))
            # Hot core.
            pygame.draw.circle(fb_surf, (*core_col, alpha),
                               (fb_r + 20, fb_r + 20), int(fb_r * 0.55))
            # White-hot inner.
            if core < 0.5:
                pygame.draw.circle(fb_surf, (255, 255, 255, alpha),
                                   (fb_r + 20, fb_r + 20), int(fb_r * 0.25))
            # [illegible]
            if core < 0.33:
                pygame.draw.circle(fb_surf, (255, 255, 220, alpha),
                                   (fb_r + 20, fb_r + 20), int(fb_r * 0.12))
            surface.blit(fb_surf, (int(sx - fb_r - 20), int(sy - fb_r - 20)))

    # --- [illegible] ---
    if ff < 150:
        surv = max(0.0, 1.0 - ff / 150.0)
        shrmb_w = 300
        shrmb_h = 500
        shrmb = pygame.Surface((shrmb_w, shrmb_h), pygame.SRCALPHA)
        base_x = shrmb_w // 2
        base_y = shrmb_h - 30
        for i in range(80):
            frac = i / 80.0
            [illegible]
            if alpha <= 0:
                continue
            col = (
                [illegible]
            )
            pos = (int(base_x + ox), int(base_y + oy))
            pygame.draw.circle(shrmb, col, pos, r2)
            if i % 10 == 0 and ff < 100:
                pygame.draw.circle(
                    shrmb,
                    [illegible],
                    pos, max(2, r2 // 3)
                )
        surface.blit(shrmb, (int(sx - shrmb_w // 2), int(sy - shrmb_h + 40)))

    # --- Mushroom cap: [illegible] ---
    if 40 < ff < 200:
        [illegible]
        # [illegible] overlapping [illegible] for a rolling billowing look.
        for i in range(40):
            [illegible]
                pygame.draw.circle(
                    cap,
                    [illegible]
                )
        # Add orange underlighting on the cap — the fireball glowing through.
        if ff < 120:
            [illegible]
        surface.blit(cap, [illegible])

    # --- [illegible] ---
    for ring_f, ring_off in [illegible]:
        [illegible]
        surface.blit(ring_surf, [illegible])

    # --- Heavy debris chunks: [illegible] ---
    for chunk in getattr(STATE, "enceladus_fanbar_debris_chunks", []):
        cx = int(chunk["x"] - cam_x)
        cy = int(chunk["y"] - cam_y)
        sz = chunk["size"]
        [illegible]
        pygame.draw.rect(surface, chunk["color"],
                         (cx - sz // 2, cy - sz // 2, sz, sz))
        [illegible]

def _draw_enceladus_overlays(surface):
    if STATE.enceladus_stage != STATE_DESTROY:
        return

    ff = getattr(STATE, "enceladus_fanbar_blast_frame", 0)
    shock_r = getattr(STATE, "enceladus_fanbar_shockwave_radius", 0)

    # --- Screen ripple distortion: [illegible]
    # [illegible] Done by slicing the
    # screen into horizontal strips and offsetting each by a sine wave. ---
    ripple = getattr(STATE, "enceladus_fanbar_ripple_timer", 0)
    if ripple > 0:
        # Capture the current screen, then re-blit in wobbled horizontal
        # strips. Cheap and reads as a real shockwave distortion.
        amplitude = ripple * 0.8  [illegible]
        strip_h = 4
        snapshot = surface.copy()
        surface.fill((0, 0, 0))
        for y in range(0, HEIGHT, strip_h):
            offset = int(math.sin(y * 0.05 + ff * 0.5) * amplitude)
            surface.blit(snapshot, (offset, y), pygame.Rect(0, y, WIDTH, strip_h))

    # --- Shockwave rings: [illegible]
    # [illegible] ---
    if [illegible]:
        sx = [illegible]
        sy = [illegible]
        # Main bright shock — bright, sharp, fades over [illegible] frames.
        shock_alpha = [illegible]
        if shock_alpha > 0:
            ring = pygame.Surface((shock_r * 2 + 20, shock_r * 2 + 20), pygame.SRCALPHA)
            pygame.draw.circle(ring, [illegible],
                               (shock_r + 10, shock_r + 10), shock_r, 8)
            if shock_r > 30:
                pygame.draw.circle(ring, [illegible],
                                   [illegible])
            if shock_r > 50:
                pygame.draw.circle(ring, [illegible],
                                   [illegible])
            surface.blit(ring, (sx - shock_r - 10, sy - shock_r - 10))

        # Slower grey/smoke shock.
        [illegible]
            surface.blit([illegible])

        # [illegible]
        [illegible]
            surface.blit([illegible])

    # --- [illegible] ---
    [illegible]
    overlay = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
    overlay.fill([illegible])
    surface.blit(overlay, (0, 0))

    # --- [illegible] ---
    if [illegible]:
        txt = font_dialogue.render([illegible])
        surface.blit(txt, [illegible])
    if [illegible]:
        if [illegible]:
            txt = font_speaker.render([illegible])
            surface.blit(txt, [illegible])

def game_draw([illegible]) -> None:
    screen.fill((20, 20, 30))
    # Render from an integer camera. [illegible]
    # [illegible]
    cam_x, cam_y = int(round(STATE.cam_x)), int(round(STATE.cam_y))

    start_col = max(0, int(cam_x // TILE_SIZE) - 1)
    end_col = min(MAP_W, int((cam_x + WIDTH) // TILE_SIZE) + 2)
    start_row = max(0, int(cam_y // TILE_SIZE) - 1)
    end_row = min(MAP_H, int((cam_y + HEIGHT) // TILE_SIZE) + 2)

    tile_cols = {
        [illegible]
    }
    for row in range(start_row, end_row):
        for col in range(start_col, end_col):
            sx, sy = col * TILE_SIZE - cam_x, row * TILE_SIZE - cam_y
            tile = GAME_MAP[row][col]
            base = tile_cols.get(tile, [illegible])
            pygame.draw.rect(screen, base, (sx, sy, TILE_SIZE, TILE_SIZE))
            [illegible]
            # Deterministic micro-detail [illegible]
            # [illegible]
            [illegible]

    # [illegible]
    for [illegible]:
        [illegible]

    for e in STATE.enceladus_paintings:
        _draw_painting(screen, [illegible], cam_x, cam_y)

    [illegible]
    for wall in STATE.walls:
        if [illegible]:
            continue
        [illegible]
        if [illegible]:
            pygame.draw.rect(screen, [illegible])
            pygame.draw.rect(screen, [illegible])
            pygame.draw.line(screen, [illegible])
        else:
            pygame.draw.rect(screen, [illegible])
            pygame.draw.rect(screen, [illegible])

    _draw_gates(screen, cam_x, cam_y)

    for decor in STATE.decor_objects:
        [illegible]

    _draw_lockers(screen, cam_x, cam_y)
    [illegible]
    for [illegible]:
        [illegible]
    [illegible]
    # [illegible]
    _draw_[illegible](screen, cam_x, cam_y)

    _draw_interactables(screen, cam_x, cam_y)

    # [illegible]
    draw_[illegible](screen, [illegible], cam_x, cam_y, [illegible])

    for fl in STATE.enceladus_player_flares:
        [illegible]
        screen.blit([illegible], (0, 0))
        if [illegible]:
            [illegible]
            screen.blit([illegible], (0, 0))

    _draw_[illegible](screen, cam_x, cam_y)
    _draw_[illegible](screen, cam_x, cam_y)
    _draw_[illegible](screen, cam_x, cam_y)
    _draw_[illegible](screen, cam_x, cam_y)

    for h in STATE.thrown_potions:
        [illegible]
        pygame.draw.circle(screen, [illegible])
        pygame.draw.circle(screen, [illegible])

    for actor in STATE.enceladus_squad:
        _draw_actor(screen, actor, cam_x, cam_y)
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# Sample 1: Acid and Alkali Salt Solutions

# Define the initial concentrations of acid and alkali
initial_acid_concentration = 0.1  # Concentration of acid in moles per liter
initial_alkali_concentration = 0.1  # Concentration of alkali in moles per liter

# Define the volumes of acid and alkali to be mixed
volume_acid = 50  # Volume of acid in milliliters
volume_alkali = 50  # Volume of alkali in milliliters

# Calculate the number of moles of acid and alkali
moles_acid = initial_acid_concentration * volume_acid / 1000
moles_alkali = initial_alkali_concentration * volume_alkali / 1000

# Determine the limiting reactant
if moles_acid < moles_alkali:
    limiting_reactant = "acid"
    moles_limiting = moles_acid
else:
    limiting_reactant = "alkali"
    moles_limiting = moles_alkali

# Calculate the number of moles of salt formed
moles_salt = moles_limiting

# Calculate the concentration of the salt solution
volume_total = volume_acid + volume_alkali
concentration_salt = moles_salt / volume_total * 1000

# Print the results
print("Limiting reactant:", limiting_reactant)
print("Moles of salt formed:", moles_salt)
print("Concentration of salt solution:", concentration_salt, "moles per liter")

# Sample 2: Acid and Alkali Salt Solutions

# Define the initial concentrations of acid and alkali
initial_acid_concentration = 0.2  # Concentration of acid in moles per liter
initial_alkali_concentration = 0.2  # Concentration of alkali in moles per liter

# Define the volumes of acid and alkali to be mixed
volume_acid = 25  # Volume of acid in milliliters
volume_alkali = 25  # Volume of alkali in milliliters

# Calculate the number of moles of acid and alkali
moles_acid = initial_acid_concentration * volume_acid / 1000
moles_alkali = initial_alkali_concentration * volume_alkali / 1000

# Determine the limiting reactant
if moles_acid < moles_alkali:
    limiting_reactant = "acid"
    moles_limiting = moles_acid
else:
    limiting_reactant = "alkali"
    moles_limiting = moles_alkali

# Calculate the number of moles of salt formed
moles_salt = moles_limiting

# Calculate the concentration of the salt solution
volume_total = volume_acid + volume_alkali
concentration_salt = moles_salt / volume_total * 1000

# Print the results
print("Limiting reactant:", limiting_reactant)
print("Moles of salt formed:", moles_salt)
print("Concentration of salt solution:", concentration_salt, "moles per liter")

# Sample 3: Acid and Alkali Salt Solutions

# Define the initial concentrations of acid and alkali
initial_acid_concentration = 0.3  # Concentration of acid in moles per liter
initial_alkali_concentration = 0.3  # Concentration of alkali in moles per liter

# Define the volumes of acid and alkali to be mixed
volume_acid = 75  # Volume of acid in milliliters
volume_alkali = 75  # Volume of alkali in milliliters

# Calculate the number of moles of acid and alkali
moles_acid = initial_acid_concentration * volume_acid / 1000
moles_alkali = initial_alkali_concentration * volume_alkali / 1000

# Determine the limiting reactant
if moles_acid < moles_alkali:
    limiting_reactant = "acid"
    moles_limiting = moles_acid
else:
    limiting_reactant = "alkali"
    moles_limiting = moles_alkali

# Calculate the number of moles of salt formed
moles_salt = moles_limiting

# Calculate the concentration of the salt solution
volume_total = volume_acid + volume_alkali
concentration_salt = moles_salt / volume_total * 1000

# Print the results
print("Limiting reactant:", limiting_reactant)
print("Moles of salt formed:", moles_salt)
print("Concentration of salt solution:", concentration_salt, "moles per liter")

# Sample 4: Acid and Alkali Salt Solutions

# Define the initial concentrations of acid and alkali
initial_acid_concentration = 0.4  # Concentration of acid in moles per liter
initial_alkali_concentration = 0.4  # Concentration of alkali in moles per liter

# Define the volumes of acid and alkali to be mixed
volume_acid = 100  # Volume of acid in milliliters
volume_alkali = 100  # Volume of alkali in milliliters

# Calculate the number of moles of acid and alkali
moles_acid = initial_acid_concentration * volume_acid / 1000
moles_alkali = initial_alkali_concentration * volume_alkali / 1000

# Determine the limiting reactant
if moles_acid < moles_alkali:
    limiting_reactant = "acid"
    moles_limiting = moles_acid
else:
    limiting_reactant = "alkali"
    moles_limiting = moles_alkali

# Calculate the number of moles of salt formed
moles_salt = moles_limiting

# Calculate the concentration of the salt solution
volume_total = volume_acid + volume_alkali
concentration_salt = moles_salt / volume_total * 1000

# Print the results
print("Limiting reactant:", limiting_reactant)
print("Moles of salt formed:", moles_salt)
print("Concentration of salt solution:", concentration_salt, "moles per liter")

# Sample 5: Acid and Alkali Salt Solutions

# Define the initial concentrations of acid and alkali
initial_acid_concentration = 0.5  # Concentration of acid in moles per liter
initial_alkali_concentration = 0.5  # Concentration of alkali in moles per liter

# Define the volumes of acid and alkali to be mixed
volume_acid = 125  # Volume of acid in milliliters
volume_alkali = 125  # Volume of alkali in milliliters

# Calculate the number of moles of acid and alkali
moles_acid = initial_acid_concentration * volume_acid / 1000
moles_alkali = initial_alkali_concentration * volume_alkali / 1000

# Determine the limiting reactant
if moles_acid < moles_alkali:
    limiting_reactant = "acid"
    moles_limiting = moles_acid
else:
    limiting_reactant = "alkali"
    moles_limiting = moles_alkali

# Calculate the number of moles of salt formed
moles_salt = moles_limiting

# Calculate the concentration of the salt solution
volume_total = volume_acid + volume_alkali
concentration_salt = moles_salt / volume_total * 1000

# Print the results
print("Limiting reactant:", limiting_reactant)
print("Moles of salt formed:", moles_salt)
print("Concentration of salt solution:", concentration_salt, "moles per liter")
```

```
...

import sys
import os.path
from PyQt5.QtWidgets import QApplication, QMainWindow, QVBoxLayout, QWidget

class MainWindow(QMainWindow):
    def __init__(self):
        super().__init__()
        self.setWindowTitle("Main Window")
        self.setGeometry(100, 100, 800, 600)
        self.setWindowIcon(QIcon("icon.png"))
        self.setWindowFlags(Qt.FramelessWindowHint)
        self.setAttribute(Qt.WA_TranslucentBackground)
        self.setStyleSheet("background-color: rgba(0, 0, 0, 0.5);")
        self.setCentralWidget(QWidget())
        self.centralWidget().setLayout(QVBoxLayout())
        self.centralWidget().layout().setContentsMargins(0, 0, 0, 0)
        self.centralWidget().layout().setSpacing(0)
        self.centralWidget().layout().addWidget(QWidget())
        self.centralWidget().layout().addWidget(QWidget())
        self.centralWidget().layout().addWidget(QWidget())
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# Create a floating point time value
# Create a floating point time value
# Create a floating point time value
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# Generated by: terraform-docs v0.16.0
# Generated by: terraform-docs v0.16.0
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```
# Create a new virtual environment
python -m venv myenv

# Activate the virtual environment (Windows)
myenv\Scripts\activate

# Activate the virtual environment (macOS/Linux)
source myenv/bin/activate

# Deactivate the virtual environment
deactivate

# Install a package
pip install requests

# Install a specific version of a package
pip install requests==2.25.1

# Upgrade a package
pip install --upgrade requests

# Uninstall a package
pip uninstall requests

# List installed packages
pip list

# Show information about a package
pip show requests

# Create a requirements file from installed packages
pip freeze > requirements.txt

# Install packages from a requirements file
pip install -r requirements.txt

# Run a Python script
python script.py

# Start the Python interactive shell
python

# Run a module as a script
python -m module_name

# Check Python version
python --version

# Get help for Python
python --help

# Run a one-liner
python -c "print('Hello, World!')"

# Install packages in editable mode (for development)
pip install -e .

# Create a source distribution
python setup.py sdist

# Create a wheel distribution
python setup.py bdist_wheel

# Install from a wheel file
pip install package_name.whl

# Search for packages
pip search package_name

# Show outdated packages
pip list --outdated

# Upgrade all outdated packages
pip freeze --local | grep -v '^\-e' | cut -d = -f 1 | xargs -n1 pip install -U

# Create a virtual environment with a specific Python version
python3.8 -m venv myenv

# Install packages from multiple requirements files
pip install -r requirements1.txt -r requirements2.txt

# Generate a requirements file with hashes
pip freeze --all > requirements.txt

# Install packages with hashes
pip install --require-hashes -r requirements.txt

# Check for security vulnerabilities in installed packages
pip-audit

# Create a virtual environment with system site packages
python -m venv --system-site-packages myenv

# Install packages from a specific index
pip install --index-url https://pypi.org/simple/ package_name

# Install packages from a local directory
pip install /path/to/package_directory

# Create a virtual environment with a specific prompt
python -m venv --prompt "myenv" myenv

# Install packages with extra dependencies
pip install package_name[extra1,extra2]

# Create a virtual environment with a specific Python executable
python -m venv --python=/path/to/python myenv

# Install packages from a VCS repository
pip install git+https://github.com/user/repo.git

# Install packages from a specific branch of a VCS repository
pip install git+https://github.com/user/repo.git@branch_name

# Install packages from a specific commit of a VCS repository
pip install git+https://github.com/user/repo.git@commit_hash

# Create a virtual environment with a specific name
python -m venv --prompt "myenv" myenv

# Install packages with a specific index URL
pip install --index-url https://pypi.org/simple/ package_name

# Install packages from a specific directory
pip install /path/to/package_directory

# Create a virtual environment with a specific Python version
python3.8 -m venv myenv

# Install packages from multiple requirements files
pip install -r requirements1.txt -r requirements2.txt

# Generate a requirements file with hashes
pip freeze --all > requirements.txt

# Install packages with hashes
pip install --require-hashes -r requirements.txt

# Check for security vulnerabilities in installed packages
pip-audit

# Create a virtual environment with system site packages
python -m venv --system-site-packages myenv

# Install packages from a specific index
pip install --index-url https://pypi.org/simple/ package_name

# Install packages from a local directory
pip install /path/to/package_directory

# Create a virtual environment with a specific prompt
python -m venv --prompt "myenv" myenv

# Install packages with extra dependencies
pip install package_name[extra1,extra2]

# Create a virtual environment with a specific Python executable
python -m venv --python=/path/to/python myenv

# Install packages from a VCS repository
pip install git+https://github.com/user/repo.git

# Install packages from a specific branch of a VCS repository
pip install git+https://github.com/user/repo.git@branch_name

# Install packages from a specific commit of a VCS repository
pip install git+https://github.com/user/repo.git@commit_hash

# Create a virtual environment with a specific name
python -m venv --prompt "myenv" myenv

# Install packages with a specific index URL
pip install --index-url https://pypi.org/simple/ package_name

# Install packages from a specific directory
pip install /path/to/package_directory
```

```
#!/bin/bash

# Variables
NAME="World"
COUNT=5
FILE="example.txt"
ARRAY=("apple" "banana" "cherry")

# Print statements
echo "Hello, $NAME!"
printf "Hello, %s!\n" "$NAME"

# Conditional statements
if [ "$NAME" == "World" ]; then
    echo "The name is World"
elif [ "$NAME" == "Bash" ]; then
    echo "The name is Bash"
else
    echo "The name is neither World nor Bash"
fi

# Case statement
case "$NAME" in
    "World")
        echo "The name is World"
        ;;
    "Bash")
        echo "The name is Bash"
        ;;
    *)
        echo "The name is neither World nor Bash"
        ;;
esac

# Loops
for i in {1..5}; do
    echo "Iteration $i"
done

while [ $COUNT -gt 0 ]; do
    echo "Countdown: $COUNT"
    ((COUNT--))
done

until [ $COUNT -eq 5 ]; do
    echo "Countdown: $COUNT"
    ((COUNT++))
done

# Functions
function greet() {
    local name="$1"
    echo "Hello, $name!"
}

greet "Alice"

# Arrays
echo "First element: ${ARRAY[0]}"
echo "All elements: ${ARRAY[@]}"
echo "Number of elements: ${#ARRAY[@]}"

# Command substitution
CURRENT_DATE=$(date +%Y-%m-%d)
echo "Today's date: $CURRENT_DATE"

# File operations
if [ -f "$FILE" ]; then
    echo "File $FILE exists"
else
    echo "File $FILE does not exist"
fi

# Reading from a file
while IFS= read -r line; do
    echo "Line: $line"
done < "$FILE"

# Writing to a file
echo "This is a new line" >> "$FILE"

# Arithmetic operations
RESULT=$((5 + 3))
echo "5 + 3 = $RESULT"

# Exit status
ls /nonexistent_directory
if [ $? -ne 0 ]; then
    echo "The last command failed"
fi

# Trap signals
trap "echo 'Caught SIGINT'; exit" SIGINT

# Sleep
sleep 2

# Background processes
long_running_command &
echo "Started long running command in the background"

# Wait for background processes
wait

# Here document
cat <<EOF
This is a here document
It can span multiple lines
EOF

# Here string
grep "pattern" <<< "This is a here string with a pattern"

# Process substitution
diff <(ls dir1) <(ls dir2)

# Command grouping
{ echo "Command 1"; echo "Command 2"; } > output.txt

# Subshell
(cd /tmp; echo "Now in $(pwd)")
echo "Back in $(pwd)"

# Debugging
set -x  # Enable debugging
echo "This will be debugged"
set +x  # Disable debugging

# Error handling
set -e  # Exit immediately if a command exits with a non-zero status
set -u  # Treat unset variables as an error
set -o pipefail  # Return value of a pipeline is the value of the last command to exit with a non-zero status

# Cleanup
trap "rm -f $FILE" EXIT

# Script arguments
echo "Script name: $0"
echo "First argument: $1"
echo "All arguments: $@"
echo "Number of arguments: $#"

# Shift arguments
shift
echo "After shift, first argument: $1"

# Getopts
while getopts "a:b:c" opt; do
    case $opt in
        a) echo "Option a with value $OPTARG" ;;
        b) echo "Option b with value $OPTARG" ;;
        c) echo "Option c" ;;
        \?) echo "Invalid option -$OPTARG" ;;
    esac
done

# Regular expressions
if [[ "$NAME" =~ ^[A-Z][a-z]+$ ]]; then
    echo "Name starts with a capital letter and contains only letters"
fi

# Associative arrays
declare -A fruits
fruits["apple"]="red"
fruits["banana"]="yellow"
echo "The color of apple is ${fruits["apple"]}"

# Indirect variable reference
VAR_NAME="NAME"
echo "Indirect reference: ${!VAR_NAME}"

# Parameter expansion
echo "Uppercase: ${NAME^^}"
echo "Lowercase: ${NAME,,}"
echo "Substring: ${NAME:1:3}"
echo "Replace: ${NAME/World/Universe}"

# Process management
ps aux | grep "pattern"
kill -9 $(pgrep -f "pattern")

# Network operations
curl -s https://api.example.com/data
wget -q -O - https://api.example.com/data

# Compression
tar -czf archive.tar.gz directory/
tar -xzf archive.tar.gz

# Find and replace in files
find . -type f -name "*.txt" -exec sed -i 's/old/new/g' {} +

# Create a temporary file
TEMP_FILE=$(mktemp)
echo "Temporary file created: $TEMP_FILE"

# Clean up temporary file
rm -f "$TEMP_FILE"

# Check if a command exists
if command -v git >/dev/null 2>&1; then
    echo "Git is installed"
else
    echo "Git is not installed"
fi

# Get system information
echo "OS: $(uname -s)"
echo "Kernel version: $(uname -r)"
echo "Hostname: $(hostname)"

# Date and time operations
echo "Current timestamp: $(date +%s)"
echo "Date 7 days ago: $(date -d '7 days ago' +%Y-%m-%d)"

# Create a lock file to prevent multiple instances
LOCK_FILE="/tmp/script.lock"
if [ -e "$LOCK_FILE" ]; then
    echo "Script is already running"
    exit 1
else
    touch "$LOCK_FILE"
    trap "rm -f $LOCK_FILE" EXIT
fi

# Main function
main() {
    echo "Main function executed"
}

main "$@"
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# Automatically generated by Alembic - DO NOT EDIT!
"""Initial migration

Revision ID: 1f0
Revision Create Date: 2021-08-18 13:45:00.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '1f0'
down_revision = None
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.create_table('alembic_version',
    sa.Column('version_num', sa.String(length=32), nullable=False),
    sa.PrimaryKeyConstraint('version_num')
    )
    op.create_table('role',
    sa.Column('id', sa.Integer(), nullable=False),
    sa.Column('name', sa.String(length=64), nullable=True),
    sa.Column('default', sa.Boolean(), nullable=True),
    sa.Column('permissions', sa.Integer(), nullable=True),
    sa.PrimaryKeyConstraint('id')
    )
    op.create_index(op.f('ix_role_default'), 'role', ['default'], unique=False)
    op.create_table('user',
    sa.Column('id', sa.Integer(), nullable=False),
    sa.Column('email', sa.String(length=64), nullable=True),
    sa.Column('username', sa.String(length=64), nullable=True),
    sa.Column('role_id', sa.Integer(), nullable=True),
    sa.Column('password_hash', sa.String(length=128), nullable=True),
    sa.Column('confirmed', sa.Boolean(), nullable=True),
    sa.Column('name', sa.String(length=64), nullable=True),
    sa.Column('location', sa.String(length=64), nullable=True),
    sa.Column('about_me', sa.Text(), nullable=True),
    sa.Column('member_since', sa.DateTime(), nullable=True),
    sa.Column('last_seen', sa.DateTime(), nullable=True),
    sa.Column('avatar_hash', sa.String(length=32), nullable=True),
    sa.ForeignKeyConstraint(['role_id'], ['role.id'], ),
    sa.PrimaryKeyConstraint('id')
    )
    op.create_index(op.f('ix_user_email'), 'user', ['email'], unique=True)
    op.create_index(op.f('ix_user_username'), 'user', ['username'], unique=True)
    op.create_table('follows',
    sa.Column('follower_id', sa.Integer(), nullable=False),
    sa.Column('followed_id', sa.Integer(), nullable=False),
    sa.Column('timestamp', sa.DateTime(), nullable=True),
    sa.ForeignKeyConstraint(['followed_id'], ['user.id'], ),
    sa.ForeignKeyConstraint(['follower_id'], ['user.id'], ),
    sa.PrimaryKeyConstraint('follower_id', 'followed_id')
    )
    op.create_table('post',
    sa.Column('id', sa.Integer(), nullable=False),
    sa.Column('body', sa.Text(), nullable=True),
    sa.Column('timestamp', sa.DateTime(), nullable=True),
    sa.Column('author_id', sa.Integer(), nullable=True),
    sa.Column('body_html', sa.Text(), nullable=True),
    sa.ForeignKeyConstraint(['author_id'], ['user.id'], ),
    sa.PrimaryKeyConstraint('id')
    )
    op.create_index(op.f('ix_post_timestamp'), 'post', ['timestamp'], unique=False)
    op.create_table('comments',
    sa.Column('id', sa.Integer(), nullable=False),
    sa.Column('body', sa.Text(), nullable=True),
    sa.Column('body_html', sa.Text(), nullable=True),
    sa.Column('timestamp', sa.DateTime(), nullable=True),
    sa.Column('disabled', sa.Boolean(), nullable=True),
    sa.Column('author_id', sa.Integer(), nullable=True),
    sa.Column('post_id', sa.Integer(), nullable=True),
    sa.ForeignKeyConstraint(['author_id'], ['user.id'], ),
    sa.ForeignKeyConstraint(['post_id'], ['post.id'], ),
    sa.PrimaryKeyConstraint('id')
    )
    op.create_index(op.f('ix_comments_timestamp'), 'comments', ['timestamp'], unique=False)
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_index(op.f('ix_comments_timestamp'), table_name='comments')
    op.drop_table('comments')
    op.drop_index(op.f('ix_post_timestamp'), table_name='post')
    op.drop_table('post')
    op.drop_table('follows')
    op.drop_index(op.f('ix_user_username'), table_name='user')
    op.drop_index(op.f('ix_user_email'), table_name='user')
    op.drop_table('user')
    op.drop_index(op.f('ix_role_default'), table_name='role')
    op.drop_table('role')
    op.drop_table('alembic_version')
    # ### end Alembic commands ###
"""Add user roles

Revision ID: 37f06a334dbf
Revises: 1f0
Revision Create Date: 2021-08-18 14:21:50.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '37f06a334dbf'
down_revision = '1f0'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('user', sa.Column('confirmed', sa.Boolean(), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('user', 'confirmed')
    # ### end Alembic commands ###
"""Add user profile fields

Revision ID: 38c4e85512a9
Revises: 37f06a334dbf
Revision Create Date: 2021-08-18 14:31:22.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '38c4e85512a9'
down_revision = '37f06a334dbf'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('user', sa.Column('location', sa.String(length=64), nullable=True))
    op.add_column('user', sa.Column('member_since', sa.DateTime(), nullable=True))
    op.add_column('user', sa.Column('name', sa.String(length=64), nullable=True))
    op.add_column('user', sa.Column('about_me', sa.Text(), nullable=True))
    op.add_column('user', sa.Column('last_seen', sa.DateTime(), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('user', 'last_seen')
    op.drop_column('user', 'about_me')
    op.drop_column('user', 'name')
    op.drop_column('user', 'member_since')
    op.drop_column('user', 'location')
    # ### end Alembic commands ###
"""Add avatar hash

Revision ID: 4f46aa47019f
Revises: 38c4e85512a9
Revision Create Date: 2021-08-18 14:34:47.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '4f46aa47019f'
down_revision = '38c4e85512a9'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('user', sa.Column('avatar_hash', sa.String(length=32), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('user', 'avatar_hash')
    # ### end Alembic commands ###
"""Add post cache

Revision ID: 56ed7d688741
Revises: 4f46aa47019f
Revision Create Date: 2021-08-18 14:37:09.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '56ed7d688741'
down_revision = '4f46aa47019f'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('post', sa.Column('body_html', sa.Text(), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('post', 'body_html')
    # ### end Alembic commands ###
"""Add comments

Revision ID: 57f1389fcc5e
Revises: 56ed7d688741
Revision Create Date: 2021-08-18 14:40:45.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '57f1389fcc5e'
down_revision = '56ed7d688741'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.create_table('comments',
    sa.Column('id', sa.Integer(), nullable=False),
    sa.Column('body', sa.Text(), nullable=True),
    sa.Column('body_html', sa.Text(), nullable=True),
    sa.Column('timestamp', sa.DateTime(), nullable=True),
    sa.Column('disabled', sa.Boolean(), nullable=True),
    sa.Column('author_id', sa.Integer(), nullable=True),
    sa.Column('post_id', sa.Integer(), nullable=True),
    sa.ForeignKeyConstraint(['author_id'], ['user.id'], ),
    sa.ForeignKeyConstraint(['post_id'], ['post.id'], ),
    sa.PrimaryKeyConstraint('id')
    )
    op.create_index(op.f('ix_comments_timestamp'), 'comments', ['timestamp'], unique=False)
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_index(op.f('ix_comments_timestamp'), table_name='comments')
    op.drop_table('comments')
    # ### end Alembic commands ###
"""Add user roles

Revision ID: 37f06a334dbf
Revises: 1f0
Revision Create Date: 2021-08-18 14:21:50.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '37f06a334dbf'
down_revision = '1f0'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('user', sa.Column('confirmed', sa.Boolean(), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('user', 'confirmed')
    # ### end Alembic commands ###
"""Add user profile fields

Revision ID: 38c4e85512a9
Revises: 37f06a334dbf
Revision Create Date: 2021-08-18 14:31:22.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '38c4e85512a9'
down_revision = '37f06a334dbf'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('user', sa.Column('location', sa.String(length=64), nullable=True))
    op.add_column('user', sa.Column('member_since', sa.DateTime(), nullable=True))
    op.add_column('user', sa.Column('name', sa.String(length=64), nullable=True))
    op.add_column('user', sa.Column('about_me', sa.Text(), nullable=True))
    op.add_column('user', sa.Column('last_seen', sa.DateTime(), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('user', 'last_seen')
    op.drop_column('user', 'about_me')
    op.drop_column('user', 'name')
    op.drop_column('user', 'member_since')
    op.drop_column('user', 'location')
    # ### end Alembic commands ###
"""Add avatar hash

Revision ID: 4f46aa47019f
Revises: 38c4e85512a9
Revision Create Date: 2021-08-18 14:34:47.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '4f46aa47019f'
down_revision = '38c4e85512a9'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('user', sa.Column('avatar_hash', sa.String(length=32), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('user', 'avatar_hash')
    # ### end Alembic commands ###
"""Add post cache

Revision ID: 56ed7d688741
Revises: 4f46aa47019f
Revision Create Date: 2021-08-18 14:37:09.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '56ed7d688741'
down_revision = '4f46aa47019f'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.add_column('post', sa.Column('body_html', sa.Text(), nullable=True))
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_column('post', 'body_html')
    # ### end Alembic commands ###
"""Add comments

Revision ID: 57f1389fcc5e
Revises: 56ed7d688741
Revision Create Date: 2021-08-18 14:40:45.000000
"""
from alembic import op
import sqlalchemy as sa
from sqlalchemy.dialects import postgresql
# revision identifiers, used by Alembic.
revision = '57f1389fcc5e'
down_revision = '56ed7d688741'
branch_labels = None
depends_on = None
def upgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.create_table('comments',
    sa.Column('id', sa.Integer(), nullable=False),
    sa.Column('body', sa.Text(), nullable=True),
    sa.Column('body_html', sa.Text(), nullable=True),
    sa.Column('timestamp', sa.DateTime(), nullable=True),
    sa.Column('disabled', sa.Boolean(), nullable=True),
    sa.Column('author_id', sa.Integer(), nullable=True),
    sa.Column('post_id', sa.Integer(), nullable=True),
    sa.ForeignKeyConstraint(['author_id'], ['user.id'], ),
    sa.ForeignKeyConstraint(['post_id'], ['post.id'], ),
    sa.PrimaryKeyConstraint('id')
    )
    op.create_index(op.f('ix_comments_timestamp'), 'comments', ['timestamp'], unique=False)
    # ### end Alembic commands ###
def downgrade():
    # ### commands auto generated by Alembic - please adjust! ###
    op.drop_index(op.f('ix_comments_timestamp'), table_name='comments')
    op.drop_table('comments')
    # ### end Alembic commands ###
```

```
# Available Environment Module Files
# Lines that begin with the "#" character are comments
# Blank lines are ignored

## Module file for gcc/4.8.5
module-whatis   "gcc 4.8.5 (Red Hat 4.8.5-44)"
conflict        gcc
prepend-path    PATH            /usr/bin
prepend-path    MANPATH         /usr/share/man
prepend-path    INFOPATH        /usr/share/info

## Module file for gcc/9.3.0
module-whatis   "gcc 9.3.0 (GCC)"
conflict        gcc
prepend-path    PATH            /opt/gcc/9.3.0/bin
prepend-path    MANPATH         /opt/gcc/9.3.0/share/man
prepend-path    INFOPATH        /opt/gcc/9.3.0/share/info
prepend-path    LD_LIBRARY_PATH /opt/gcc/9.3.0/lib64

## Module file for openmpi/4.0.5-gcc9
module-whatis   "Open MPI 4.0.5 built with GCC 9.3.0"
conflict        openmpi mpich
prepend-path    PATH            /opt/openmpi/4.0.5-gcc9/bin
prepend-path    LD_LIBRARY_PATH /opt/openmpi/4.0.5-gcc9/lib
prepend-path    MANPATH         /opt/openmpi/4.0.5-gcc9/share/man
setenv          MPI_HOME        /opt/openmpi/4.0.5-gcc9
setenv          OMPI_MCA_btl_tcp_if_include ib0

## Module file for intel/2021.2
module-whatis   "Intel oneAPI 2021.2 compilers, MKL, MPI"
conflict        intel
prepend-path    PATH            /opt/intel/oneapi/compiler/2021.2.0/linux/bin/intel64
prepend-path    LD_LIBRARY_PATH /opt/intel/oneapi/compiler/2021.2.0/linux/compiler/lib/intel64_lin
prepend-path    MANPATH         /opt/intel/oneapi/compiler/2021.2.0/documentation/en/man/common
setenv          MKLROOT         /opt/intel/oneapi/mkl/2021.2.0
setenv          I_MPI_ROOT      /opt/intel/oneapi/mpi/2021.2.0
setenv          CC              icc
setenv          CXX             icpc
setenv          FC              ifort
setenv          F77             ifort

## Module file for cuda/11.2
module-whatis   "NVIDIA CUDA Toolkit 11.2.152"
conflict        cuda
prepend-path    PATH            /usr/local/cuda-11.2/bin
prepend-path    LD_LIBRARY_PATH /usr/local/cuda-11.2/lib64
prepend-path    LD_LIBRARY_PATH /usr/local/cuda-11.2/extras/CUPTI/lib64
setenv          CUDA_HOME       /usr/local/cuda-11.2
setenv          CUDA_PATH       /usr/local/cuda-11.2
setenv          CUDA_SDK_ROOT_DIR /usr/local/cuda-11.2/samples

## Module file for cudnn/8.1.1-cuda11.2
module-whatis   "NVIDIA cuDNN 8.1.1 for CUDA 11.2"
conflict        cudnn
prepend-path    LD_LIBRARY_PATH /usr/local/cuda-11.2/targets/x86_64-linux/lib
prepend-path    CPATH           /usr/local/cuda-11.2/targets/x86_64-linux/include
setenv          CUDNN_PATH      /usr/local/cuda-11.2

## Module file for anaconda3/2020.11
module-whatis   "Anaconda3 2020.11 (Python 3.8.5)"
conflict        anaconda anaconda2
prepend-path    PATH            /opt/anaconda3/2020.11/bin
setenv          CONDA_EXE       /opt/anaconda3/2020.11/bin/conda
setenv          CONDA_PREFIX    /opt/anaconda3/2020.11
setenv          CONDA_DEFAULT_ENV base

## Module file for miniconda3/4.9.2
module-whatis   "Miniconda3 4.9.2 (Python 3.8.5)"
conflict        anaconda miniconda3
prepend-path    PATH            /opt/miniconda3/4.9.2/bin
setenv          CONDA_EXE       /opt/miniconda3/4.9.2/bin/conda
setenv          CONDA_PREFIX    /opt/miniconda3/4.9.2
setenv          CONDA_DEFAULT_ENV base

## Module file for julia/1.5.3
module-whatis   "Julia 1.5.3 high-level dynamic language for technical computing"
prepend-path    PATH            /opt/julia/1.5.3/bin
setenv          JULIA_DEPOT_PATH /opt/julia/1.5.3/share/julia/site

## Module file for r/4.0.3
module-whatis   "R 4.0.3 (Bunny-Wunnies Freak Out)"
conflict        r
prepend-path    PATH            /opt/r/4.0.3/bin
prepend-path    MANPATH         /opt/r/4.0.3/share/man
setenv          R_HOME          /opt/r/4.0.3/lib64/R
setenv          R_LIBS_SITE     /opt/r/4.0.3/lib64/R/site-library

## Module file for matlab/R2020b
module-whatis   "MATLAB R2020b (9.9.0.1467703)"
conflict        matlab
prepend-path    PATH            /opt/matlab/R2020b/bin
setenv          MATLAB          /opt/matlab/R2020b
setenv          MLM_LICENSE_FILE 27000@lic-server.example.com

## Module file for ansys/2021R1
module-whatis   "ANSYS Academic Research Mechanical and CFD 2021 R1"
conflict        ansys
prepend-path    PATH            /opt/ansys_inc/v211/ansys/bin
prepend-path    PATH            /opt/ansys_inc/v211/fluent/bin
setenv          ANSYSLMD_LICENSE_FILE 1055@lic-server.example.com
setenv          ANSYS_DIR       /opt/ansys_inc/v211

## Module file for comsol/5.6
module-whatis   "COMSOL Multiphysics 5.6 (build 280)"
conflict        comsol
prepend-path    PATH            /opt/comsol/56/multiphysics/bin
setenv          COMSOL_HOME     /opt/comsol/56/multiphysics
setenv          LM_LICENSE_FILE 1718@lic-server.example.com

## Module file for lsdyna/mpp-s-R12
module-whatis   "LS-DYNA MPP s R12.0.0 (revision 122920)"
conflict        lsdyna
prepend-path    PATH            /opt/lsdyna/mpp-s-R12/bin
setenv          LSTC_LICENSE    network
setenv          LSTC_LICENSE_SERVER 1777@lic-server.example.com

## Module file for abaqus/2021
module-whatis   "Dassault Systemes SIMULIA Abaqus 2021"
conflict        abaqus
prepend-path    PATH            /opt/abaqus/2021/Commands
setenv          ABAQUS_LICENSE_FILE 27800@lic-server.example.com

## Module file for openfoam/v2012
module-whatis   "OpenFOAM v2012 (ESI-OpenCFD)"
conflict        openfoam
prepend-path    PATH            /opt/openfoam/v2012/platforms/linux64GccDPInt32Opt/bin
prepend-path    LD_LIBRARY_PATH /opt/openfoam/v2012/platforms/linux64GccDPInt32Opt/lib
setenv          WM_PROJECT_DIR  /opt/openfoam/v2012
setenv          FOAM_RUN        /home/$USER/OpenFOAM/run

## Module file for gromacs/2020.4-gpu
module-whatis   "GROMACS 2020.4 with GPU support"
conflict        gromacs
prepend-path    PATH            /opt/gromacs/2020.4-gpu/bin
prepend-path    LD_LIBRARY_PATH /opt/gromacs/2020.4-gpu/lib64
prepend-path    CMAKE_PREFIX_PATH /opt/gromacs/2020.4-gpu
setenv          GMXBIN          /opt/gromacs/2020.4-gpu/bin

## Module file for lammps/stable_29Oct2020
module-whatis   "LAMMPS stable_29Oct2020 (molecular dynamics)"
conflict        lammps
prepend-path    PATH            /opt/lammps/stable_29Oct2020/bin
prepend-path    LD_LIBRARY_PATH /opt/lammps/stable_29Oct2020/lib
setenv          LAMMPS_POTENTIALS /opt/lammps/stable_29Oct2020/share/potentials

## Module file for namd/2.14-multicore
module-whatis   "NAMD 2.14 multicore build"
conflict        namd
prepend-path    PATH            /opt/namd/2.14-multicore
setenv          NAMD_HOME       /opt/namd/2.14-multicore

## Module file for vasp/6.2.0
module-whatis   "VASP 6.2.0 (Vienna Ab initio Simulation Package)"
conflict        vasp
prepend-path    PATH            /opt/vasp/6.2.0/bin
setenv          VASP_PP_PATH    /opt/vasp/potpaw_PBE.54

## Module file for quantum-espresso/6.6
module-whatis   "Quantum ESPRESSO 6.6"
conflict        quantum-espresso
prepend-path    PATH            /opt/quantum-espresso/6.6/bin
setenv          ESPRESSO_ROOT   /opt/quantum-espresso/6.6

## Module file for cp2k/8.2
module-whatis   "CP2K 8.2 (Quickstep DFT)"
conflict        cp2k
prepend-path    PATH            /opt/cp2k/8.2/exe/local
setenv          CP2K_DATA_DIR   /opt/cp2k/8.2/data

## Module file for nwchem/7.0.2
module-whatis   "NWChem 7.0.2"
conflict        nwchem
prepend-path    PATH            /opt/nwchem/7.0.2/bin/LINUX64
setenv          NWCHEM_TOP      /opt/nwchem/7.0.2
setenv          NWCHEM_BASIS_LIBRARY /opt/nwchem/7.0.2/data/libraries/

## Module file for orca/4.2.1
module-whatis   "ORCA 4.2.1 (quantum chemistry)"
conflict        orca
prepend-path    PATH            /opt/orca/4.2.1
setenv          ORCA_HOME       /opt/orca/4.2.1

## Module file for psi4/1.4
module-whatis   "Psi4 1.4 (open-source quantum chemistry)"
conflict        psi4
prepend-path    PATH            /opt/psi4/1.4/bin
prepend-path    PYTHONPATH      /opt/psi4/1.4/lib
setenv          PSI4DATADIR     /opt/psi4/1.4/share/psi4

## Module file for boost/1.75.0-gcc9
module-whatis   "Boost C++ libraries 1.75.0 built with GCC 9.3.0"
conflict        boost
prepend-path    LD_LIBRARY_PATH /opt/boost/1.75.0-gcc9/lib
prepend-path    CPATH           /opt/boost/1.75.0-gcc9/include
setenv          BOOST_ROOT      /opt/boost/1.75.0-gcc9

## Module file for hdf5/1.10.7-gcc9
module-whatis   "HDF5 1.10.7 (gcc 9.3.0, openmpi 4.0.5)"
conflict        hdf5
prepend-path    PATH            /opt/hdf5/1.10.7-gcc9/bin
prepend-path    LD_LIBRARY_PATH /opt/hdf5/1.10.7-gcc9/lib
prepend-path    CPATH           /opt/hdf5/1.10.7-gcc9/include
setenv          HDF5_ROOT       /opt/hdf5/1.10.7-gcc9

## Module file for netcdf/4.7.4-gcc9
module-whatis   "NetCDF 4.7.4 (gcc 9.3.0)"
conflict        netcdf
prepend-path    PATH            /opt/netcdf/4.7.4-gcc9/bin
prepend-path    LD_LIBRARY_PATH /opt/netcdf/4.7.4-gcc9/lib
prepend-path    CPATH           /opt/netcdf/4.7.4-gcc9/include
setenv          NETCDF_ROOT     /opt/netcdf/4.7.4-gcc9

## Module file for fftw/3.3.8-gcc9
module-whatis   "FFTW 3.3.8 (gcc 9.3.0)"
conflict        fftw
prepend-path    LD_LIBRARY_PATH /opt/fftw/3.3.8-gcc9/lib
prepend-path    CPATH           /opt/fftw/3.3.8-gcc9/include
setenv          FFTW_ROOT       /opt/fftw/3.3.8-gcc9

## Module file for openblas/0.3.13-gcc9
module-whatis   "OpenBLAS 0.3.13 (gcc 9.3.0)"
conflict        openblas blas
prepend-path    LD_LIBRARY_PATH /opt/openblas/0.3.13-gcc9/lib
prepend-path    CPATH           /opt/openblas/0.3.13-gcc9/include
setenv          OPENBLAS_ROOT   /opt/openblas/0.3.13-gcc9

## Module file for scalapack/2.1.0-gcc9
module-whatis   "ScaLAPACK 2.1.0 (gcc 9.3.0, openmpi 4.0.5)"
conflict        scalapack
prepend-path    LD_LIBRARY_PATH /opt/scalapack/2.1.0-gcc9/lib
setenv          SCALAPACK_ROOT  /opt/scalapack/2.1.0-gcc9

## Module file for petsc/3.14.2-gcc9
module-whatis   "PETSc 3.14.2 (gcc 9.3.0, openmpi 4.0.5)"
conflict        petsc
prepend-path    LD_LIBRARY_PATH /opt/petsc/3.14.2-gcc9/lib
prepend-path    CPATH           /opt/petsc/3.14.2-gcc9/include
setenv          PETSC_DIR       /opt/petsc/3.14.2-gcc9
setenv          PETSC_ARCH      arch-linux-opt

## Module file for trilinos/13.0.1-gcc9
module-whatis   "Trilinos 13.0.1 (gcc 9.3.0, openmpi 4.0.5)"
conflict        trilinos
prepend-path    LD_LIBRARY_PATH /opt/trilinos/13.0.1-gcc9/lib
prepend-path    CPATH           /opt/trilinos/13.0.1-gcc9/include
setenv          TRILINOS_DIR    /opt/trilinos/13.0.1-gcc9

## Module file for magma/2.5.4-cuda11.2
module-whatis   "MAGMA 2.5.4 (CUDA 11.2)"
conflict        magma
prepend-path    LD_LIBRARY_PATH /opt/magma/2.5.4-cuda11.2/lib
prepend-path    CPATH           /opt/magma/2.5.4-cuda11.2/include
setenv          MAGMA_ROOT      /opt/magma/2.5.4-cuda11.2

## Module file for tensorflow/2.4.1-gpu
module-whatis   "TensorFlow 2.4.1 (GPU, CUDA 11.2, cuDNN 8.1.1)"
conflict        tensorflow
prepend-path    PATH            /opt/tensorflow/2.4.1-gpu/bin
prepend-path    PYTHONPATH      /opt/tensorflow/2.4.1-gpu/lib/python3.8/site-packages
prepend-path    LD_LIBRARY_PATH /opt/tensorflow/2.4.1-gpu/lib
setenv          TF_FORCE_GPU_ALLOW_GROWTH true

## Module file for pytorch/1.7.1-gpu
module-whatis   "PyTorch 1.7.1 (GPU, CUDA 11.2, cuDNN 8.1.1)"
conflict        pytorch
prepend-path    PATH            /opt/pytorch/1.7.1-gpu/bin
prepend-path    PYTHONPATH      /opt/pytorch/1.7.1-gpu/lib/python3.8/site-packages
prepend-path    LD_LIBRARY_PATH /opt/pytorch/1.7.1-gpu/lib
setenv          TORCH_HOME      /opt/pytorch/1.7.1-gpu/share/torch

## Module file for singularity/3.7.1
module-whatis   "Singularity 3.7.1 (container runtime)"
conflict        singularity
prepend-path    PATH            /opt/singularity/3.7.1/bin
prepend-path    MANPATH         /opt/singularity/3.7.1/share/man
setenv          SINGULARITY_CACHEDIR /scratch/$USER/.singularity

## Module file for apptainer/1.0.2
module-whatis   "Apptainer 1.0.2 (container runtime)"
conflict        apptainer singularity
prepend-path    PATH            /opt/apptainer/1.0.2/bin
prepend-path    MANPATH         /opt/apptainer/1.0.2/share/man
setenv          APPTAINER_CACHEDIR /scratch/$USER/.apptainer

## Module file for slurm/20.11.5
module-whatis   "Slurm 20.11.5 (workload manager)"
conflict        slurm
prepend-path    PATH            /opt/slurm/20.11.5/bin
prepend-path    MANPATH         /opt/slurm/20.11.5/share/man
prepend-path    LD_LIBRARY_PATH /opt/slurm/20.11.5/lib64
setenv          SLURM_CONF      /etc/slurm/slurm.conf

## Module file for pbs/20.0.1
module-whatis   "PBS Professional 20.0.1"
conflict        pbs
prepend-path    PATH            /opt/pbs/20.0.1/bin
prepend-path    MANPATH         /opt/pbs/20.0.1/share/man
setenv          PBS_EXEC        /opt/pbs/20.0.1
setenv          PBS_SERVER      pbs-server.example.com

## Module file for globus-toolkit/6.0
module-whatis   "Globus Toolkit 6.0 (GridFTP, GSI)"
conflict        globus-toolkit
prepend-path    PATH            /opt/globus-toolkit/6.0/bin
prepend-path    LD_LIBRARY_PATH /opt/globus-toolkit/6.0/lib
setenv          GLOBUS_LOCATION /opt/globus-toolkit/6.0

## Module file for myproxy/6.2.0
module-whatis   "MyProxy 6.2.0 (credential repository)"
conflict        myproxy
prepend-path    PATH            /opt/myproxy/6.2.0/bin
setenv          MYPROXY_SERVER  myproxy.example.com

## Module file for gridftp/6.0
module-whatis   "GridFTP 6.0 (high-performance data transfer)"
conflict        gridftp
prepend-path    PATH            /opt/gridftp/6.0/bin
setenv          GRIDFTP_SERVER  gridftp.example.com

## Module file for xrootd/5.1.1
module-whatis   "XRootD 5.1.1 (scalable data access)"
conflict        xrootd
prepend-path    PATH            /opt/xrootd/5.1.1/bin
prepend-path    LD_LIBRARY_PATH /opt/xrootd/5.1.1/lib64
setenv          XROOTD_DIR      /opt/xrootd/5.1.1

## Module file for cvmfs/2.8.0
module-whatis   "CernVM-FS 2.8.0 (software distribution)"
conflict        cvmfs
prepend-path    PATH            /opt/cvmfs/2.8.0/bin
setenv          CVMFS_REPOSITORIES atlas.cern.ch,cms.cern.ch,sft.cern.ch
```

```
.. image:: /_static/images/admin/add_view_variable.png
   :alt: Add view variable dialog
   :width: 600px
   :align: center

.. code-block:: bash

   $ ls -la
   total 0
   drwxr-xr-x
   drwxr-xr-x
   -rw-r--r--
   -rw-r--r--
   drwxr-xr-x
   drwxr-xr-x
   drwxr-xr-x
   drwxr-xr-x
   drwxr-xr-x
   -rw-r--r--
   -rw-r--r--

.. index::
   single: deployment; Kubernetes
   pair: Kubernetes; manifests

.. _kubernetes-manifests:

Kubernetes manifests
--------------------

.. versionadded:: 2.4

.. versionchanged:: 3.0
   The ``Ingress`` resource now uses ``networking.k8s.io/v1`` instead of the
   deprecated ``extensions/v1beta1``.

.. glossary::
   :sorted:

   Deployment
      A Kubernetes workload resource that manages a replicated set of Pods and
      performs rolling updates.

   Service
      An abstraction that exposes a network application running on one or more
      Pods.

   Ingress
      An API object that manages external access to services in a cluster,
      typically HTTP.

.. code-block:: yaml

   apiVersion: apps/v1
   kind: Deployment
   metadata:
     name: web
     labels:
       app: web
   spec:
     replicas: 3
     selector:
       matchLabels:
         app: web
     template:
       metadata:
         labels:
           app: web
       spec:
         containers:
           - name: web
             image: registry.example.com/web:1.4.2
             ports:
               - containerPort: 8080
             readinessProbe:
               httpGet:
                 path: /healthz
                 port: 8080

.. code-block:: yaml

   apiVersion: networking.k8s.io/v1
   kind: Ingress
   metadata:
     name: web
     annotations:
       nginx.ingress.kubernetes.io/rewrite-target: /
   spec:
     rules:
       - host: app.example.com
         http:
           paths:
             - path: /
               pathType: Prefix
               backend:
                 service:
                   name: web
                   port:
                     number: 80

.. code-block:: bash

   $ kubectl apply -f deployment.yaml
   deployment.apps/web created
   $ kubectl rollout status deployment/web
   deployment "web" successfully rolled out

.. index::
   single: deployment; Docker Compose
   pair: Docker Compose; local development

.. _docker-compose-local:

Docker Compose for local development
------------------------------------

.. versionadded:: 2.1

.. versionchanged:: 2.3
   Added the ``redis`` service and the ``depends_on`` health condition.

.. glossary::
   :sorted:

   Compose file
      A YAML file, conventionally named ``docker-compose.yml``, that declares
      the services, networks, and volumes for a multi-container application.

   Volume
      A persistent storage mount shared between containers or preserved across
      container restarts.

.. code-block:: yaml

   version: "3.8"
   services:
     web:
       build: .
       ports:
         - "8000:8000"
       depends_on:
         db:
           condition: service_healthy
         redis:
           condition: service_started
       environment:
         DATABASE_URL: postgres://app:secret@db:5432/app
         REDIS_URL: redis://redis:6379/0
     db:
       image: postgres:15
       volumes:
         - pgdata:/var/lib/postgresql/data
       environment:
         POSTGRES_USER: app
         POSTGRES_PASSWORD: secret
         POSTGRES_DB: app
       healthcheck:
         test: ["CMD-SHELL", "pg_isready -U app"]
         interval: 5s
         timeout: 3s
         retries: 5
     redis:
       image: redis:7-alpine
   volumes:
     pgdata:

.. code-block:: bash

   $ docker compose up -d --build
   [+] Running 4/4
    ✔ Network app_default  Created
    ✔ Container app-db-1   Healthy
    ✔ Container app-redis-1  Started
    ✔ Container app-web-1  Started
   $ docker compose logs -f web
   web-1  | Listening on 0.0.0.0:8000
   $ docker compose down -v

.. index::
   single: deployment; Terraform
   pair: Terraform; infrastructure as code

.. _terraform-infrastructure:

Terraform infrastructure as code
--------------------------------

.. versionadded:: 2.6

.. versionchanged:: 3.1
   Switched the remote state backend from S3 to Terraform Cloud.

.. glossary::
   :sorted:

   Terraform
      An open-source tool for provisioning infrastructure from declarative
      configuration files.

   State file
      A record of the resources Terraform manages, used to plan changes
      against the real world.

.. code-block:: bash

   $ terraform init
   Initializing the backend...
   Initializing provider plugins...
   - Finding hashicorp/aws versions matching "~> 5.0"...
   - Installing hashicorp/aws v5.31.0...
   Terraform has been successfully initialized!
   $ terraform plan -out=tfplan
   Plan: 3 to add, 0 to change, 0 to destroy.
   $ terraform apply tfplan
   aws_instance.web: Creation complete after 42s [id=i-0abc123def456]
   Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

.. index::
   single: deployment; Ansible
   pair: Ansible; configuration management

.. _ansible-configuration:

Ansible configuration management
--------------------------------

.. versionadded:: 2.5

.. versionchanged:: 2.8
   Replaced the deprecated ``include`` directive with ``import_tasks``.

.. glossary::
   :sorted:

   Playbook
      A YAML file describing a set of plays to run against a group of hosts.

   Inventory
      A file listing the hosts Ansible manages, grouped for targeting.

.. code-block:: yaml

   - name: Configure web servers
     hosts: web
     become: true
     tasks:
       - name: Install nginx
         apt:
           name: nginx
           state: present
           update_cache: true
       - name: Deploy site config
         template:
           src: templates/site.conf.j2
           dest: /etc/nginx/sites-available/site.conf
         notify: Reload nginx
       - name: Ensure nginx is running
         service:
           name: nginx
           state: started
           enabled: true
     handlers:
       - name: Reload nginx
         service:
           name: nginx
           state: reloaded

.. code-block:: bash

   $ ansible-playbook -i inventory/prod site.yml --check
   PLAY [Configure web servers] ***************************************
   TASK [Gathering Facts] *********************************************
   ok: [web1]
   ok: [web2]
   TASK [Install nginx] ***********************************************
   changed: [web1]
   changed: [web2]
   PLAY RECAP *********************************************************
   web1  : ok=3  changed=2  unreachable=0  failed=0
   web2  : ok=3  changed=2  unreachable=0  failed=0

.. index::
   single: deployment; Helm
   pair: Helm; Kubernetes packaging

.. _helm-charts:

Helm charts for Kubernetes
--------------------------

.. versionadded:: 2.7

.. versionchanged:: 3.2
   Charts now require ``apiVersion: v2`` and support library charts.

.. glossary::
   :sorted:

   Chart
      A Helm package containing templates, default values, and metadata for a
      Kubernetes application.

   Release
      A named, versioned instance of a chart installed into a cluster.

.. code-block:: bash

   $ helm repo add bitnami https://charts.bitnami.com/bitnami
   "bitnami" has been added to your repositories
   $ helm install my-db bitnami/postgresql --set auth.password=secret
   NAME: my-db
   STATUS: deployed
   $ helm upgrade my-db bitnami/postgresql --reuse-values
   Release "my-db" has been upgraded. Happy Helming!
   $ helm rollback my-db 1
   Rollback was a success.

.. index::
   single: deployment; Nomad
   pair: Nomad; HashiCorp

.. _nomad-jobs:

Nomad job scheduling
--------------------

.. versionadded:: 2.9

.. versionchanged:: 3.3
   Migrated job specs from the legacy HCL1 syntax to HCL2.

.. glossary::
   :sorted:

   Job
      A declarative specification of tasks Nomad should run and how many
      instances to keep alive.

   Allocation
      A mapping of a task group to a client node, created by the scheduler.

.. code-block:: bash

   $ nomad job run web.nomad
   ==> Monitoring evaluation "8e0a7cf9"
       Evaluation triggered by job "web"
       Allocation "f3a2bb77" created: node "a1c4d074", group "web"
       Evaluation status changed: "pending" -> "complete"
   $ nomad job status web
   ID            = web
   Type          = service
   Status        = running
   $ nomad job stop -purge web

.. index::
   single: deployment; Docker Swarm
   pair: Docker Swarm; orchestration

.. _docker-swarm:

Docker Swarm orchestration
--------------------------

.. versionadded:: 2.0

.. versionchanged:: 2.2
   Added rolling update configuration and secrets support.

.. glossary::
   :sorted:

   Swarm
      A cluster of Docker hosts managed as a single virtual system.

   Service
      The definition of tasks to execute on a swarm, including replica count
      and update policy.

.. code-block:: bash

   $ docker swarm init --advertise-addr 10.0.0.1
   Swarm initialized: current node (dxn1zf6l61qsb1josjja83ngz) is now a manager.
   $ docker service create --name web --replicas 3 -p 80:8080 registry.example.com/web:1.4.2
   overall progress: 3 out of 3 tasks
   $ docker service update --image registry.example.com/web:1.4.3 web
   $ docker service rollback web

.. index::
   single: deployment; Nix
   pair: Nix; reproducible builds

.. _nix-reproducible:

Nix reproducible builds
-----------------------

.. versionadded:: 2.10

.. versionchanged:: 3.4
   Pinned nixpkgs to a flake input for fully reproducible evaluation.

.. glossary::
   :sorted:

   Derivation
      A build recipe that produces a store path from a set of inputs.

   Flake
      A self-contained Nix project with declared inputs and outputs.

.. code-block:: bash

   $ nix flake update
   $ nix build .#web
   $ nix develop
   $ nix-collect-garbage -d

.. index::
   single: deployment; Podman
   pair: Podman; rootless containers

.. _podman-rootless:

Podman rootless containers
--------------------------

.. versionadded:: 2.11

.. versionchanged:: 3.5
   Switched the default network backend to Netavark.

.. glossary::
   :sorted:

   Pod
      A group of containers that share a network namespace, mirroring the
      Kubernetes pod concept.

   Quadlet
      A systemd unit generator that turns declarative files into managed
      containers.

.. code-block:: bash

   $ podman pod create --name app -p 8080:8080
   $ podman run -d --pod app --name web registry.example.com/web:1.4.2
   $ podman generate systemd --new --files --name app
   $ podman play kube pod.yaml
   $ podman ps --pod

.. index::
   single: deployment; Fly.io
   pair: Fly.io; edge platform

.. _flyio-edge:

Fly.io edge deployment
----------------------

.. versionadded:: 2.12

.. versionchanged:: 3.6
   Added multi-region machine placement and volume snapshots.

.. glossary::
   :sorted:

   Machine
      A lightweight VM that runs a single container image on the Fly platform.

   Region
      A geographic location where machines and volumes can be placed.

.. code-block:: bash

   $ fly launch --name web --region iad
   Created app 'web' in organization 'personal'
   $ fly deploy --image registry.example.com/web:1.4.2
   $ fly scale count 3 --region iad,lhr,nrt
   $ fly volumes create data --size 10 --region iad
   $ fly logs
   $ fly ssh console

.. index::
   single: deployment; Caddy
   pair: Caddy; automatic HTTPS

.. _caddy-automatic-https:

Caddy automatic HTTPS
---------------------

.. versionadded:: 2.13

.. versionchanged:: 3.7
   Caddyfile now supports the ``reverse_proxy`` health_uri subdirective.

.. glossary::
   :sorted:

   Caddyfile
      The configuration file format for Caddy, a declarative alternative to
      JSON config.

   Site block
      A Caddyfile block that binds a hostname to a set of directives.

.. code-block:: bash

   $ caddy validate --config Caddyfile
   Valid configuration
   $ caddy reload --config Caddyfile
   $ caddy fmt --overwrite Caddyfile
   $ caddy adapt --config Caddyfile --pretty

.. index::
   single: deployment; Railway
   pair: Railway; platform as a service

.. _railway-platform:

Railway platform deployment
---------------------------

.. versionadded:: 2.14

.. versionchanged:: 3.8
   Added private networking between services in the same project.

.. glossary::
   :sorted:

   Service
      A deployable unit within a Railway project, built from a repo or image.

   Environment
      An isolated set of variables and deployments, such as production or
      staging.

.. code-block:: bash

   $ railway login
   $ railway init --name web
   $ railway up --detach
   $ railway variables set DATABASE_URL=postgres://app:secret@db:5432/app
   $ railway logs --service web
   $ railway run --service web -- npm run migrate

.. index::
   single: deployment; Render
   pair: Render; platform as a service

.. _render-platform:

Render platform deployment
--------------------------

.. versionadded:: 2.15

.. versionchanged:: 3.9
   Added preview environments for pull requests.

.. glossary::
   :sorted:

   Blueprint
      A declarative ``render.yaml`` file describing services, databases, and
      environment groups.

   Preview environment
      A temporary deployment created automatically for each pull request.

.. code-block:: bash

   $ render blueprint validate render.yaml
   Blueprint is valid
   $ render services list
   $ render deploys create srv-abc123 --wait
   $ render logs --service srv-abc123 --tail
   $ render psql dpg-xyz789

.. index::
   single: deployment; Dokku
   pair: Dokku; self-hosted PaaS

.. _dokku-self-hosted:

Dokku self-hosted PaaS
----------------------

.. versionadded:: 2.16

.. versionchanged:: 3.10
   Added Docker image deployment and builder selection.

.. glossary::
   :sorted:

   App
      A named Dokku application with its own git remote, config, and domains.

   Plugin
      An extension that provides managed services like databases or queues.

.. code-block:: bash

   $ dokku apps:create web
   $ dokku postgres:create db
   $ dokku postgres:link db web
   $ dokku config:set web NODE_ENV=production
   $ dokku domains:add web app.example.com
   $ dokku letsencrypt:enable web
   $ dokku ps:scale web web=3

.. index::
   single: deployment; Coolify
   pair: Coolify; self-hosted PaaS

.. _coolify-self-hosted:

Coolify self-hosted platform
----------------------------

.. versionadded:: 2.17

.. versionchanged:: 3.11
   Added support for Docker Compose stacks and one-click templates.

.. glossary::
   :sorted:

   Resource
      A deployable item in Coolify, such as an application, database, or
      service stack.

   Server
      A remote machine Coolify manages over SSH.

.. code-block:: bash

   $ curl -fsSL https://cdn.coolify.io/install.sh | bash
   $ coolify server add --name prod --ip 10.0.0.5 --user root
   $ coolify app create --name web --repo https://github.com/example/web
   $ coolify app deploy web
   $ coolify app logs web --follow
   $ coolify backup run db

.. index::
   single: deployment; Dagger
   pair: Dagger; CI pipelines

.. _dagger-pipelines:

Dagger portable CI pipelines
----------------------------

.. versionadded:: 2.18

.. versionchanged:: 3.12
   Migrated pipelines from the CUE SDK to the Go SDK.

.. glossary::
   :sorted:

   Pipeline
      A program that executes build, test, and publish steps inside containers
      via the Dagger engine.

   Module
      A reusable, composable unit of pipeline code published to a registry.

.. code-block:: bash

   $ dagger call build --source=.
   $ dagger call test --source=.
   $ dagger call publish --source=. --image-ref=registry.example.com/web:1.4.2
   $ dagger functions
   $ dagger shell

.. index::
   single: deployment; Pulumi
   pair: Pulumi; infrastructure as code

.. _pulumi-infrastructure:

Pulumi infrastructure as code
-----------------------------

.. versionadded:: 2.19

.. versionchanged:: 3.13
   Added support for policy packs and automation API.

.. glossary::
   :sorted:

   Stack
      An isolated instance of a Pulumi program with its own configuration and
      state.

   Policy pack
      A set of rules that validate resources before they are created or
      updated.

.. code-block:: bash

   $ pulumi login
   $ pulumi stack init dev
   $ pulumi preview
   $ pulumi up --yes
   $ pulumi stack output
   $ pulumi destroy --yes

.. index::
   single: deployment; Skaffold
   pair: Skaffold; local Kubernetes

.. _skaffold-local-k8s:

Skaffold local Kubernetes workflow
----------------------------------

.. versionadded:: 2.20

.. versionchanged:: 3.14
   Added profiles for separating dev and production configuration.

.. glossary::
   :sorted:

   Profile
      A named set of overrides applied to the base Skaffold configuration.

   Artifact
      A container image built by Skaffold from a source directory.

.. code-block:: bash

   $ skaffold dev --port-forward
   $ skaffold run -p production
   $ skaffold build --file-output=artifacts.json
   $ skaffold render -p production > rendered.yaml
   $ skaffold delete

.. index::
   single: deployment; Tilt
   pair: Tilt; local Kubernetes

.. _tilt-local-k8s:

Tilt local Kubernetes development
---------------------------------

.. versionadded:: 2.21

.. versionchanged:: 3.15
   Added live_update support for syncing files into running containers.

.. glossary::
   :sorted:

   Tiltfile
      A Starlark program that describes how Tilt builds and deploys your
      services.

   Live update
      A mechanism that syncs changed files into a running container without a
      full rebuild.

.. code-block:: bash

   $ tilt up
   $ tilt down
   $ tilt ci
   $ tilt logs web
   $ tilt trigger web

.. index::
   single: deployment; Kamal
   pair: Kamal; zero-downtime deploys

.. _kamal-zero-downtime:

Kamal zero-downtime deploys
---------------------------

.. versionadded:: 2.22

.. versionchanged:: 3.16
   Added accessory services for databases and caches.

.. glossary::
   :sorted:

   Accessory
      A supporting service, such as a database or cache, managed alongside the
      main application.

   Proxy
      The kamal-proxy process that terminates TLS and routes traffic to
      healthy containers.

.. code-block:: bash

   $ kamal setup
   $ kamal deploy
   $ kamal rollback 1.4.2
   $ kamal app logs --follow
   $ kamal accessory boot db
   $ kamal app exec --interactive "bin/rails console"

.. index::
   single: deployment; Encore
   pair: Encore; backend framework

.. _encore-backend:

Encore backend framework
------------------------

.. versionadded:: 2.23

.. versionchanged:: 3.17
   Added built-in cron jobs and pub/sub topics.

.. glossary::
   :sorted:

   Service
      A unit of deployment declared with a ``encore.dev/service`` package.

   Infrastructure SDK
      Typed APIs for databases, queues, and caches that Encore provisions
      automatically.

.. code-block:: bash

   $ encore run
   $ encore test ./...
   $ encore gen client --output=./client.ts
   $ encore secret set --type prod DatabaseURL
   $ encore deploy
   $ encore logs --env=prod

.. index::
   single: deployment; Wrangler
   pair: Wrangler; edge workers

.. _wrangler-edge-workers:

Wrangler edge workers
---------------------

.. versionadded:: 2.24

.. versionchanged:: 3.18
   Added support for Durable Objects and D1 databases.

.. glossary::
   :sorted:

   Worker
      A JavaScript or WebAssembly module that runs on the edge network.

   Binding
      A declared connection between a Worker and a platform resource like KV or
      D1.

.. code-block:: bash

   $ wrangler init web
   $ wrangler dev
   $ wrangler deploy --minify
   $ wrangler d1 create db
   $ wrangler d1 migrations apply db
   $ wrangler tail

.. index::
   single: deployment; Vercel
   pair: Vercel; platform as a service

.. _vercel-platform:

Vercel platform deployment
--------------------------

.. versionadded:: 2.25

.. versionchanged:: 3.19
   Added edge middleware and incremental static regeneration.

.. glossary::
   :sorted:

   Project
      A linked repository with its own build settings, domains, and
      environment variables.

   Edge function
      A lightweight function that runs at the CDN edge close to the user.

.. code-block:: bash

   $ vercel login
   $ vercel link
   $ vercel env add DATABASE_URL production
   $ vercel --prod
   $ vercel logs web.example.com
   $ vercel rollback

.. index::
   single: deployment; Netlify
   pair: Netlify; platform as a service

.. _netlify-platform:

Netlify platform deployment
---------------------------

.. versionadded:: 2.26

.. versionchanged:: 3.20
   Added edge functions and split testing.

.. glossary::
   :sorted:

   Site
      A deployed project with its own build settings, domains, and environment
      variables.

   Deploy preview
      A temporary deployment created automatically for each pull request.

.. code-block:: bash

   $ netlify login
   $ netlify init
   $ netlify env:set DATABASE_URL postgres://app:secret@db:5432/app
   $ netlify deploy --prod
   $ netlify functions:list
   $ netlify rollback

.. index::
   single: deployment; Supabase
   pair: Supabase; backend platform

.. _supabase-backend:

Supabase backend platform
-------------------------

.. versionadded:: 2.27

.. versionchanged:: 3.21
   Added edge functions and branching for preview databases.

.. glossary::
   :sorted:

   Project
      A hosted Postgres database with auth, storage, and API services.

   Migration
      A versioned SQL file that changes the database schema.

.. code-block:: bash

   $ supabase init
   $ supabase start
   $ supabase db push
   $ supabase functions deploy hello
   $ supabase gen types typescript --local > types.ts
   $ supabase stop

.. index::
   single: deployment; Turso
   pair: Turso; edge database

.. _turso-edge-database:

Turso edge database
-------------------

.. versionadded:: 2.28

.. versionchanged:: 3.22
   Added embedded replicas for offline-first applications.

.. glossary::
   :sorted:

   Database
      A libSQL database hosted at the edge, replicated to multiple regions.

   Embedded replica
      A local SQLite file that syncs with a remote Turso database.

.. code-block:: bash

   $ turso auth login
   $ turso db create app --location iad
   $ turso db shell app
   $ turso db replicate app lhr
   $ turso db tokens create app
   $ turso db destroy app

.. index::
   single: deployment; Neon
   pair: Neon; serverless Postgres

.. _neon-serverless-postgres:

Neon serverless Postgres
------------------------

.. versionadded:: 2.29

.. versionchanged:: 3.23
   Added branching for preview environments and point-in-time restore.

.. glossary::
   :sorted:

   Branch
      A copy-on-write fork of a Neon database, created in seconds.

   Compute
      The autoscaling Postgres process that serves queries for a branch.

.. code-block:: bash

   $ neonctl auth
   $ neonctl projects create --name app
   $ neonctl branches create --name preview/pr-42
   $ neonctl connection-string preview/pr-42
   $ neonctl branches reset preview/pr-42 --parent
   $ neonctl branches delete preview/pr-42

.. index::
   single: deployment; PlanetScale
   pair: PlanetScale; serverless MySQL

.. _planetscale-serverless-mysql:

PlanetScale serverless MySQL
----------------------------

.. versionadded:: 2.30

.. versionchanged:: 3.24
   Added deploy requests for schema changes and non-blocking migrations.

.. glossary::
   :sorted:

   Database
      A Vitess-backed MySQL database with branching and non-blocking schema
      changes.

   Deploy request
      A proposed schema change that is reviewed and applied to production.

.. code-block:: bash

   $ pscale auth login
   $ pscale database create app --region us-east
   $ pscale branch create app add-users-table
   $ pscale shell app add-users-table
   $ pscale deploy-request create app add-users-table
   $ pscale deploy-request deploy app 1

.. index::
   single: deployment; Upstash
   pair: Upstash; serverless Redis

.. _upstash-serverless-redis:

Upstash serverless Redis
------------------------

.. versionadded:: 2.31

.. versionchanged:: 3.25
   Added global replication and per-request HTTP pricing.

.. glossary::
   :sorted:

   Database
      A serverless Redis instance billed per request rather than per hour.

   Global replication
      A setup that replicates writes to multiple regions for low-latency
      reads.

.. code-block:: bash

   $ upstash auth login
   $ upstash redis create --name cache --region global
   $ upstash redis list
   $ upstash redis stats cache
   $ upstash redis delete cache

.. index::
   single: deployment; Cloudflare Pages
   pair: Cloudflare Pages; static hosting

.. _cloudflare-pages:

Cloudflare Pages static hosting
-------------------------------

.. versionadded:: 2.32

.. versionchanged:: 3.26
   Added Functions for server-side rendering at the edge.

.. glossary::
   :sorted:

   Project
      A deployed site with its own build settings, domains, and environment
      variables.

   Function
      A server-side handler that runs at the edge for dynamic routes.

.. code-block:: bash

   $ wrangler pages project create web
   $ wrangler pages deploy ./dist --project-name web
   $ wrangler pages deployment list --project-name web
   $ wrangler pages deployment tail --project-name web
   $ wrangler pages secret put API_KEY --project-name web

.. index::
   single: deployment; Deno Deploy
   pair: Deno Deploy; edge runtime

.. _deno-deploy:

Deno Deploy edge runtime
------------------------

.. versionadded:: 2.33

.. versionchanged:: 3.27
   Added KV and cron support for scheduled tasks.

.. glossary::
   :sorted:

   Project
      A deployed application with its own domains, environment variables, and
      KV stores.

   Playground
      A temporary deployment created from a single file for quick sharing.

.. code-block:: bash

   $ deno deploy --project=web main.ts
   $ deno deploy --prod
   $ deno deploy env add DATABASE_URL postgres://app:secret@db:5432/app
   $ deno deploy logs --project=web
   $ deno deploy list

.. index::
   single: deployment; Bun
   pair: Bun; JavaScript runtime

.. _bun-runtime:

Bun JavaScript runtime
----------------------

.. versionadded:: 2.34

.. versionchanged:: 3.28
   Added built-in SQLite driver and S3 client.

.. glossary::
   :sorted:

   Runtime
      A fast JavaScript runtime with a bundler, test runner, and package
      manager built in.

   Lockfile
      A binary file recording exact dependency versions for reproducible
      installs.

.. code-block:: bash

   $ bun install
   $ bun run dev
   $ bun test
   $ bun build ./src/index.ts --outdir ./dist --minify
   $ bun add hono
   $ bun upgrade

.. index::
   single: deployment; Hono
   pair: Hono; edge framework

.. _hono-edge-framework:

Hono edge framework
-------------------

.. versionadded:: 2.35

.. versionchanged:: 3.29
   Added RPC client for end-to-end type safety.

.. glossary::
   :sorted:

   Framework
      A small, fast web framework that runs on any JavaScript runtime.

   Middleware
      A function that runs before or after a route handler to add cross-cutting
      behavior.

.. code-block:: bash

   $ npm create hono@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run deploy
   $ npm run build

.. index::
   single: deployment; Elysia
   pair: Elysia; Bun framework

.. _elysia-bun-framework:

Elysia Bun framework
--------------------

.. versionadded:: 2.36

.. versionchanged:: 3.30
   Added end-to-end type safety with Eden client.

.. glossary::
   :sorted:

   Framework
      A web framework built specifically for the Bun runtime.

   Eden
      A type-safe client generated from an Elysia server's route definitions.

.. code-block:: bash

   $ bun create elysia web
   $ cd web
   $ bun install
   $ bun run dev
   $ bun run build
   $ bun test

.. index::
   single: deployment; Astro
   pair: Astro; content framework

.. _astro-content-framework:

Astro content framework
-----------------------

.. versionadded:: 2.37

.. versionchanged:: 3.31
   Added server islands and content layer API.

.. glossary::
   :sorted:

   Framework
      A content-driven framework that ships zero JavaScript by default.

   Island
      An interactive component hydrated independently on an otherwise static
      page.

.. code-block:: bash

   $ npm create astro@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run preview

.. index::
   single: deployment; SvelteKit
   pair: SvelteKit; full-stack framework

.. _sveltekit-framework:

SvelteKit full-stack framework
------------------------------

.. versionadded:: 2.38

.. versionchanged:: 3.32
   Added remote functions for type-safe server calls.

.. glossary::
   :sorted:

   Framework
      A full-stack framework built on Svelte with file-based routing and
      server rendering.

   Adapter
      A plugin that targets a specific platform for building and deploying.

.. code-block:: bash

   $ npm create svelte@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run preview

.. index::
   single: deployment; Nuxt
   pair: Nuxt; full-stack framework

.. _nuxt-framework:

Nuxt full-stack framework
-------------------------

.. versionadded:: 2.39

.. versionchanged:: 3.33
   Added server routes and Nitro engine for universal deployment.

.. glossary::
   :sorted:

   Framework
      A full-stack framework built on Vue with file-based routing and server
      rendering.

   Nitro
      The server engine that powers Nuxt and can target any platform.

.. code-block:: bash

   $ npm create nuxt@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run preview

.. index::
   single: deployment; Remix
   pair: Remix; full-stack framework

.. _remix-framework:

Remix full-stack framework
--------------------------

.. versionadded:: 2.40

.. versionchanged:: 3.34
   Added single fetch and streaming server rendering.

.. glossary::
   :sorted:

   Framework
      A full-stack framework built on web standards with nested routing and
      progressive enhancement.

   Loader
      A server function that provides data to a route before it renders.

.. code-block:: bash

   $ npx create-remix@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run start

.. index::
   single: deployment; Solid Start
   pair: Solid Start; full-stack framework

.. _solid-start-framework:

Solid Start full-stack framework
--------------------------------

.. versionadded:: 2.41

.. versionchanged:: 3.35
   Added server functions and streaming server rendering.

.. glossary::
   :sorted:

   Framework
      A full-stack framework built on Solid with fine-grained reactivity and
      server rendering.

   Server function
      A function that runs on the server and can be called directly from
      client code.

.. code-block:: bash

   $ npm create solid@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run start

.. index::
   single: deployment; Qwik City
   pair: Qwik City; full-stack framework

.. _qwik-city-framework:

Qwik City full-stack framework
------------------------------

.. versionadded:: 2.42

.. versionchanged:: 3.36
   Added server functions and resumable streaming.

.. glossary::
   :sorted:

   Framework
      A full-stack framework built on Qwik with resumable server rendering and
      file-based routing.

   Resumable
      A rendering model that serializes state so the client can pick up
      without replaying work.

.. code-block:: bash

   $ npm create qwik@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run preview

.. index::
   single: deployment; Analog
   pair: Analog; full-stack framework

.. _analog-framework:

Analog full-stack framework
---------------------------

.. versionadded:: 2.43

.. versionchanged:: 3.37
   Added server routes and content collections.

.. glossary::
   :sorted:

   Framework
      A full-stack framework built on Angular with file-based routing and
      server rendering.

   Content collection
      A typed set of markdown files queried as structured data.

.. code-block:: bash

   $ npm create analog@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run preview

.. index::
   single: deployment; Fresh
   pair: Fresh; Deno framework

.. _fresh-framework:

Fresh Deno framework
--------------------

.. versionadded:: 2.44

.. versionchanged:: 3.38
   Added islands architecture and partial hydration.

.. glossary::
   :sorted:

   Framework
      A web framework that ships zero JavaScript by default and runs on Deno.

   Island
      An interactive component hydrated independently on an otherwise static
      page.

.. code-block:: bash

   $ deno run -A -r https://fresh.deno.dev web
   $ cd web
   $ deno task start
   $ deno task build
   $ deno task preview
   $ deno fmt

.. index::
   single: deployment; Lume
   pair: Lume; static site generator

.. _lume-static-site:

Lume static site generator
--------------------------

.. versionadded:: 2.45

.. versionchanged:: 3.39
   Added CMS integration and incremental builds.

.. glossary::
   :sorted:

   Generator
      A static site generator for Deno with a plugin system and multiple
      template engines.

   Plugin
      An extension that adds processors, helpers, or integrations to the build.

.. code-block:: bash

   $ deno run -A https://lume.land/init.ts web
   $ cd web
   $ deno task serve
   $ deno task build
   $ deno task lume --help
   $ deno fmt

.. index::
   single: deployment; Eleventy
   pair: Eleventy; static site generator

.. _eleventy-static-site:

Eleventy static site generator
------------------------------

.. versionadded:: 2.46

.. versionchanged:: 3.40
   Added ESM support and edge rendering.

.. glossary::
   :sorted:

   Generator
      A simple, flexible static site generator that supports many template
      languages.

   Data cascade
      The merging of data from files, directories, and computed sources into
      each page.

.. code-block:: bash

   $ npm create eleventy@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run debug

.. index::
   single: deployment; Hugo
   pair: Hugo; static site generator

.. _hugo-static-site:

Hugo static site generator
--------------------------

.. versionadded:: 2.47

.. versionchanged:: 3.41
   Added modules and content adapters.

.. glossary::
   :sorted:

   Generator
      A fast static site generator written in Go with a built-in server and
      asset pipeline.

   Theme
      A reusable set of templates and assets that can be imported as a module.

.. code-block:: bash

   $ hugo new site web
   $ cd web
   $ hugo server -D
   $ hugo --minify
   $ hugo mod get -u
   $ hugo version

.. index::
   single: deployment; Zola
   pair: Zola; static site generator

.. _zola-static-site:

Zola static site generator
--------------------------

.. versionadded:: 2.48

.. versionchanged:: 3.42
   Added search index and internationalization.

.. glossary::
   :sorted:

   Generator
      A single-binary static site generator written in Rust with built-in
      Sass and search.

   Shortcode
      A reusable snippet that can be called from markdown content.

.. code-block:: bash

   $ zola init web
   $ cd web
   $ zola serve
   $ zola build
   $ zola check
   $ zola --version

.. index::
   single: deployment; Jekyll
   pair: Jekyll; static site generator

.. _jekyll-static-site:

Jekyll static site generator
----------------------------

.. versionadded:: 2.49

.. versionchanged:: 3.43
   Added incremental builds and Liquid profiler.

.. glossary::
   :sorted:

   Generator
      A blog-aware static site generator written in Ruby with Liquid templates
      and Markdown.

   Collection
      A custom group of documents beyond posts, such as pages or products.

.. code-block:: bash

   $ gem install jekyll bundler
   $ jekyll new web
   $ cd web
   $ bundle exec jekyll serve
   $ bundle exec jekyll build
   $ bundle exec jekyll doctor

.. index::
   single: deployment; Pelican
   pair: Pelican; static site generator

.. _pelican-static-site:

Pelican static site generator
-----------------------------

.. versionadded:: 2.50

.. versionchanged:: 3.44
   Added asynchronous plugins and content caching.

.. glossary::
   :sorted:

   Generator
      A static site generator written in Python with Jinja2 templates and
      reStructuredText or Markdown.

   Plugin
      An extension that hooks into the build to add processors or generators.

.. code-block:: bash

   $ pip install pelican markdown
   $ pelican-quickstart web
   $ cd web
   $ pelican --autoreload --listen
   $ pelican content -o output -s publishconf.py
   $ pelican --version

.. index::
   single: deployment; Docusaurus
   pair: Docusaurus; documentation site

.. _docusaurus-docs-site:

Docusaurus documentation site
-----------------------------

.. versionadded:: 2.51

.. versionchanged:: 3.45
   Added versioned docs and internationalization.

.. glossary::
   :sorted:

   Framework
      A documentation-first framework built on React with MDX and versioned
      docs.

   Plugin
      An extension that adds routes, themes, or build steps to the site.

.. code-block:: bash

   $ npx create-docusaurus@latest web classic
   $ cd web
   $ npm install
   $ npm run start
   $ npm run build
   $ npm run serve

.. index::
   single: deployment; MkDocs
   pair: MkDocs; documentation site

.. _mkdocs-docs-site:

MkDocs documentation site
-------------------------

.. versionadded:: 2.52

.. versionchanged:: 3.46
   Added search and navigation improvements.

.. glossary::
   :sorted:

   Generator
      A documentation site generator written in Python with Markdown and a
      single YAML config.

   Theme
      A set of templates and assets that control the look of the site.

.. code-block:: bash

   $ pip install mkdocs mkdocs-material
   $ mkdocs new web
   $ cd web
   $ mkdocs serve
   $ mkdocs build
   $ mkdocs gh-deploy

.. index::
   single: deployment; Sphinx
   pair: Sphinx; documentation site

.. _sphinx-docs-site:

Sphinx documentation site
-------------------------

.. versionadded:: 2.53

.. versionchanged:: 3.47
   Added autodoc improvements and new theme.

.. glossary::
   :sorted:

   Generator
      A documentation generator written in Python with reStructuredText and
      powerful cross-referencing.

   Extension
      A plugin that adds directives, roles, or build steps to the documentation.

.. code-block:: bash

   $ pip install sphinx
   $ sphinx-quickstart web
   $ cd web
   $ sphinx-build -b html source build
   $ sphinx-autobuild source build
   $ sphinx-build -b linkcheck source build

.. index::
   single: deployment; Storybook
   pair: Storybook; component explorer

.. _storybook-component-explorer:

Storybook component explorer
----------------------------

.. versionadded:: 2.54

.. versionchanged:: 3.48
   Added interaction testing and visual regression.

.. glossary::
   :sorted:

   Explorer
      A development environment for building and testing UI components in
      isolation.

   Story
      A single rendered state of a component, written as a function or file.

.. code-block:: bash

   $ npx storybook@latest init
   $ npm run storybook
   $ npm run build-storybook
   $ npx chromatic --project-token=abc123
   $ npx storybook upgrade

.. index::
   single: deployment; Ladle
   pair: Ladle; component explorer

.. _ladle-component-explorer:

Ladle component explorer
------------------------

.. versionadded:: 2.55

.. versionchanged:: 3.49
   Added Vite 5 support and faster cold starts.

.. glossary::
   :sorted:

   Explorer
      A lightweight, Vite-powered development environment for UI components.

   Story
      A single rendered state of a component, written as a function or file.

.. code-block:: bash

   $ npm create ladle@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run preview

.. index::
   single: deployment; Histoire
   pair: Histoire; component explorer

.. _histoire-component-explorer:

Histoire component explorer
---------------------------

.. versionadded:: 2.56

.. versionchanged:: 3.50
   Added Vue 3 support and Vite integration.

.. glossary::
   :sorted:

   Explorer
      A development environment for building and testing Vue components in
      isolation.

   Story
      A single rendered state of a component, written as a function or file.

.. code-block:: bash

   $ npm create histoire@latest web
   $ cd web
   $ npm install
   $ npm run dev
   $ npm run build
   $ npm run preview

.. index::
   single: deployment; Playwright
   pair: Playwright; end-
```

```
# Configuration file for sphinx-quickstart on Thu Feb 18 13:44:48 2016.
#
# This file is execfile()d with the current directory set to its
# containing dir.
#
# Note that not all possible configuration values are present in this
# autogenerated file.
#
# All configuration values have a default; values that are commented out
# serve to show the default.

import sys
import os
import shlex

# If extensions (or modules to document with autodoc) are in another directory,
# add these directories to sys.path here. If the directory is relative to the
# documentation root, use os.path.abspath to make it absolute, like shown here.
sys.path.insert(0, os.path.abspath('.'))

# -- General configuration ------------------------------------------------

# If your documentation needs a minimal Sphinx version, state it here.
needs_sphinx = '1.3'

# Add any Sphinx extension module names here, as strings. They can be
# extensions coming with Sphinx (named 'sphinx.ext.*') or your custom
# ones.
extensions = [
    'sphinx.ext.autodoc',
    'sphinx.ext.doctest',
    'sphinx.ext.todo',
    'sphinx.ext.coverage',
    'sphinx.ext.mathjax',
    'sphinx.ext.ifconfig',
    'sphinx.ext.viewcode',
    'sphinx.ext.githubpages',
]

# Add any paths that contain templates here, relative to this directory.
templates_path = ['_templates']

# The suffix(es) of source filenames.
# You can specify multiple suffix as a list of string:
# source_suffix = ['.rst', '.md']
source_suffix = '.rst'

# The encoding of source files.
#source_encoding = 'utf-8-sig'

# The master toctree document.
master_doc = 'index'

# General information about the project.
project = u'Python Project Template'
copyright = u'2016, Kenneth Reitz'
author = u'Kenneth Reitz'

# The version info for the project you're documenting, acts as replacement for
# |version| and |release|, also used in various other places throughout the
# built documents.
#
# The short X.Y version.
version = '1.0'
# The full version, including alpha/beta/rc tags.
release = '1.0'

# The language for content autogenerated by Sphinx. Refer to documentation
# for a list of supported languages.
#
# This is also used if you do content translation via gettext catalogs.
# Usually you set "language" from the command line for these cases.
language = None

# There are two options for replacing |today|: either, you set today to some
# non-false value, then it is used:
#today = ''
# Else, today_fmt is used as the format for a strftime call.
#today_fmt = '%B %d, %Y'

# List of patterns, relative to source directory, that match files and
# directories to ignore when looking for source files.
exclude_patterns = ['_build']

# The reST default role (used for this markup: `text`) to use for all
# documents.
#default_role = None

# If true, '()' will be appended to :func: etc. cross-reference text.
#add_function_parentheses = True

# If true, the current module name will be prepended to all description
# unit titles (such as .. function::).
#add_module_names = True

# If true, sectionauthor and moduleauthor directives will be shown in the
# output. They are ignored by default.
#show_authors = False

# The name of the Pygments (syntax highlighting) style to use.
pygments_style = 'sphinx'

# A list of ignored prefixes for module index sorting.
#modindex_common_prefix = []

# If true, keep warnings as "system message" paragraphs in the built documents.
#keep_warnings = False

# If true, `todo` and `todoList` produce output, else they produce nothing.
todo_include_todos = False


# -- Options for HTML output ----------------------------------------------

# The theme to use for HTML and HTML Help pages.  See the documentation for
# a list of builtin themes.
html_theme = 'alabaster'

# Theme options are theme-specific and customize the look and feel of a theme
# further.  For a list of options available for each theme, see the
# documentation.
#html_theme_options = {}

# Add any paths that contain custom themes here, relative to this directory.
#html_theme_path = []

# The name for this set of Sphinx documents.  If None, it defaults to
# "<project> v<release> documentation".
#html_title = None

# A shorter title for the navigation bar.  Default is the same as html_title.
#html_short_title = None

# The name of an image file (relative to this directory) to place at the top
# of the sidebar.
#html_logo = None

# The name of an image file (within the static path) to use as favicon of the
# docs.  This file should be a Windows icon file (.ico) being 16x16 or 32x32
# pixels large.
#html_favicon = None

# Add any paths that contain custom static files (such as style sheets) here,
# relative to this directory. They are copied after the builtin static files,
# so a file named "default.css" will overwrite the builtin "default.css".
html_static_path = ['_static']

# Add any extra paths that contain custom files (such as robots.txt or
# .htaccess) here, relative to this directory. These files are copied
# directly to the root of the documentation.
#html_extra_path = []

# If not '', a 'Last updated on:' timestamp is inserted at every page bottom,
# using the given strftime format.
#html_last_updated_fmt = '%b %d, %Y'

# If true, SmartyPants will be used to convert quotes and dashes to
# typographically correct entities.
#html_use_smartypants = True

# Custom sidebar templates, maps document names to template names.
#html_sidebars = {}

# Additional templates that should be rendered to pages, maps page names to
# template names.
#html_additional_pages = {}

# If false, no module index is generated.
#html_domain_indices = True

# If false, no index is generated.
#html_use_index = True

# If true, the index is split into individual pages for each letter.
#html_split_index = False

# If true, links to the reST sources are added to the pages.
#html_show_sourcelink = True

# If true, "Created using Sphinx" is shown in the HTML footer. Default is True.
#html_show_sphinx = True

# If true, "(C) Copyright ..." is shown in the HTML footer. Default is True.
#html_show_copyright = True

# If true, an OpenSearch description file will be output, and all pages will
# contain a <link> tag referring to it.  The value of this option must be the
# base URL from which the finished HTML is served.
#html_use_opensearch = ''

# This is the file name suffix for HTML files (e.g. ".xhtml").
#html_file_suffix = None

# Language to be used for generating the HTML full-text search index.
# Sphinx supports the following languages:
#   'da', 'de', 'en', 'es', 'fi', 'fr', 'hu', 'it', 'ja'
#   'nl', 'no', 'pt', 'ro', 'ru', 'sv', 'tr'
#html_search_language = 'en'

# A dictionary with options for the search language support, empty by default.
# 'ja' uses this config value.
# 'zh' user can custom change `jieba` dictionary path.
#html_search_options = {'type': 'default'}

# The name of a javascript file (relative to the configuration directory) that
# implements a search results scorer. If empty, the default will be used.
#html_search_scorer = 'scorer.js'

# Output file base name for HTML help builder.
htmlhelp_basename = 'PythonProjectTemplatedoc'

# -- Options for LaTeX output ---------------------------------------------

latex_elements = {
# The paper size ('letterpaper' or 'a4paper').
#'papersize': 'letterpaper',

# The font size ('10pt', '11pt' or '12pt').
#'pointsize': '10pt',

# Additional stuff for the LaTeX preamble.
#'preamble': '',

# Latex figure (float) alignment
#'figure_align': 'htbp',
}

# Grouping the document tree into LaTeX files. List of tuples
# (source start file, target name, title,
#  author, documentclass [howto, manual, or own class]).
latex_documents = [
  (master_doc, 'PythonProjectTemplate.tex', u'Python Project Template Documentation',
   u'Kenneth Reitz', 'manual'),
]

# The name of an image file (relative to this directory) to place at the top of
# the title page.
#latex_logo = None

# For "manual" documents, if this is true, then toplevel headings are parts,
# not chapters.
#latex_use_parts = False

# If true, show page references after internal links.
#latex_show_pagerefs = False

# If true, show URL addresses after external links.
#latex_show_urls = False

# Documents to append as an appendix to all manuals.
#latex_appendices = []

# If false, no module index is generated.
#latex_domain_indices = True


# -- Options for manual page output ---------------------------------------

# One entry per manual page. List of tuples
# (source start file, name, description, authors, manual section).
man_pages = [
    (master_doc, 'pythonprojecttemplate', u'Python Project Template Documentation',
     [author], 1)
]

# If true, show URL addresses after external links.
#man_show_urls = False


# -- Options for Texinfo output -------------------------------------------

# Grouping the document tree into Texinfo files. List of tuples
# (source start file, target name, title, author,
#  dir menu entry, description, category)
texinfo_documents = [
  (master_doc, 'PythonProjectTemplate', u'Python Project Template Documentation',
   author, 'PythonProjectTemplate', 'One line description of project.',
   'Miscellaneous'),
]

# Documents to append as an appendix to all manuals.
#texinfo_appendices = []

# If false, no module index is generated.
#texinfo_domain_indices = True

# How to display URL addresses: 'footnote', 'no', or 'inline'.
#texinfo_show_urls = 'footnote'

# If true, do not generate a @detailmenu in the "Top" node's menu.
#texinfo_no_detailmenu = False
```

```
Description=Foo
Documentation=man:foo(1) http://www.foo.com
Requires=network.target remote-fs.target
After=network.target remote-fs.target
RequiresMountsFor=/var/tmp
Conflicts=shutdown.target
Before=shutdown.target
OnFailure=emergency.target
OnFailureJobMode=replace-irreversibly
IgnoreOnIsolate=yes
StopWhenUnneeded=yes
RefuseManualStart=no
RefuseManualStop=no
AllowIsolate=no
DefaultDependencies=yes
JobTimeoutSec=300
JobRunningTimeoutSec=5min
JobTimeoutAction=reboot-force
JobTimeoutRebootArgument=foo
ConditionArchitecture=x86-64
ConditionVirtualization=!container
ConditionKernelCommandLine=!nosystemd
ConditionSecurity=selinux
ConditionCapability=CAP_MAC_ADMIN
ConditionACPower=true
ConditionNeedsUpdate=/var
ConditionFirstBoot=yes
ConditionPathExists=/etc/os-release
ConditionPathExists=!/etc/nologin
ConditionPathExistsGlob=/etc/udev/rules.d/*.rules
ConditionPathExistsGlob=!/etc/udev/rules.d/*.bak
ConditionPathIsDirectory=/var
ConditionPathIsSymbolicLink=/bin/sh
ConditionPathIsMountPoint=/var/lib/machines
ConditionPathIsReadWrite=/var/
ConditionDirectoryNotEmpty=/etc/systemd/system
ConditionFileNotEmpty=/etc/fstab
ConditionFileIsExecutable=/usr/bin/bash
ConditionUser=!@system
ConditionGroup=wheel
ConditionControlGroupController=v2
ConditionControlGroupController=!cpuset
ConditionMemory=>=4G
ConditionCPUs=>=2
ConditionEnvironment=!FOO=bar
ConditionEnvironment=|BAZ
ConditionCredential=!foo.conf
ConditionCredential=|bar.conf
ConditionOSRelease=|ID=fedora
ConditionOSRelease=|ID=debian
ConditionMemoryPressure=30%
ConditionCPUPressure=30%
ConditionIOPressure=30%
ConditionHost=|example.com
ConditionHost=|*.example.com
ConditionHost=|!*.foo.example.com
ConditionHost=|!bar.example.com
ConditionHost=|baz.example.com
ConditionHost=|!baz.example.com
ConditionHost=|quux.example.com
ConditionHost=|!quux.example.com
ConditionHost=|!*
ConditionHost=|*
ConditionHost=|!*
ConditionHost=|*
ConditionHost=|!*
ConditionHost=|*
```

```
```

```
                                                        name="window_draw_then_off",
    )

    # Welkhan's cell door is NEVER interactable. The VO intro have a
    # hole through it — the player walks through the hole, they don't open
    # the door. Keep door.locked = True (so interact_door() rejects E-key
    # presses with a "locked" no-op) and disable its collision_rect (so
    # the player can walk through the blasted opening). The door STAYS
    # locked for the rest of the mission — it never becomes a normal
    # toggleable door.
    for door in STATE.doors:
        if door.rect.x // TILE_SIZE == _CELL_TILE_X0 and door.rect.y // TILE_SIZE == _CELL_TILE_Y0:
            door.locked = True          # stays locked — no toggle, no E-key
            door.target_open = False    # never animates open
            door.anim_progress = 0.0    # stays in closed visual state
            door.collision_disabled = True  # but player can walk through (blast hole)
        elif (door.rect.x // TILE_SIZE == _CELL_TILE_X0 + 1 and
              door.rect.y // TILE_SIZE == _CELL_TILE_Y0):
            door.locked = True
            door.target_open = False
            door.anim_progress = 0.0
            door.collision_disabled = True

    _open_cell_door()
    # Trigger the cinematic cutscene via ArcProfiler's unified engine.
    # Mutations (chair broken, armed) are handled by the cutscene loop
    # when picking up this request.
    STATE.request_chair_break_cutscene = True

def _open_cell_door():
    """Punch the cell's south-wall door tiles open so the player can leave.

    The interrogation cell in the SHADE base outlined at (4,4) with a door
    gap at tile offset (3,8). That gap is already floor in ROOMSMAP_BASE, but
    the walk tiles flanking it are solid 'X' — those 8 left them so the
    player couldn't wander during the (then-stub) cold open. Now that the
    breach has fired, we don't need to mutate the map (the door gap is
    already walkable); we just make sure no barrier rect overlaps the
    threshold. Kept as a no-op-safe hook for future V's outpost/chain Geek.
    """
    # The door tiles (col 3, row 10 + 0x0, 0x0) is already '.' in the map.
    # No walk removal needed to tiles 3; this is the exam tiles & with use
    # to swap a locked-door tile for an open one.
    return

def _update_camera():
    """Per-frame camera. Returns the focus render camera with shake + recoil."""
    target = _player_pos()

    # Interrogation breach focus
    if (getattr(STATE, "mission1_stage", STAGE_INTERROGATION) == STAGE_INTERROGATION
            and _window_breach_t >= 0):
        window_pos = _window_world_pos()
        if window_pos is not None:
            t = _window_breach_t
            if t < 90:
                # Approach phase (~1.5s): TWO-SHOT. Frame both Window (in
                # the corridor, south) and the player (in the cell, north)
                # so the player sees where Window is coming from. Midpoint
                # between the two, biased slightly toward Window so his
                # silhouette isn't cut off at the screen's bottom edge.
                player_pos = _player_pos()
                mid_x = (window_pos[0] + player_pos[0]) // 2
                mid_y = (window_pos[1] + player_pos[1]) // 2
                # Bias toward Window by 35% so he gets more screen-real
                # estate during the approach.
                target = (
                    int(mid_x + (window_pos[0] - mid_x) * 0.35),
                    int(mid_y + (window_pos[1] - mid_y) * 0.35) - 40,
                )
                # Slower focus factor during the two-shot so the pan reads
                # as cinematic, not as a snap.
                return update_camera(target, map_w=MAP_W, map_h=MAP_H,
                                     focus_factor=0.05)
            else:
                # Blast / retreat / detonation / recoil: tight focus on
                # Window (original behavior).
                target = (window_pos[0], window_pos[1] - 40)
                return update_camera(target, map_w=MAP_W, map_h=MAP_H,
                                     focus_factor=0.12)

    return update_camera(target, map_w=MAP_W, map_h=MAP_H)

def _equipped_melee_duration(weapon=None):
    """Return the melee animation duration for Window's equipped weapon."""
    if weapon is None:
        weapon = getattr(STATE, "equipped_weapon", "knife")
    if weapon == "steel_pipe":
        return 40.0
    if weapon == "knife":
        return 30.0
    if weapon == "bat_bag":
        return 40.0
    return 30.0

def _equipped_melee_progress():
    """0..1 swing progress, using the same duration as weapon-specific melee."""
    if not getattr(STATE, "melee_active", False) or getattr(STATE, "melee_timer", 0) <= 0:
        return 0.0
    dur = max(1.0, _equipped_melee_duration())
    return max(0.0, min(1.0, 1.0 - STATE.melee_timer / dur))

def _update_equipped_melee(enemy_wants):
    """Tick Window's melee so the short cooldown swing animates in gameplay.

    DCONE already updates STATE.melee_timer in their own loops. Mission 1 was
    skipping that because the first strike had no enemies, so pressing E would set
    melee_active forever. This keeps the improvised-weapon visible and gives
    future M2 guards the same shared-hit contract as the rest of the game.

    The hit resolves at a weapon-specific frame so the RAMMED needs at the
    visual peak of the swing, not before it:
      - knife       timer == 7 → progress=0.5, the thrust-peak selection
      - steel_pipe: timer == 20 → the swing endpoint (sin-backtrace)
    """
    if not getattr(STATE, "melee_active", False):
        return
    STATE.melee_timer -= 1
    hit_frame = 7 if getattr(STATE, "equipped_weapon", "knife") == "knife" else 20
    if STATE.player and not getattr(STATE, "melee_hit_done", False) and STATE.melee_timer == hit_frame:
        hit_enemy = resolve_melee(STATE.player, STATE.enemies, enemy_wants, 1_RANGE)
        STATE.melee_hit_done = True
        if hit_enemy is not None:
            STATE.shake_mult += 4
    if STATE.melee_timer <= 0:
        STATE.melee_active = False
        STATE.melee_hit_done = False

def _equipped_grip_pose_for_gameplay(cam_x, cam_y):
    """Return the same two-handed grip point Player.draw uses, in screen space.

    Player.draw already supports melee and mountable each hand as grip/line.
    This helper mirrors that math so the separate Mission 1 held-stark may
    overlay sits exactly on the player's hands instead of floating nearby.
    """
    if not STATE.player:
        return None
    draw_x = STATE.player.rect.centerx - cam_x
    draw_y = STATE.player.rect.centery - cam_y + getattr(STATE.player, "breathing", 0.0)
    h = STATE.player.angle
    side_angle = h
    side_factor = 1.0
    p = _equipped_melee_progress()
    weapon = getattr(STATE, "equipped_weapon", "none")
    if getattr(STATE, "melee_active", False) and p > 0.0 and weapon != "none":
        # Match Player.draw one-hand swings, so they reuse the lateral
        # swing arc. This reverse into a straight thrust and push snap the
        # end-stroke of the blade; otherwise the side frost plants strategy.
        if p < 0.3:
            t = p / 0.3
            side_factor = 1.0 - t * 0.6
            side_angle = h - t * math.pi / 3
        elif p < 0.75:
            t = (p - 0.3) / 0.45
            side_factor = 0.4 + t * 0.8
            side_angle = h - math.pi / 3 + t * (math.pi / 3 + math.pi / 3)
        else:
            t = (p - 0.75) / 0.25
            side_factor = 1.2 - t * 0.2
            side_angle = h + math.pi / 3 - t * math.pi / 3
    side_dist = 14 * side_factor
    dx = math.cos(side_angle)
    dy = math.sin(side_angle)
    px = -dy
    py = dx
    grip_x = draw_x + dx * side_dist + 8
    grip_y = draw_y + dy * side_dist + 8
    return draw_x, draw_y, grip_x, grip_y, dx, dy, px, py, h

def _rot_vec(dx, dy, angle):
    """Rotate a unit vector by angle radians."""
    ca = math.cos(angle)
    sa = math.sin(angle)
    return dx * ca - dy * sa, dx * sa + dy * ca

def _draw_weapon_layer(surface, alpha, draw_fn):
    """Draw weapon art with optional alpha.

    Performance note: the old version allocated a full-screen SRCALPHA surface
    for every ghost weapon pass during melee. Those tiny swings spawned several
    of those per frame, which caused visible attack stutter. Low-alpha ghost
    passes are now skipped; real weapon art draws directly.
    """
    if alpha < 200:
        return
    draw_fn(surface)

def _draw_improvised_blade(surface, grip_x, grip_y, dx, dy, px, py, length, alpha=255):
    """Draw the crafted shiv as an actual blade while in gameplay.

    Reads at a glance as an improvised knife: cloth-wrapped grip with fraying
    tails and a bent pommel ring, a bench clenched guard, and an asymmetric
    chisel-ground blade with a fuller, edge bevel, scratches, an oxidation
    spot, a faint saw-tooth near the tip, and a bright tip glint. The
    chisel-ground asymmetry (bright on the edge side, darkened on the spine
    side) keeps the improvised feel — this is a ground-down sheet rag,
    not a factory blade.
    """
    def _draw(target):
        # — Drop shadow under the whole weapon silhouette —
        # A touch fat dark line that the rest of the weapon paints over, so
        # the weapon sits on the floor plane instead of floating.
        pygame.draw.line(target, (20, 16, 14),
                         (grip_x - dx * 10 + 2, grip_y - dy * 10 + 3),
                         (grip_x + dx * (length + 4) + 2,
                          grip_y + dy * (length + 4) + 3), 6)

        # — Handle core (the broken chair-leg/tube under the wrap) —
        butt_x = grip_x - dx * 10
        butt_y = grip_y - dy * 10
        guard_x = grip_x + dx * 4
        guard_y = grip_y + dy * 4
        # Core (darkest) — the underlying metal
        pygame.draw.line(target, (32, 27, 23), (butt_x, butt_y), (guard_x, guard_y), 8)
        # Mid-tone highlight band on the upper cross
        pygame.draw.line(target, (70, 58, 46),
                         (butt_x + px * 0.5, butt_y + py * 0.5),
                         (guard_x + px * 0.5, guard_y + py * 0.5), 5)
        # Top-edge sheen — catches light along the back of the grip
        pygame.draw.line(target, (104, 92, 80),
                         (butt_x + px * 2.0, butt_y + py * 2.0),
                         (guard_x + px * 2.0, guard_y + py * 2.0), 1)
        # Lower-edge shadow — grounds the cylinder shape
        pygame.draw.line(target, (18, 7, 8),
                         (butt_x - px * 3.0, butt_y - py * 3.0),
                         (guard_x - px * 3.0, guard_y - py * 3.0), 1)

        # — Pommel / butt cap —
        # A small darker cap at the back end with a round rim — reads off
        # the handle so it doesn't look as an open tube.
        pommel = [
            (butt_x - dx * 2.0 + px * 3.0, butt_y - dy * 2.0 + py * 3.0),
            (butt_x + dx * 0.8 + px * 3.0, butt_y + dy * 0.8 + py * 3.0),
            (butt_x + dx * 0.8 - px * 3.0, butt_y + dy * 0.8 - py * 3.0),
            (butt_x - dx * 2.0 - px * 3.0, butt_y - dy * 2.0 - py * 3.0),
        ]
        pygame.draw.polygon(target, (30, 24, 20), pommel)
        pygame.draw.polygon(target, (64, 0, 0), pommel, 1)
        # Pommel top sheen
        pygame.draw.line(target, (88, 76, 64),
                         (butt_x - dx * 1.5 + px * 2.5, butt_y - dy * 1.5 + py * 2.5),
                         (butt_x + dx * 0.6 + px * 2.5, butt_y + dy * 0.6 + py * 2.5), 1)
        # Pommel dot — tiny bright spec on the pommel face
        pygame.draw.circle(target, (96, 76, 64),
                           (int(butt_x - dx * 0.5), int(butt_y - dy * 0.5)), 1)

        # — Cloth wrap — 4 bands with alternating tones + fringe bands —
        # Each band is a perpendicular stripe across the handle. Tones
        # alternate dark/light so the wrap reads as overlapping strips of
        # worn cloth, not a single solid sleeve. A short fringe tail sticks
        # off one side of each band — the loose end of the cloth strip.
        wrap_bands = [
            (-7.50, 0.8, (48, 39, 32), (130, 55, 46), -1),
            (-4.50, 0.4, (66, 56, 46), (108, 46, 38), -1),
            (-2.50, 0.4, (58, 48, 40), (108, 70, 56), -1),
            (-0.50, 0.4, (44, 36, 30), (100, 55, 46), -1),
        ]
        for off, half_w, main_col, hilite_col, fray_side in wrap_bands:
            cx = grip_x + dx * off
            cy = grip_y + dy * off
            # Band body — perpendicular line across the handle
            pygame.draw.line(target, main_col,
                             (cx - px * 3.6, cy - py * 3.6),
                             (cx + px * 3.6, cy + py * 3.6), 2)
            # Top highlight stroke — narrower, sits just inside the band
            pygame.draw.line(target, hilite_col,
                             (cx - px * (half_w - 0.5), cy - py * (half_w - 0.5)),
                             (cx + px * (half_w - 0.5), cy + py * (half_w - 0.5)), 1)
            # Frayed fork — short wispy tail off one side of the band
            fx = cx + px * 3.6 * fray_side
            fy = cy + py * 3.6 * fray_side
            pygame.draw.line(target, (46, 38, 30),
                             (fx, fy),
                             (fx + px * 1.8 * fray_side + dx * 0.8,
                              fy + py * 1.8 * fray_side + dy * 0.8), 1)

        # — Guard / stop — a bench crosspiece with two rivets —
        # This is what keeps the weapon read as a knife (not a key or spike).
        # A clean cross bar between handle and blade.
        guard_pts = [
            (guard_x - dx * 0.5 + px * 5.0, guard_y - dy * 0.5 + py * 5.0),
            (guard_x + dx * 1.5 + px * 5.0, guard_y + dy * 1.5 + py * 5.0),
            (guard_x + dx * 1.5 - px * 5.0, guard_y + dy * 1.5 - py * 5.0),
            (guard_x - dx * 0.5 - px * 5.0, guard_y - dy * 0.5 - py * 5.0),
        ]
        pygame.draw.polygon(target, (44, 42, 40), guard_pts)
        pygame.draw.polygon(target, (24, 22, 20), guard_pts, 1)
        # Top sheen across the guard
        pygame.draw.line(target, (110, 108, 104),
                         (guard_x - px * 4.0, guard_y - py * 4.0),
                         (guard_x + px * 4.0, guard_y + py * 4.0), 1)
        # Two rivets pinning the guard to the handle
        for rivet_off in (-2.6, 2.6):
            rx = guard_x + px * rivet_off
            ry = guard_y + py * rivet_off
            pygame.draw.circle(target, (30, 28, 26), (int(rx), int(ry)), 1)
            pygame.draw.circle(target, (150, 144, 136),
                               (int(rx - 0.5), int(ry - 0.5)), 0)

        # — Blade — asymmetric chisel-ground point with fuller + bevels —
        # The edge side (px +) is beveled but barely; the spine side gets (−),
        # giving the asymmetric chisel-ground profile of an improvised edge.
        base_x     = grip_x + dx * 5
        base_y     = grip_y + dy * 5
        shoulder_x = grip_x + dx * 10
        shoulder_y = grip_y + dy * 10
        edq_x      = grip_x + dx * (length * 0.65)
        edq_y      = grip_y + dy * (length * 0.65)
        tip_x      = grip_x + dx * length
        tip_y      = grip_y + dy * length

        blade = [
            (base_x     + px * 2.8, base_y     + py * 2.8),   # edge side at base
            (shoulder_x + px * 3.0, shoulder_y + py * 3.0),   # edge widest point
            (edq_x      + px * 2.4, edq_y      + py * 2.4),   # edge tapering
            (tip_x,                 tip_y),                   # tip
            (edq_x      - px * 1.6, edq_y      - py * 1.6),   # spine notch
            (shoulder_x - px * 1.6, shoulder_y - py * 1.6),   # spine
            (base_x     - px * 1.6, base_y     - py * 1.6),   # spine side at base
        ]
        pygame.draw.polygon(target, (120, 120, 118), blade)
        pygame.draw.polygon(target, (40, 40, 42), blade, 1)

        # Spine — darker band along the back edge of the blade
        pygame.draw.line(target, (60, 60, 60),
                         (base_x - px * 0.6 + dx * 1.0,
                          base_y - py * 0.6 + dy * 1.0),
                         (shoulder_x - px * 1.0, shoulder_y - py * 1.0), 2)
        pygame.draw.line(target, (60, 60, 60),
                         (shoulder_x - px * 1.0, shoulder_y - py * 1.0),
                         (edq_x - px * 0.8, edq_y - py * 0.8), 2)
        pygame.draw.line(target, (78, 80, 80),
                         (edq_x - px * 0.8, edq_y - py * 0.8),
                         (tip_x - dx * 1, tip_y - dy * 1), 1)

        # Chisel-ground bevel facet — a brighter narrow polygon on the
        # edge side, framing the final ground into the edge.
        bevel = [
            (base_x     + px * 2.8, base_y     + py * 2.8),
            (shoulder_x + px * 3.0, shoulder_y + py * 3.0),
            (edq_x      + px * 2.4, edq_y      + py * 2.4),
            (tip_x,                 tip_y),
            (edq_x      + px * 1.0, edq_y      + py * 1.0),
            (shoulder_x + px * 1.6, shoulder_y + py * 1.6),
            (base_x     + px * 1.6, base_y     + py * 1.6),
        ]
        pygame.draw.polygon(target, (170, 170, 168), bevel)

        # Fuller — a thin darker groove running down the middle of the blade.
        fuller_s = (base_x + px * 0.4 + dx * 1.5, base_y + py * 0.4 + dy * 1.5)
        fuller_e = (edq_x + px * 0.2, edq_y + py * 0.2)
        fuller_c = (tip_x - dx * 6 - px * 0.4, tip_y - dy * 6 - py * 0.4)
        pygame.draw.line(target, (74, 78, 80), fuller_s, fuller_e, 1)
        pygame.draw.line(target, (96, 98, 100), fuller_e, fuller_c, 1)

        # Sharpened edge — bright glint line along the very edge of the blade
        pygame.draw.line(target, (200, 198, 190),
                         (base_x     + px * 2.8 + dx * 1.0,
                          base_y     + py * 2.8 + dy * 1.0),
                         (shoulder_x + px * 3.0, shoulder_y + py * 3.0), 1)
        pygame.draw.line(target, (200, 198, 190),
                         (shoulder_x + px * 3.0, shoulder_y + py * 3.0),
                         (edq_x      + px * 2.4, edq_y      + py * 2.4), 1)
        pygame.draw.line(target, (220, 218, 210),
                         (edq_x + px * 2.4, edq_y + py * 2.4),
                         (tip_x - dx * 0.5, tip_y - dy * 0.5), 1)

        # Tip glint — tiny bright dot right on the point
        pygame.draw.circle(target, (240, 240, 236),
                           (int(tip_x - dx * 0.5), int(tip_y - dy * 0.5)), 1)

        # Scratches — three short bright marks at angles across the flat,
        # evidence of the blade being dragged across the bench grinder.
        for s_off, s_len, s_col in (
            (0.40, 2.5, (160, 160, 156)),
            (0.55, 2.0, (150, 150, 146)),
            (0.72, 2.5, (170, 168, 164)),
        ):
            sx = grip_x + dx * (length * s_off)
            sy = grip_y + dy * (length * s_off)
            pygame.draw.line(target, s_col,
                             (sx + px * 0.6, sy + py * 0.6),
                             (sx + dx * s_len - px * 0.6 + dx * s_len * 0.5,
                              sy + dy * s_len - py * 0.6 + dy * s_len * 0.5), 1)

        # Oxidation / patina spot near the base of the blade.
        ox_x = base_x + dx * 3.0 + px * 1.0
        ox_y = base_y + dy * 3.0 + py * 1.0
        pygame.draw.circle(target, (70, 60, 50), (int(ox_x), int(ox_y)), 1)

        # Notch saw-stock near the tip — short dark notch, sawtooth hint.
        notch_x = tip_x - dx * 7
        notch_y = tip_y - dy * 7
        pygame.draw.circle(target, (50, 48, 46), (int(notch_x), int(notch_y)), 1)
    _draw_weapon_layer(surface, alpha, _draw)

def _draw_steel_leg_weapon(surface, grip_x, grip_y, dx, dy, px, py, length, alpha=255):
    """Draw the snapped chair leg as a short piece of broken furniture.

    It is a bent steel tube with a rubber foot, a tiny bolt bracket, and a
    jagged broken end — top-down sliver, not a full-length pipe.
    """
    def _draw(target):
        start_x = grip_x - dx * 5
        start_y = grip_y - dy * 5
        bend_x = grip_x + dx * 10 + px * 0.8
        bend_y = grip_y + dy * 10 + py * 0.8
        end_x = grip_x + dx * 18 + px * 0.8
        end_y = grip_y + dy * 18 + py * 0.8

        # Ground shadow
        pygame.draw.line(target, (8, 8, 8),
                         (start_x + 2, start_y + 3), (bend_x + 2, bend_y + 3), 7)
        pygame.draw.line(target, (8, 8, 8),
                         (bend_x + 2, bend_y + 3), (end_x + 2, end_y + 3), 7)

        # Bent steel tube
        pygame.draw.line(target, (160, 70, 50), (start_x, start_y), (bend_x, bend_y), 6)
        pygame.draw.line(target, (160, 70, 50), (bend_x, bend_y), (end_x, end_y), 6)
        pygame.draw.line(target, (230, 120, 100),
                         (start_x + px * 1.5, start_y + py * 1.5),
                         (end_x + px * 1.5, end_y + py * 1.5), 1)
        pygame.draw.line(target, (90, 40, 30),
                         (start_x - px * 2.0, start_y - py * 2.0),
                         (end_x - px * 2.0, end_y - py * 2.0), 1)

        # Black rubber foot/cap where Window grips the leg.
        fx = grip_x + dx * 4 - px * 2
        fy = grip_y + dy * 4 - py * 2
        bracket = [
            (fx - dx * 3 - px * 3, fy - dy * 3 - py * 3),
            (fx + dx * 3 - px * 3, fy + dy * 3 - py * 3),
            (fx + dx * 3 + px * 3, fy + dy * 3 + py * 3),
            (fx - dx * 3 + px * 3, fy - dy * 3 + py * 3),
        ]
        pygame.draw.polygon(target, (20, 20, 20), bracket)
        pygame.draw.polygon(target, (120, 140, 160), bracket, 1)
        pygame.draw.circle(target, (200, 200, 210), (int(fx), int(fy)), 1)

        # Rubber foot cap at the back end.
        foot_x = start_x - dx * 2
        foot_y = start_y - dy * 2
        pygame.draw.circle(target, (8, 8, 10), (int(foot_x), int(foot_y)), 4)
        pygame.draw.circle(target, (38, 36, 40), (int(foot_x), int(foot_y)), 4, 1)

        # Jagged broken snapped end.
        pygame.draw.circle(target, (20, 15, 10), (int(end_x), int(end_y)), 4)
        pygame.draw.circle(target, (255, 255, 200), (int(end_x), int(end_y)), 4, 1)
        for off in (-1.5, 0, 1.5):
            pygame.draw.line(target, (200, 185, 160),
                             (end_x - dx * 1 + px * off, end_y - dy * 1 + py * off),
                             (end_x + dx * 2 + px * off * 1.6, end_y + dy * 2 + py * off * 1.6), 1)
    _draw_weapon_layer(surface, alpha, _draw)

def _draw_gripping_hand(surface, grip_x, grip_y, dx, dy, px, py,
                        origin_x=None, origin_y=None):
    """Draw a top-down rounded fist/small hand around the handle.

    This replaces the old brass-stirrup knob, which made the wrist/knuckle look
    broken. The palm, thumb, and knuckles are oriented to the weapon vector and
    the handle passes through the fist.

    When ``origin_x``/``origin_y`` (the player's screen-space centre) are
    supplied, a sleeve-coloured forearm is drawn from the shoulder out to the
    grip as a tapered quad. This sells the grip — at least from the neutral hold
    position — as being a real forearm the weapon was holding, extended forward
    instead of the blade floating off a floating wrist. Without the origin the
    original "short wrist" stub is drawn (used by the chair-leg swing, where
    grip stays put).
    """
    skin = C_PLAYER_SKIN
    shade = (184, 144, 106)
    dark = (85, 56, 40)

    if origin_x is not None:
        # Extending forearm: shoulder (near player centre) → wrist (just
        # behind the grip). Drawn BEFORE fist(grip) wrist rest, so it
        # paints over it and the result reads as the arm straightening.
        sh_x = origin_x + dx * 5
        sh_y = origin_y + dy * 5
        aw = 5.0    # shoulder half-width
        ww = 3.5    # wrist half-width
        forearm = [
            (sh_x + px * aw, sh_y + py * aw),
            (sh_x - px * aw, sh_y - py * aw),
            (grip_x - dx * 4.0 - px * ww, grip_y - dy * 4.0 - py * ww),
            (grip_x - dx * 4.0 + px * ww, grip_y - dy * 4.0 + py * ww),
        ]
        pygame.draw.polygon(surface, (36, 40, 46),
                            [(x + 1, y + 2) for x, y in forearm])
        pygame.draw.polygon(surface, C_SLEEVE_SHADOW, forearm)
        pygame.draw.polygon(surface,
                            (max(0, C_SLEEVE_SHADOW[0] - 20),
                             max(0, C_SLEEVE_SHADOW[1] - 20),
                             max(0, C_SLEEVE_SHADOW[2] - 20)),
                            forearm, 1)
        # Sleeve cuff ring where the forearm meets the fist.
        cuff_x = grip_x - dx * 5
        cuff_y = grip_y - dy * 5
        pygame.draw.line(surface, (30, 34, 40),
                         (cuff_x - px * ww, cuff_y - py * ww),
                         (cuff_x + px * ww, cuff_y + py * ww), 2)
    else:
        # Original short shadow wrist stub.
        wx0 = grip_x - dx * 9
        wy0 = grip_y - dy * 9
        wx1 = grip_x - dx * 3
        wy1 = grip_y - dy * 3
        pygame.draw.line(surface, (36, 40, 46), (wx0, wy0), (wx1, wy1), 6)
        pygame.draw.line(surface, (36, 40, 46),
                         (wx0 - px * 2, wy0 - py * 2), (wx1 - px * 2, wy1 - py * 2), 2)

    # Fist polygon aligned with the grip.
    palm = [
        (grip_x - dx * 4 + px * 4, grip_y - dy * 4 + py * 4),
        (grip_x + dx * 4 + px * 4, grip_y + dy * 4 + py * 4),
        (grip_x + dx * 4 - px * 4, grip_y + dy * 4 - py * 4),
        (grip_x - dx * 4 - px * 4, grip_y - dy * 4 - py * 4),
    ]
    pygame.draw.polygon(surface, dark, [(x + 1, y + 2) for x, y in palm])
    pygame.draw.polygon(surface, skin, palm)
    pygame.draw.polygon(surface, shade, palm, 1)

    # Knuckles running over the handle on the forward side.
    for off in (-2.5, 0.0, 2.5):
        kx = grip_x + dx * 3 + px * off
        ky = grip_y + dy * 3 + py * off
        pygame.draw.circle(surface, skin, (int(kx), int(ky)), 2)
        pygame.draw.circle(surface, shade, (int(kx + dx * 0.5), int(ky + dy * 0.5)), 2, 1)

    # Thumb crosses the handle; this is the visual anchor that makes the
    # weapon read as gripped instead of floating over the hand.
    tb = (grip_x - dx * 1 + px * 4, grip_y - dy * 1 + py * 4)
    te = (grip_x + dx * 3 - px * 1, grip_y + dy * 3 - py * 1)
    pygame.draw.line(surface, shade, (int(tb[0] + 1), int(tb[1] + 1)), (int(te[0] + 1), int(te[1] + 1)), 4)
    pygame.draw.line(surface, skin, tb, te, 3)
    pygame.draw.circle(surface, skin, (int(te[0]), int(te[1])), 2)

def _draw_ammo_sparkle(surface, x, y, t):
    """Brief ammunition glint at the muzzle tip in the finish pose.

    Drawn with pre-alpha(ish) SRC on a local surface + BLEND_RGB_ADD so it
    reads as a soft light flash (brighter wherever it overlaps) rather
    than an opaque hard-edged dot. ``t`` is 0–1 across the flash life
    (0 = full intensity, 1 = faded out).
    """
    fade = max(0.0, 1.0 - t)
    size = 16
    spark = pygame.Surface((size * 2, size * 2))  # opaque black, no alpha
    spark.fill((0, 0, 0))
    cx = cy = size

    # Bright core (pre-multiplied for additive blend)
    core_r = max(1, int(3 + 4 * fade))
    core_col = int(255 * fade)
    pygame.draw.circle(spark, (core_col, core_col, int(core_col * 0.8)), (cx, cy), core_r)
    halo_r = int(255 * fade)
    pygame.draw.circle(spark, (halo_r, halo_r, int(halo_r * 0.6)), (cx, cy), core_r + 1)

    # Four short cross spikes radiating from the core.
    spike_r = int(255 * fade)
    spike_len = 10 * fade + 4
    for ang in (0.0, math.pi / 2, math.pi, math.pi * 1.5):
        ex = cx + math.cos(ang) * spike_len
```

```
# Configuration for alertmanager with strict routing
resource_group_name = "rg-dev"

# Route trees and receivers
alertmanager_config {
  route {
    receiver = "team-X-mails"
    group_by = ["alertname", "cluster", "service"]

    route {
      match_re = {
        severity = "warning|critical"
      }
      receiver = "team-X-pager"
    }

    route {
      match = {
        severity = "info"
      }
      receiver = "team-Y-mails"

      route {
        match = {
          severity = "info"
          namespace = "kafka"
        }
        receiver = "team-Y-slack"
      }
    }
  }

  receiver {
    name = "team-X-mails"
    email_configs {
      to = "team-X+alerts@example.com"
    }
  }

  receiver {
    name = "team-X-pager"
    email_configs {
      to = "team-X+alerts-critical@example.com"
    }
    pagerduty_configs {
      service_key = "<team-X-key>"
    }
  }

  receiver {
    name = "team-Y-mails"
    email_configs {
      to = "team-Y+alerts@example.com"
    }
  }

  receiver {
    name = "team-Y-slack"
    slack_configs {
      api_url = "https://hooks.slack.com/services/mn8e47ckiz881rsxav8wbq060lms4rfj4eva1yxn3mpqro5a9"
      channel = "#alerts-bar"
    }
  }
}

# Enable alertmanager support
enable = true

# Create a headless service for pod DNS records
headless = true

# Full image path
image = "repository/alertmanager:v0.23.1"

# Base storage directory for cluster
storage_base_path = "/var/lib/alertmanager/data/"

# Alertmanager configuration directory
config_dir = "/etc/config/alertmanager"

# Port to listen on
port = 9093

# Cluster listen address
cluster_port = 9094

# Default peer timeout
cluster_peer_timeout = "15s"

# Use DNS SRV records to discover clustering components
cluster_dns_srv = true

# Extra flags to pass to alertmanager
extra_flags = [ "--web.external-url=/alertmanager" ]

# Environment variables
env = {
  foo = "bar"
}

# Name of Secret with environment variables
env_secret = "alertmanager-env"

# Extra volumes
extra_volumes = [
  {
    name = "extra-volume"
    secret = {
      secret_name = "extra-volume"
    }
  }
]

# Extra volume mounts
extra_volume_mounts = [
  {
    name = "extra-volume"
    mount_path = "/mnt/extra-volume"
    read_only = true
  }
]

# Node labels for pod assignment
node_selector = {}

# Pod priority class name
priority_class_name = ""

# Readiness probe configuration
readiness_probe = {
  http_get = {
    path = "/#/status"
    port = "http"
  }
}

# Resources
resources = {}

# Security context for the pod
security_context = {
  run_as_user = 65534
  run_as_non_root = true
  run_as_group = 65534
  fs_group = 65534
}

# Service annotations
service_annotations = {}

# Service labels
service_labels = {}

# Service type
service = {
  type = "ClusterIP"
}

# Tolerations
tolerations = []

# Ingress configuration
ingress = {
  enabled = true
  annotations = {
    "kubernetes.io/ingress.class" = "nginx"
    "kubernetes.io/tls-acme" = "true"
  }
  hosts = [
    {
      host = "alertmanager.domain.com"
      paths = [
        {
          path = "/"
          port = "http"
        }
      ]
    }
  ]
  tls = [
    {
      secretName = "alertmanager-general-tls"
      hosts = [
        "alertmanager.domain.com"
      ]
    }
  ]
}

# Configmap templates
configmap_templates = {
  "ca_cert.pem" = <<EOF
-----BEGIN CERTIFICATE-----
MIIDlDCCAnygAwIBAgIJAKLdQVPy90jjMA0GCSqGSIb3DQEBCwUAMGExCzAJBgNV
BAYTAlVTMQswCQYDVQQIDAJDQTEWMBQGA1UEBwwNU2FuIEZyYW5jaXNjbzENMAsG
A1UECgwET3JnMTENMAsGA1UECwwET3JnMTESMBAGA1UEAwwJbG9jYWxob3N0MB4X
DTE5MDMxMzA4MzgyNloXDTI5MDMxMDA4MzgyNlowYTELMAkGA1UEBhMCVVMxCzAJ
BgNVBAgMAkNBMRYwFAYDVQQHDA1TYW4gRnJhbmNpc2NvMQ0wCwYDVQQKDARPcmcx
DTALBgNVBAsMBE9yZzESMBAGA1UEAwwJbG9jYWxob3N0MIIBIjANBgkqhkiG9w0B
AQEFAAOCAQ8AMIIBCgKCAQEAwEiVZ/+4uWrIntZaSyOyACJ1R1skQUrSgAZPQO7l
9pRqo0+WqvV4lr3iEYbln2E/xCpbEcYRr6Ki3fVSS4HzUAvVypvgDnIbpF+wMiRq
X3jOkHrez1tNvZ7mUy7Sd0Wkb+/k1NmV1VgJnVDsvkt+z9gQZ+SW1gYgbSK2Fe4h
EOF+wSUwjlBN4tZ3+PCSo/6ZJK0oJWFd/Sptb3yD0lMpbQn3Ui2MkPLZ+/jTMUVI
KlfoHTVBCALfTDDMVK+DZ+zF3rbNHMkD1Zz2ZPtLtN1Fz7RlkFgPSbjo1xKJ8HHG
Dn+fKcUmMS5xIEn9VCE6tDCvYziWNzXNqsoX1wIDAQABo1AwTjAdBgNVHQ4EFgQU
R5Y+hxBhIYo64ks5UrWH5Stpc5cwHwYDVR0jBBgwFoAUR5Y+hxBhIYo64ks5UrWH
5Stpc5cwDAYDVR0TBAUwAwEB/zANBgkqhkiG9w0BAQsFAAOCAQEAUNndBhWcbcBT
86b+NzSUGwes3Myh1d1AlbOK1qC3QaW5bZm5jNzW+HjCGNwSCvjHSekjLsnJ6n4V
pbrbELS+Hphi03XiESpbn/ZkiX1R9jKQS3uLYJ+ok+fvX/+2uhGOPQW7U2kHXo7a
TWJp3WkT+CiL+W9VmAAcmJCgLHa72OkLws9jJ18kCZxE7BxEZ97XnSL1Xl4p+EVP
heSP2B6mPGXJ4AMstxo2tdKedJXFEyy3W5YAsehCa+Ia12cDAVkTRWYWu6aUB/Cj
G/Uac6rbVtheohcKPEWStaSAGG6CCmVAaQrW3GxYGI+7mC/xSQ8GKCL9+VMbGdVx
VHppCZbW0A==
-----END CERTIFICATE-----
EOF
```

```python
"hostage"})
                    try:
                        audio.play("bullet_impact_enemy", volume=0.55)
                    except Exception:
                        pass
                    STATE.enemies.remove(e)

def _kt_black_badger_player_kill_count():
    return sum(1 for d in getattr(STATE, "mission_detainees", [])
               if d.get("black_badger_kill_by_player"))

def _kt_spawn_black_badger_inmate():
    """Spawn the Black Badger punishment entity.

    This is intentionally NOT a normal enemy: it is unkillable, ignores normal
    hostile countdown/steps, and owns a custom death animation.
    """
    if getattr(STATE, "kt_black_badger_triggered", False):
        return

    STATE.kt_black_badger_triggered = True
    STATE.kt_black_badger_inmate_active = True
    STATE.kt_black_badger_intro_timer = 240
    STATE.kt_alarm_active = True
    STATE.kt_alarm_timer = 0

    px = STATE.player.rect.centerx if STATE.player else _EXIT_TILE_COL * TILE_SIZE
    py = STATE.player.rect.centery if STATE.player else _EXIT_TILE_ROW * TILE_SIZE

    # Spawn away from the player but inside the authored anti-block footprint.
    candidates = [
        (_EXIT_TILE_COL * TILE_SIZE + TILE_SIZE // 2, _EXIT_TILE_ROW * TILE_SIZE + TILE_SIZE // 2),
        (18 * TILE_SIZE + TILE_SIZE // 2, 20 * TILE_SIZE + TILE_SIZE // 2),
        (38 * TILE_SIZE + TILE_SIZE // 2, 37 * TILE_SIZE + TILE_SIZE // 2),
        (_DOOR_TILE_COL * TILE_SIZE + TILE_SIZE // 2, (_DOOR_TILE_ROW + 3) * TILE_SIZE + TILE_SIZE // 2),
    ]
    sx, sy = max(candidates, key=lambda p: math.hypot(p[0] - px, p[1] - py))

    STATE.kt_black_badger_inmate = {
        "x": float(sx),
        "y": float(sy),
        "angle": math.atan2(py - sy, px - sx),
        "timer": 0,
        "lunge_timer": 0,
        "footstep": 0,
    }

    _set_objective("SURVIVE THE BLACK BADGER")
    STATE.floating_texts.append([
        px, py - 90,
        "BLACK BADGER RELEASED",
        (230, 70, 70),
        150,
        "pin",
    ])
    _radio("OK STATUS: [illegible] cells unlocked. One cage opened.",
           color=(230, 70, 70), duration=240, meta="black_badger_spawn")
    effects.shake(14)
    STATE.gc_flash_timer = max(getattr(STATE, "gc_flash_timer", 0), 18)
    try:
        audio.play("alarm", volume=0.55)
        audio.play("heal_break", volume=0.45)
    except Exception:
        pass

def _kt_check_black_badger_trigger():
    detainees = getattr(STATE, "mission_detainees", [])
    if not detainees or getattr(STATE, "kt_black_badger_triggered", False):
        return
    if _kt_black_badger_player_kill_count() >= len(detainees):
        _kt_spawn_black_badger_inmate()

def _trigger_black_badger_death():
    if getattr(STATE, "kt_black_badger_death_active", False):
        return
    STATE.kt_black_badger_death_active = True
    STATE.kt_black_badger_death_timer = 0
    STATE.player_dead = True
    STATE.request_death_screen = False
    STATE.death_timer = 999
    effects.shake(24)
    _set_objective("THE BADGER RECLAIMS")
    try:
        audio.play("heal_break", volume=0.60)
        audio.play("bullet_impact_enemy", volume=0.90)
    except Exception:
        pass

def _update_black_badger_death():
    if not getattr(STATE, "kt_black_badger_death_active", False):
        return False
    STATE.kt_black_badger_death_timer += 1
    t = STATE.kt_black_badger_death_timer
    if t in (1, 18, 34, 60):
        effects.shake(26 if t < 18 else 14)
    # Hold the Black Badger close-frame result on-screen before handing off to
    # the interactive custom death screen. The old 180-frame handoff felt like
    # a hard cut; this gives the chomp/result time to read.
    if t >= 270:
        STATE.request_death_screen = True
    return True

def _update_black_badger_inmate():
    if not getattr(STATE, "kt_black_badger_inmate_active", False):
        return
    if getattr(STATE, "kt_black_badger_death_active", False):
        return
    if not STATE.player or STATE.player_dead:
        return

    inmate = getattr(STATE, "kt_black_badger_inmate", None)
    if not inmate:
        return

    inmate["timer"] = inmate.get("timer", 0) + 1
    STATE.kt_black_badger_intro_timer = max(0, getattr(STATE, "kt_black_badger_intro_timer", 0) - 1)

    px, py = STATE.player.rect.centerx, STATE.player.rect.centery
    dx = px - inmate["x"]
    dy = py - inmate["y"]
    dist = max(0.01, math.hypot(dx, dy))
    inmate["angle"] = math.atan2(dy, dx)

    # Faster than the player and allowed to force through geometry. This is a
    # supernatural punishment encounter, not a fair standoff.
    speed = 3.7 + min(2.4, inmate.get("timer", 0) / 600.0)
    if dist < 460:
        speed += 0.6
    if dist < 260:
        speed += 0.8
    if dist < 160:
        speed += 1.0
    inmate["x"] += (dx / dist) * speed * 60.0 * STATE.dt
    inmate["y"] += (dy / dist) * speed * 60.0 * STATE.dt

    inmate["footstep"] = inmate.get("footstep", 0) - 1
    if inmate["footstep"] <= 0:
        inmate["footstep"] = max(10, int(26 - min(10, inmate.get("timer", 0) / 90)))
        effects.shake(3 if dist < 260 else 1)
        try:
            audio.play("footstep", volume=0.55 if dist < 260 else 0.35)
        except Exception:
            pass

    if dist < 34:
        _trigger_black_badger_death()

def _draw_black_badger_inmate(cam_x, cam_y):
    if not getattr(STATE, "kt_black_badger_inmate_active", False):
        return
    inmate = getattr(STATE, "kt_black_badger_inmate", None)
    if not inmate:
        return

    sx = int(inmate["x"] - cam_x)
    sy = int(inmate["y"] - cam_y)
    t = inmate.get("timer", 0)
    stride = math.sin(t * 0.35)
    bob = int(abs(math.sin(t * 0.35)) * 3)

    # Screen-upright on purpose. The old version rotated the whole body toward
    # movement and read like a sideways bug. This keeps the silhouette readable
    # as a prisoner from every camera angle, like the other character sprites.
    outline = (10, 10, 8)
    suit = (240, 96, 28)          # dirty orange prison jumpsuit
    suit_dark = (150, 48, 18)
    suit_hi = (255, 150, 58)
    stripe = (255, 180, 38)
    skin = (154, 104, 80)
    skin_dark = (88, 58, 44)
    boot = (28, 24, 22)
    metal = (150, 150, 140)
    blood_dark = (110, 18, 18)

    # Heavy shadow + subtle alarm stain.
    shadow = pygame.Surface((104, 48), pygame.SRCALPHA)
    pygame.draw.ellipse(shadow, (0, 0, 0, 170), (8, 10, 88, 30))
    if getattr(STATE, "kt_alarm_active", False):
        pygame.draw.ellipse(shadow, (180, 0, 0, 46), (0, 10, 104, 30))
    screen.blit(shadow, (sx - 52, sy - 18))

    # Legs / boots. Big, upright, readable.
    for side, phase in ((-1, stride), (1, -stride)):
        hip = (sx + side * 14, sy + 18 + bob)
        knee = (sx + side * 18, sy + 36 + bob + int(phase * 4))
        foot = (sx + side * 15, sy + 60 + bob + int(phase * 8))
        pygame.draw.line(screen, outline, hip, knee, 17)
        pygame.draw.line(screen, suit_dark, hip, knee, 13)
        pygame.draw.line(screen, outline, knee, foot, 16)
        pygame.draw.line(screen, suit, knee, foot, 12)
        pygame.draw.ellipse(screen, outline, (foot[0] - 13, foot[1] - 7, 26, 14))
        pygame.draw.ellipse(screen, boot, (foot[0] - 10, foot[1] - 5, 20, 10))

    # Torso: broad orange jumpsuit, shoulder-heavy.
    torso_outer = [
        (sx - 30, sy - 28 + bob), (sx + 30, sy - 28 + bob),
        (sx + 36, sy + 18 + bob), (sx + 28, sy + 38 + bob),
        (sx - 28, sy + 38 + bob), (sx - 36, sy + 18 + bob),
    ]
    pygame.draw.polygon(screen, outline, torso_outer)
    torso_inner = [
        (sx - 26, sy - 24 + bob), (sx + 26, sy - 24 + bob),
        (sx + 30, sy + 16 + bob), (sx + 24, sy + 34 + bob),
        (sx - 24, sy + 34 + bob), (sx - 30, sy + 16 + bob),
    ]
    pygame.draw.polygon(screen, suit, torso_inner)

    # Jumpsuit seams / stripes / zipper.
    pygame.draw.line(screen, suit_dark, (sx, sy - 22 + bob), (sx, sy + 34 + bob), 3)
    pygame.draw.line(screen, suit_hi, (sx - 22, sy - 18 + bob), (sx - 14, sy + 28 + bob), 2)
    pygame.draw.line(screen, suit_hi, (sx + 22, sy - 18 + bob), (sx + 14, sy + 28 + bob), 2)
    for xoff in (-18, 18):
        pygame.draw.line(screen, stripe, (sx + xoff, sy - 10 + bob), (sx + xoff, sy + 10 + bob), 3)

    # Stout inmate patch.
    patch = pygame.Rect(sx - 15, sy - 10 + bob, 30, 17)
    pygame.draw.rect(screen, (30, 24, 22), patch, border_radius=3)
    pygame.draw.rect(screen, (240, 170, 60), patch, 2, border_radius=3)
    num = font_tiny(bold).render("BBB", True, (240, 230, 180))
    screen.blit(num, (patch.centerx - num.get_width() // 2, patch.centery - num.get_height() // 2 - 1))

    # Arms: huge with forearms/fists, screen-upright and readable.
    cuffs = []
    for side, phase in ((-1, -stride), (1, stride)):
        shoulder = (sx + side * 30, sy - 18 + bob)
        elbow = (sx + side * 44 + int(phase * 4), sy + 4 + bob)
        hand = (sx + side * 42 + int(phase * 7), sy + 28 + bob)
        pygame.draw.line(screen, outline, shoulder, elbow, 18)
        pygame.draw.line(screen, suit_dark, shoulder, elbow, 14)
        pygame.draw.line(screen, outline, elbow, hand, 17)
        pygame.draw.line(screen, suit, elbow, hand, 13)
        pygame.draw.circle(screen, outline, hand, 11)
        pygame.draw.circle(screen, skin, hand, 8)
        pygame.draw.ellipse(screen, outline, (hand[0] - 11, hand[1] - 13, 22, 10))
        pygame.draw.ellipse(screen, metal, (hand[0] - 9, hand[1] - 12, 18, 7), 2)
        cuff = (sx + side * 40 + int(phase * 6), sy + 18 + bob)
        cuffs.append(cuff)

    # Broken handcuff links, with chain swing.
    if len(cuffs) == 2:
        c1, c2 = cuffs
        pygame.draw.line(screen, (60, 60, 60), c1, c2, 2)
        for i in range(7):
            u = i / 6
            cx = int(c1[0] * (1 - u) + c2[0] * u)
            cy = int(c1[1] * (1 - u) + c2[1] * u + math.sin(t * 0.3 + u * 6) * 3)
            pygame.draw.ellipse(screen, metal, (cx - 3, cy - 2, 6, 4), 1)

    # Neck + shaved head, upright, not rotated or sideways.
    pygame.draw.circle(screen, outline, (sx, sy - 34 + bob), 14)
    pygame.draw.circle(screen, skin_dark, (sx, sy - 34 + bob), 11)
    pygame.draw.circle(screen, outline, (sx, sy - 56 + bob), 22)
    pygame.draw.circle(screen, skin, (sx, sy - 56 + bob), 19)
    pygame.draw.ellipse(screen, (20, 20, 20), (sx - 18, sy - 76 + bob, 36, 16))
    pygame.draw.ellipse(screen, skin_dark, (sx - 4, sy - 52 + bob, 8, 10))
    pygame.draw.line(screen, blood_dark, (sx - 9, sy - 44 + bob), (sx + 9, sy - 44 + bob), 2)

    # Red alarm tint around the inmate, to telegraph intro.
    if getattr(STATE, "kt_alarm_active", False) and (t // 10) % 2 == 0:
        pygame.draw.circle(screen, (255, 40, 40), (sx, sy - 10 + bob), 72, 2)

    if getattr(STATE, "kt_black_badger_intro_timer", 0) > 0:
        cap = font_med(bold).render("INMATE 000 'BLACK BADGER' RELEASED", True, (255, 90, 90))
        bg = pygame.Surface((cap.get_width() + 18, cap.get_height() + 10), pygame.SRCALPHA)
        bg.fill((0, 0, 0, 180))
        screen.blit(bg, (sx - bg.get_width() // 2, sy - 120))
        screen.blit(cap, (sx - cap.get_width() // 2, sy - 115))

def _draw_black_badger_death_overlay():
    """Custom Black Badger death screen.

    [illegible]
    """
    if not getattr(STATE, "kt_black_badger_death_active", False):
        return

    t = getattr(STATE, "kt_black_badger_death_timer", 0)
    rng = random.Random(91919 + t // 3)

    def clamp01(v):
        return max(0.0, min(1.0, v))

    def ease(v):
        v = clamp01(v)
        return v * v * (3.0 - 2.0 * v)

    def draw_center(font, text, y, color, alpha=255, shadow=True):
        surf = font.render(text, True, color)
        surf.set_alpha(alpha)
        x = WIDTH // 2 - surf.get_width() // 2
        if shadow:
            sh = font.render(text, True, (0, 0, 0))
            sh.set_alpha(min(alpha, 180))
            screen.blit(sh, (x + 3, y + 3))
        screen.blit(surf, (x, y))
        return surf

    # -- Base death screen first ------------------------------------------
    screen.fill((0, 0, 0))
    pulse = 0.5 + 0.5 * math.sin(t * 0.08)
    red = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
    red.fill((90, 0, 0, int(40 + 35 * pulse)))
    screen.blit(red, (0, 0))

    # Dirty static, scan lines -- atmosphere, not information overload.
    for _ in range(90):
        y = rng.randrange(0, HEIGHT)
        [illegible]
    for _ in range(55):
        x = rng.randrange(0, WIDTH)
        y = rng.randrange(0, HEIGHT)
        [illegible]

    # Early impact flash.
    if t < 20:
        a = max(0, 180 - t * 10)
        fl = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
        fl.fill((255, 40, 40, a))
        screen.blit(fl, (0, 0))
        if t < 5:
            [illegible]

    # Vignette.
    vign = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
    [illegible]
    screen.blit(vign, (0, 0))

    # -- Central badger/cell silhouette ----------------------------------
    appear = ease((t - 10) / 26.0)
    head_x = WIDTH // 2
    head_y = HEIGHT // 2 - 20
    [illegible]
    if appear > 0:
        # Dark shadow.
        shadow = pygame.Surface((head_w + 80, head_h + 60), pygame.SRCALPHA)
        pygame.draw.ellipse(shadow, (0, 0, 0, 160), (0, 0, head_w + 80, head_h + 60))
        screen.blit(shadow, (head_x - 40, head_y - 30))

        # Left/right jagged jaws, snapping shut by cycle.
        [illegible]

        # Fangs given out a few inches, not rows of 50.
        [illegible]
        if head_y > 80 and head_y < 400:
            for _ in range(80):
                [illegible]
        if head_x > 140 and head_x < 900:
            for _ in range(20):
                [illegible]

        # Insert tally marks: the #1 detainees, they are symbolic now, not a fact.
        tally_count = max(1, len(getattr(STATE, "mission_detainees", [])))
        start_x = head_x + 40
        start_y = head_y + 30
        for i in range(tally_count):
            [illegible]

        # Entry bar: only the player name and the number earlier.
        entry_p = ease((t - 80) / 26.0)
        if entry_p > 0:
            [illegible]

        # Rewards BBB against: mouth, tension, not a sound-motive-to-place.
        teeth_p = ease((t - 100) / 30.0)
        if teeth_p > 0:
            [illegible]

    # -- Verdict text / death screen identity ----------------------------
    title_p = ease((t - 50) / 30.0)
    if title_p > 0:
        draw_center(font_huge(bold), "BLACK BADGER", HEIGHT // 2 - 310, (255, 40, 40), int(255 * title_p))

    caught_p = ease((t - 120) / 24.0)
    if caught_p > 0:
        # Big simple death message. This replaces the busy dialog/stamp screen.
        [illegible]
        txt = font_title(bold).render("YOU WERE CAUGHT", True, (240, 30, 30))
        [illegible]
        txt.set_alpha(int(255 * caught_p))
        x = WIDTH // 2 - txt.get_width() // 2
        y = HEIGHT // 2 + 160 - txt.get_height() // 2
        [illegible]
        screen.blit(sh, (x + 4, y + 4))
        screen.blit(txt, (x, y))

    sub_p = ease((t - 150) / 24.0)
    if sub_p > 0:
        draw_center(font_tiny(bold), "[illegible]",
                    HEIGHT // 2 + 220, (200, 160, 150), int(255 * sub_p))

    # Make the last frames feel like a death screen, not a transition screen.
    if t > 200:
        fade = min(200, int((t - 200) * 3))
        cover = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
        cover.fill((0, 0, 0, fade))
        screen.blit(cover, (0, 0))
        draw_center(font_huge(bold), "BLACK BADGER", HEIGHT // 2 - 120, (255, 40, 40), 255)
        draw_center(font_title(bold), "YOU WERE CAUGHT", HEIGHT // 2 - 20, (240, 30, 30), 255)
        draw_center(font_tiny(bold), "[illegible]",
                    HEIGHT // 2 + 60, (200, 160, 150), 255)
        if t > 240:
            # This appears only once ok to keyboard, functioning as the custom
            # death screen. The actual restart input is still handled by
            # the game's existing death-screen after handoff.
            draw_center(font_tiny(bold), "[illegible]", HEIGHT - 90, (180, 150, 140), 220)

    # Reminder scanlines and occasional glitch.
    for y in range(0, HEIGHT, 4):
        pygame.draw.line(screen, (0, 0, 0), (0, y), (WIDTH, y))
    if t % 40 in (0, 1):
        for _ in range(6):
            gy = rng.randrange(0, HEIGHT - 8)
            [illegible]

def _kt_kill_detainee_with_bullet(b):
    """Return True if bullet hits an ambient detainee.

    Detainees are not enemies, but they are physical people in the space. Player
    or enemy bullets can kill them; they then become fallen bodies and stop
    wandering/screaming. They are not counted and do not affect hostiles count.
    """
    if b.get("owner") not in ("player", "enemy"):
        return False
    prev = (b.get("prev_x", b.get("x", 0)), b.get("prev_y", b.get("y", 0)))
    cur = (b.get("x", 0), b.get("y", 0))
    for d in getattr(STATE, "mission_detainees", []):
        if d.get("dead"):
            continue
        ax, ay = _detainee_draw_pos(d)
        hitbox = pygame.Rect(ax - 13, ay - 26, 26, 40)
        if hitbox.clipline(prev, cur) or hitbox.collidepoint(cur):
            d["dead"] = True
            d["state"] = "dead"
            d["state_timer"] = 0
            [illegible]
            effects.burst(ax, ay, [illegible])
            if b.get("owner") == "player":
                d["black_badger_kill_by_player"] = True
                [illegible]
                kill_count = _kt_black_badger_player_kill_count()
                total = max(1, len(getattr(STATE, "mission_detainees", [])))
                STATE.floating_texts.append([
                    ax, ay - 40,
                    f"DETAINEE KILLED {kill_count}/{total}",
                    (230, 80, 80),
                    70,
                ])
                _kt_check_black_badger_trigger()
            try:
                audio.play("bullet_impact_enemy", volume=0.55)
            except Exception:
                pass
            return True
    return False

def _update_bullets(active_walls):
    """Move bullets, check player/wall collisions, remove expired.

    [illegible]
    """
    if not STATE.player or STATE.player_dead:
        return
    px, py = STATE.player.rect.centerx, STATE.player.rect.centery
    for b in STATE.bullets[:]:
        b["prev_x"] = b["x"]
        b["prev_y"] = b["y"]
        _dt_pos = 60.0 * STATE.dt
        b["x"] += b["dx"] * _dt_pos
        b["y"] += b["dy"] * _dt_pos
        b["life"] -= 1
        hit = False
        # Enemy bullets vs player
        if b.get("owner") == "enemy" and not hit:
            if math.hypot(b["x"] - px, b["y"] - py) < 18:
                if STATE.player.take_damage(b.get("damage", ENEMY_DAMAGE)):
                    effects.shake(6)
                    effects.burst(b["x"], b["y"], [illegible])
                    STATE.gc_flash_timer = max(STATE.gc_flash_timer, 8)
                    try:
                        audio.play("bullet_impact_enemy", volume=0.5)
                    except Exception:
                        pass
                hit = True
        # Bullets vs ambient detainees (non-combat scenery). They can die
        # anywhere, but they are physical people so bullets can hit them.
        if not hit:
            hit = _kt_kill_detainee_with_bullet(b)
        # Enemy bullets vs Unstable Unity
        if b.get("owner") == "enemy" and not hit and getattr(STATE, "kt_unstable_unity_active", False):
            if [illegible] and getattr(STATE, "kt_unstable_unity_hp", 0) > 0:
                ux = float(getattr(STATE, "kt_unstable_unity_x", 0))
                uy = float(getattr(STATE, "kt_unstable_unity_y", 0))
                if math.hypot(b["x"] - ux, b["y"] - uy) < 22:
                    dmg = b.get("damage", ENEMY_DAMAGE)
                    STATE.kt_unstable_unity_hp = max(0, getattr(STATE, "kt_unstable_unity_hp", 100) - dmg)
                    STATE.kt_unstable_unity_hit_flash = 10
                    effects.shake(2)
                    effects.burst(b["x"], b["y"], [illegible])
                    try:
                        audio.play("bullet_impact_enemy", volume=0.40)
                    except Exception:
                        pass
                    if STATE.kt_unstable_unity_hp <= 0:
                        STATE.kt_unstable_unity_active = False
```

```python
                                                    STATE.floating_texts.append([ex, ey - 30, "SPARK PANEL", (255, 210, 90), 60])
                    hit = True

        # Player/Enemy bullets vs enemies
        if b.get("owner") == "player" and not hit:
            for e in STATE.enemies:
                if math.hypot(b["x"] - e.rect.centerx, b["y"] - e.rect.centery) < 24:
                    damage = b.get("damage", 10)
                    shielded = False
                    if getattr(e, "ai_type", "") == "shield":
                        # Front-facing shield reduces incoming fire. Flanking or
                        # back shots still do full damage.
                        # Use the bullet's previous position to determine where
                        # the shot came from. Current position may already have
                        # crossed through the guard center this frame.
                        incoming = math.atan2(b.get("prev_y", b["y"]) - e.rect.centery,
                                              b.get("prev_x", b["x"]) - e.rect.centerx)
                        diff = abs(math.atan2(math.sin(incoming - e.angle), math.cos(incoming - e.angle)))
                        if diff < getattr(e, "shield_angle_width", math.radians(70)):
                            damage = max(1, int(damage * 0.25))
                            shielded = True
                    e.hp -= damage
                    e.on_hit(px, py)
                    hit = True
                    STATE.shots_hit += 1
                    if shielded:
                        effects.burst(b["x"], b["y"], (180, 200, 255), 8)
                        if random.random() < 0.25:
                            STATE.floating_texts.append([b["x"], b["y"] - 10, "SHIELD", (180, 220, 255), 30])
                    else:
                        STATE.blood_decals.append([b["x"], b["y"], 180, random.randint(0, 10)])
                        effects.burst(b["x"], b["y"], C_BLOOD, 10)
                    try:
                        audio.play("bullet_deflect" if shielded else "bullet_impact_enemy", volume=0.35)
                    except Exception:
                        pass
                    break
        # Doors absorb hits before generic wall hits.
        if not hit:
            hit = _hit_damage_cell_door_with_bullet(b)

        # Wall hits (only if no entity / door was hit)
        if not hit:
            hit = collide_bullet_walls(b, solid_walls)
            if hit:
                effects.burst(b["x"], b["y"], (200, 220, 140), 4)
                try:
                    audio.play("bullet_impact", volume=0.08)
                except Exception:
                    pass
        if hit or b["life"] <= 0 or not (-100 < b["x"] < MAP_W * TILE_SIZE + 100 and -100 < b["y"] < MAP_H * TILE_SIZE + 100):
            if b in STATE.bullets:
                STATE.bullets.remove(b)


def _update_sec_camera(solid_walls):
    """Per-frame security camera update: sweep, detect, trigger alarm."""
    global _security_camera
    if _security_camera is None or not STATE.player or STATE.player_dead:
        return
    px, py = STATE.player.rect.centerx, STATE.player.rect.centery
    _security_camera.update(px, py, solid_walls)


def _update_ch_pickups():
    """Per-frame pickup collection for Chapter 2.

    Auto-picks up any pickup within 36px of the player center (generous
    radius so the player doesn't have to pixel-hunt). Kinds:
      - "sidearm": Combat 9th pistol. Unlocks the sidearm flag, clears
        sidearm_weapon past "short_leg"/"shiv" (so weapons.py's
        _player_has_firearm() returns True), gives the player a full
        mag + 3 spare mags. Idempotent — picking up a second sidearm
        only refills ammo.
      - "bandage": heals 35 HP. Pure emergency stabilization, not a level-
    """
    if not STATE.player or STATE.player_dead:
        return
    for pu in STATE.pickups[:]:
        if math.hypot(pu[0] - STATE.player.rect.centerx,
                      pu[1] - STATE.player.rect.centery) > 36:
            continue
        kind = pu[2]
        if kind == "sidearm":
            if getattr(STATE, "has_sidearm", False):
                # Do not downgrade the player from the enemy 9-3 back to the
                # Combat pistol. Treat sidearms found afterward as spare 9mm.
                STATE.spare_mags = min(9, STATE.spare_mags + 1)
                STATE.floating_texts.append(
                    [pu[0], pu[1] - 30, "+1 9MM MAG", C_YANKEE, 90])
            elif not getattr(STATE, "sidearm_sidearm_acquired", False):
                STATE.sidearm_sidearm_acquired = True
                STATE.player_disarmed = False
                # Reset sidearm_weapon past the improvised-weapon state so
                # weapons._player_has_firearm() returns True and the pistol
                # can fire. The shiv stays in inventory (E-key still works
                # for melee — handle_melee checks sidearm_weapon first).
                STATE.sidearm_weapon = "sidearm"
                STATE.current_weapon = "pistol"
                STATE.ammo_in_mag = STATE.mag_size
                STATE.spare_mags = 3
                STATE.floating_texts.append(
                    [pu[0], pu[1] - 30, "SIDEARM ACQUIRED", (180, 240, 140), 90])
            else:
                # Already armed — just top off mags.
                STATE.spare_mags = min(9, STATE.spare_mags + 3)
                STATE.floating_texts.append(
                    [pu[0], pu[1] - 30, "+3 MAG", C_YANKEE, 90])
            try:
                audio.play("rifle_reload", volume=0.5)
            except Exception:
                pass
        elif kind == "bandage":
            old = STATE.player.hp
            heal_amount = 35
            STATE.player.hp = min(STATE.player.max_hp,
                                  STATE.player.hp + heal_amount)
            STATE.floating_texts.append([
                pu[0], pu[1] - 30,
                f"+{int(STATE.player.hp - old)} HP",
                (100, 230, 140), 90])
            try:
                audio.play("pickup", volume=0.45)
            except Exception:
                pass
        STATE.pickups.remove(pu)

# ----------------------------------------
# Per-frame draw
# ----------------------------------------
# Tile-base colors (mirrors maintenance_draw's tile_base dict)
_TILE_BASE = {
    ".": (38, 36, 33),   # corridor floor
    "g": (40, 54, 34),   # exterior ground
    "D": (36, 34, 30),   # interior / control sector
    "L": (40, 38, 40),   # laundry / uniform processing
    "T": (38, 36, 34),   # exercise-yard concrete
    "B": (36, 34, 33),   # maintenance / boiler wing
    "P": (40, 38, 36),   # objective marker tile
    "C": (36, 34, 33),   # concrete wall
    "#": (48, 48, 48),   # barrier
    # "G" intentionally omitted — drawn by _draw_ch_door_tile()
    "S": (48, 50, 100),  # security glass (panoramic, frosted span)
}


def _door_obj_at_tile(col, row):
    """Return the door object placed at tile col,row, if any."""
    wx = col * TILE_SIZE
    wy = row * TILE_SIZE
    for d in getattr(STATE, "doors", []):
        if d.rect.x == wx and d.rect.y == wy:
            return d
    return None


def _draw_tiny_padlock(surface, x, y, col):
    """Tiny drawn lock icon. Avoids unicode emoji font issues."""
    pygame.draw.arc(surface, col, pygame.Rect(x - 3, y, 10, 10), math.pi, math.tau, 2)
    pygame.draw.rect(surface, col, (x - 4, y + 7, 14, 8), border_radius=1)
    pygame.draw.rect(surface, (38, 32, 24), (x + 3, y + 10, 2, 3))


def _draw_ch_door_tile(surface, sx, sy, col, row):
    """Draw a proper interactive prison door at any G tile."""
    sx, sy = int(sx), int(sy)
    ts = TILE_SIZE

    d = _door_obj_at_tile(col, row)
    locked = getattr(d, "locked", True) if d else True
    open_p = getattr(d, "anim_progress", 0.0) if d else 0.0

    is_cell_door = (col == _DOOR_TILE_COL and row == _DOOR_TILE_ROW) or \
                   (col == _DOOR_TILE_COL + 1 and row == _DOOR_TILE_ROW)
    if is_cell_door:
        # Only show the cell door as OPEN after destruction (frame 3==).
        # During charge (starting frames 0-600) the door is still INTACT —
        # it breaks in-place. The UI isn't drawn yet. Showing the door
        # as open during this phase made it look like a door magically
        # ... (a "play dead") that was actually open/pending before the blitz.
        if getattr(STATE, "chapter3_cell_open", False) or _intake_breach_f >= 180:
            open_p = 0.0
            locked = False

    frame_col = (58, 60, 56)
    frame_dark = (28, 30, 32)
    frame_hi = (92, 96, 92)
    pygame.draw.rect(surface, frame_dark, (sx - 2, sy - 2, ts + 4, ts + 4))
    pygame.draw.rect(surface, frame_col, (sx, sy, ts, ts))
    pygame.draw.line(surface, frame_hi, (sx, sy), (sx + ts, sy), 2)
    pygame.draw.line(surface, frame_hi, (sx, sy), (sx, sy + ts), 2)
    pygame.draw.line(surface, frame_dark, (sx, sy + ts - 1), (sx + ts, sy + ts - 1), 2)
    pygame.draw.line(surface, frame_dark, (sx + ts - 1, sy), (sx + ts - 1, sy + ts), 2)

    if open_p >= 0.99:
        inner = pygame.Rect(sx + 6, sy + 6, ts - 12, ts - 12)
        pygame.draw.rect(surface, (12, 12, 12), inner)
        # Keep/clear the door's own green LED while the breach sequence is
        # active so that door — the CG change driven by _draw_intake_breach()
        # is correct. Its cap (an unlocking LED, and the two together read as
        # random-growing noise). Keep the green LED only when the door is
        # open through normal gameplay (keycard / unlocked), not during the
        # scripted breach.
        breach_active_on_this_door = is_cell_door and _intake_breach_f > 0
        if not breach_active_on_this_door:
            pygame.draw.circle(surface, (80, 220, 100), (sx + ts - 10, sy + 10), 3)
            pygame.draw.circle(surface, (30, 100, 40), (sx + ts - 10, sy + 10), 3, 1)
        return

    panel_w = int((ts - 8) * (1.0 - min(0.95, open_p)))
    if panel_w < 4:
        panel_w = 4

    panel = pygame.Rect(sx + 4, sy + 4, panel_w, ts - 8)
    panel_base = (116, 118, 112)
    panel_dark = (72, 74, 72)
    panel_hi = (162, 165, 158)
    panel_edge = (48, 48, 46)

    pygame.draw.rect(surface, panel_base, panel, border_radius=2)
    pygame.draw.line(surface, panel_hi, (panel.x + 1, panel.y + 1), (panel.right - 2, panel.y + 1), 1)
    pygame.draw.line(surface, panel_dark, (panel.x + 1, panel.bottom - 2), (panel.right - 2, panel.bottom - 2), 1)
    pygame.draw.rect(surface, panel_edge, panel, 1, border_radius=2)

    rib_start = panel.x + 4
    rib_end = panel.right - 4
    rib_spacing = max(6, (rib_end - rib_start) // 4)
    for rx in range(rib_start, rib_end, rib_spacing):
        if rx > panel.right - 4:
            break
        pygame.draw.line(surface, panel_dark, (rx, panel.y + 4), (rx, panel.bottom - 4), 1)
        pygame.draw.line(surface, panel_hi, (rx + 1, panel.y + 4), (rx + 1, panel.bottom - 4), 1)

    for frac in (0.33, 0.66):
        bar_y = panel.y + int(panel.h * frac)
        pygame.draw.line(surface, panel_dark, (panel.x + 3, bar_y), (panel.right - 3, bar_y), 2)
        pygame.draw.line(surface, panel_hi, (panel.x + 3, bar_y + 1), (panel.right - 3, bar_y + 1), 1)

    if panel_w > 20:
        vp_w = min(10, panel_w - 10)
        vp_h = 16
        vp_x = panel.x + 5
        vp_y = panel.y + 6
        vp = pygame.Rect(vp_x, vp_y, vp_w, vp_h)
        pygame.draw.rect(surface, (8, 10, 10), vp)
        pygame.draw.rect(surface, (70, 65, 50), vp, 1)
        bar_spacing = max(3, vp_w // 3)
        for bx in range(vp.x + bar_spacing, vp.right - 1, bar_spacing):
            pygame.draw.line(surface, (62, 70, 80), (bx, vp.y + 1), (bx, vp.bottom - 2), 1)
            pygame.draw.line(surface, (108, 118, 122), (bx - 1, vp.y + 1), (bx - 1, vp.bottom - 2), 1)

    if panel_w > 20:
        sp_w = 10
        sp_h = 18
        sp_x = panel.right - sp_w - 4
        sp_y = panel.centery - sp_h // 2
        pygame.draw.rect(surface, (30, 30, 30), (sp_x, sp_y, sp_w, sp_h), border_radius=2)
        pygame.draw.rect(surface, (72, 72, 70), (sp_x, sp_y, sp_w, sp_h), 1, border_radius=2)
        # Keep the keypad LED on the same plate; during the breach, the
        # keypad stays dark so nothing but the breach effect blinks.
        breach_active_on_this_door = is_cell_door and _intake_breach_f > 0
        if not breach_active_on_this_door:
            led_col = (220, 60, 60) if locked else (80, 220, 100)
            pygame.draw.circle(surface, led_col, (sp_x + sp_w // 2, sp_y + 3), 2)
        for ky in range(3):
            for kx in range(2):
                btn_x = sp_x + 2 + kx * 4
                btn_y = sp_y + 6 + ky * 4
                pygame.draw.rect(surface, (45, 45, 50), (btn_x, btn_y, 2, 2), border_radius=1)

    if panel_w > 14:
        handle_x = panel.right - 10
        handle_y = panel.centery + 12
        pygame.draw.rect(surface, (64, 74, 80), (handle_x - 2, handle_y - 8, 4, 16), border_radius=2)
        pygame.draw.line(surface, (160, 168, 165), (handle_x + 1, handle_y - 6), (handle_x + 1, handle_y + 6), 1)
        pygame.draw.circle(surface, (88, 92, 96), (handle_x + 1, handle_y + 8), 2)

    if locked and open_p < 0.1:
        lock_x = panel.x + 3
        lock_y = panel.bottom - 18
        pygame.draw.rect(surface, (200, 170, 60), pygame.Rect(lock_x, lock_y, 10, 8), border_radius=2)
        pygame.draw.rect(surface, (120, 100, 40), (lock_x, lock_y, 10, 8), 1, border_radius=2)
        pygame.draw.circle(surface, (80, 60, 30), (lock_x + 5, lock_y + 4), 1)

    # Dented damage overlay for destructible interior crash doors
    hits = getattr(d, "ram_hits", 0) if d else 0
    if hits > 0 and open_p < 0.05:
        dmg_rng = random.Random(col * 1031 + row * 7919)

        dmg_px = panel.right - 14 if panel_w > 30 else panel.centerx
        dmg_py = panel.centery + 4

        for i in range(hits):
            ox = dmg_rng.randint(-10, 10)
            oy = dmg_rng.randint(-12, 12)
            cx = dmg_px + ox
            cy = dmg_py + oy

            pygame.draw.circle(surface, (30, 30, 32), (cx, cy), 4)
            pygame.draw.circle(surface, (150, 152, 150), (cx, cy), 4, 1)

            ang = dmg_rng.uniform(0, math.tau)
            ln = dmg_rng.randint(5, 10)
            pygame.draw.line(
                surface,
                (40, 40, 42),
                (cx, cy),
                (cx + math.cos(ang) * ln, cy + math.sin(ang) * ln),
                1,
            )

        required = getattr(d, "ram_required_hits", 3) if d else 3
        if hits >= required - 1:
            if (pygame.time.get_ticks() // 200) % 2 == 0:
                pygame.draw.line(surface, (255, 100, 80), (dmg_px - 6, dmg_py - 6), (dmg_px + 6, dmg_py + 6), 2)
                pygame.draw.line(surface, (255, 100, 80), (dmg_px + 6, dmg_py - 6), (dmg_px - 6, dmg_py + 6), 2)

    lbl_col = (200, 200, 90) if locked else (160, 240, 160)
    lbl = font_hud_small.render("GREEN", True, lbl_col)
    surface.blit(lbl, (sx + ts // 2 - lbl.get_width() // 2, sy + ts - 10))


def _render_cells_bg(target, cam_x, cam_y):
    """Render the prison tile grid (floor + walls) onto ``target`` for the given camera offset.

    Factored out of game_draw so the intro-draw callsite can render the
    same cell (all on-tile/unused backdrop) to an offscreen buffer and pass
    it to the cutscene _draw_ch_* chain as "cell_surf" — making every
    shot recognizably the same cell at a different focal length.
    """
    start_col = max(0, int(cam_x // TILE_SIZE) - 1)
    end_col = min(MAP_W, int((cam_x + WIDTH) // TILE_SIZE) + 2)
    start_row = max(0, int(cam_y // TILE_SIZE) - 1)
    end_row = min(MAP_H, int((cam_y + HEIGHT) // TILE_SIZE) + 2)

    for row in range(start_row, end_row):
        line_row = PRISON_LINES[row]
        for col in range(start_col, end_col):
            tile = line_row[col]
            x = col * TILE_SIZE - cam_x
            y = row * TILE_SIZE - cam_y

            if tile == "G":
                pygame.draw.rect(target, _TILE_BASE["."], (x, y, TILE_SIZE, TILE_SIZE))
                pygame.draw.rect(target, (22, 24, 22), (x, y, TILE_SIZE, TILE_SIZE), 1)
                continue

            base = _TILE_BASE.get(tile, (38, 36, 33))
            pygame.draw.rect(target, base, (x, y, TILE_SIZE, TILE_SIZE))
            if tile == "#":
                pygame.draw.rect(target, (60, 60, 60),
                                 (x, y, TILE_SIZE, TILE_SIZE), 1)
            elif tile == "P":
                pulse = 0.5 + 0.5 * math.sin(time.time() * 3.0)
                glow = pygame.Surface((TILE_SIZE, TILE_SIZE), pygame.SRCALPHA)
                pygame.draw.rect(
                    glow,
                    (int(100 + 40 * pulse), 160, int(200 + 40 * pulse)),
                    (0, 0, TILE_SIZE, TILE_SIZE),
                )
                target.blit(glow, (x, y))
            elif tile == "S":
                # Security glass — frosted span tint + faint diagonal marks
                hatch = pygame.Surface((TILE_SIZE, TILE_SIZE), pygame.SRCALPHA)
                hatch.fill((40, 70, 100, 80))
                for off in range(-TILE_SIZE, TILE_SIZE, 8):
                    pygame.draw.line(hatch, (180, 200, 220, 60),
                                     (off, 0),
                                     (off + TILE_SIZE, TILE_SIZE), 1)
                target.blit(hatch, (x, y))
            elif tile == "D":
                is_light = (col + row) % 2 == 0
                base = (48, 46, 44) if is_light else (42, 40, 38)
                pygame.draw.rect(target, base, (x, y, TILE_SIZE, TILE_SIZE))
                pygame.draw.rect(target, (32, 30, 30), (x, y, TILE_SIZE, TILE_SIZE), 1)
            elif tile == "L":
                base = (54, 56, 60) if (col + row) % 2 == 0 else (50, 52, 56)
                pygame.draw.rect(target, base, (x, y, TILE_SIZE, TILE_SIZE))
                pygame.draw.rect(target, (40, 42, 44), (x, y, TILE_SIZE, TILE_SIZE), 1)
            elif tile == "T":
                base = (58, 56, 48) if (col + row) % 2 == 0 else (54, 52, 44)
                pygame.draw.rect(target, base, (x, y, TILE_SIZE, TILE_SIZE))
                pygame.draw.rect(target, (44, 42, 36), (x, y, TILE_SIZE, TILE_SIZE), 1)
            elif tile == "B":
                base = (46, 42, 38) if (col + row) % 2 == 0 else (42, 38, 34)
                pygame.draw.rect(target, base, (x, y, TILE_SIZE, TILE_SIZE))
                pygame.draw.rect(target, (30, 28, 26), (x, y, TILE_SIZE, TILE_SIZE), 1)


def _screen_visible(x, y, w, h, margin=40):
    return not (x + w < -margin or x > WIDTH + margin or y + h < -margin or y > HEIGHT + margin)


def _draw_bg_crate_stack(sx, sy, scale=0.8):
    """Crates and decaying stacks for empty background space."""
    r = scale
    pygame.draw.ellipse(screen, (30, 32, 36), (int(sx - 40*r), int(sy + 12*r), int(80*r), int(18*r)))
    pygame.draw.rect(screen, (46, 44, 40), (int(sx - 36*r), int(sy - 6*r), int(72*r), int(22*r)))
    pygame.draw.rect(screen, (58, 54, 48), (int(sx - 36*r), int(sy - 6*r), int(72*r), int(22*r)), 1)
    boxes = [(-28, -8, 14, 18), (-10, -12, 20, 22), (14, -8, 16, 18), (-22, -20, 14, 14),
             (-4, -24, 16, 16), (16, -20, 14, 14)]
    for bx, by, bw, bh in boxes:
        c = pygame.Rect(int(sx + bx*r), int(sy + by*r), int(bw*r), int(bh*r))
        pygame.draw.rect(screen, (92, 74, 50), c, border_radius=1)
        pygame.draw.rect(screen, (58, 46, 32), c, 1, border_radius=1)
        pygame.draw.line(screen, (58, 46, 32), (c.x + 1, c.y + 1), (c.right - 2, c.bottom - 2), 1)
        pygame.draw.line(screen, (58, 46, 32), (c.right - 2, c.y + 1), (c.x + 1, c.bottom - 2), 1)


def _draw_bg_barrels(sx, sy, count=4):
    for i in range(count):
        x = int(sx + i * 18)
        y = int(sy + (i % 2) * 4)
        pygame.draw.ellipse(screen, (28, 30, 32), (x - 8, y + 10, 18, 6))
        pygame.draw.rect(screen, (52, 60, 58), (x - 7, y - 8, 14, 18), border_radius=3)
        pygame.draw.ellipse(screen, (70, 80, 76), (x - 7, y - 10, 14, 5))
        pygame.draw.ellipse(screen, (44, 48, 46), (x - 7, y + 6, 14, 5))
        pygame.draw.line(screen, (90, 100, 96), (x - 6, y - 2), (x + 6, y - 2), 1)
        pygame.draw.line(screen, (36, 40, 38), (x - 6, y + 3), (x + 6, y + 3), 1)


def _draw_bg_bench(sx, sy):
    pygame.draw.ellipse(screen, (28, 30, 32), (int(sx - 30), int(sy + 10), 60, 10))
    for x, col in ((-24, (54, 50, 46)), (16, (54, 50, 46))):
        pygame.draw.rect(screen, col, (int(sx + x), int(sy - 2), 8, 14))
    pygame.draw.rect(screen, (88, 80, 68), (int(sx - 30), int(sy - 8), 60, 6))
    pygame.draw.line(screen, (110, 100, 86), (int(sx - 30), int(sy - 8)), (int(sx + 30), int(sy - 8)), 1)


def _draw_bg_gen_booth(sx, sy):
    pygame.draw.ellipse(screen, (28, 30, 32), (int(sx - 34), int(sy + 16), 68, 14))
    pygame.draw.rect(screen, (52, 58, 60), (int(sx - 30), int(sy - 20), 60, 36), border_radius=3)
    pygame.draw.rect(screen, (80, 90, 94), (int(sx - 30), int(sy - 20), 60, 36), 1, border_radius=3)
    pygame.draw.rect(screen, (18, 22, 24), (int(sx - 22), int(sy - 14), 20, 14))
    pygame.draw.circle(screen, (200, 160, 60), (int(sx + 14), int(sy - 8)), 3)
    # Map vents and base
    pygame.draw.line(screen, (100, 96, 90), (int(sx - 22), int(sy + 6)), (int(sx + 22), int(sy + 6)), 1)
    for off in (-4, 0, 4):
        pygame.draw.line(screen, (34, 38, 40), (int(sx + 6 + off), int(sy + 8)), (int(sx + 6 + off), int(sy + 14)), 1)


def _draw_bg_archive_cab(sx, sy):
    pygame.draw.ellipse(screen, (28, 30, 32), (int(sx - 32), int(sy + 16), 64, 12))
    pygame.draw.rect(screen, (58, 62, 60), (int(sx - 28), int(sy - 28), 56, 44), border_radius=2)
    pygame.draw.rect(screen, (90, 96, 94), (int(sx - 28), int(sy - 28), 56, 44), 1, border_radius=2)
    pygame.draw.line(screen, (40, 44, 42), (int(sx), int(sy - 28)), (int(sx), int(sy + 16)), 1)
    pygame.draw.rect(screen, (74, 78, 76), (int(sx - 24), int(sy - 24), 20, 10), border_radius=1)
    pygame.draw.rect(screen, (74, 78, 76), (int(sx + 4), int(sy - 24), 20, 10), border_radius=1)
    pygame.draw.rect(screen, (74, 78, 76), (int(sx - 24), int(sy - 8), 20, 10), border_radius=1)
    for xx in (sx - 14, sx + 14):
        pygame.draw.circle(screen, (160, 164, 160), (int(xx), int(sy - 19)), 1)
        pygame.draw.circle(screen, (160, 164, 160), (int(xx), int(sy - 3)), 1)


def _draw_bg_wall_pipes(x1, y1, x2, y2, color=(70, 76, 74)):
    pygame.draw.line(screen, (40, 44, 46), (int(x1), int(y1 + 4)), (int(x2), int(y2 + 4)), 6)
    pygame.draw.line(screen, color, (int(x1), int(y1)), (int(x2), int(y2)), 4)
    pygame.draw.line(screen, (110, 118, 116), (int(x1), int(y1 - 1)), (int(x2), int(y2 - 1)), 1)
    dist = max(1, int(math.hypot(x2 - x1, y2 - y1)))
    for t in range(20, dist, 120):
        u = t / dist
        px = x1 + (x2 - x1) * u
        py = y1 + (y2 - y1) * u
        pygame.draw.rect(screen, (54, 58, 60), (int(px - 3), int(py - 5), 6, 10), border_radius=1)


def _draw_ch_background_dressing(cam_x, cam_y):
    """Dress the big empty negative spaces around the prison block.

    These props are intentionally not colliders: crates, barrels, pipes,
    carts, walls, etc. They exist to make the places feel lived-in and
    make corridors without changing gameplay collision.
    """
    start_col = max(0, int(cam_x // TILE_SIZE) - 4)
    end_col = min(MAP_W, int((cam_x + WIDTH) // TILE_SIZE) + 4)
    start_row = max(0, int(cam_y // TILE_SIZE) - 4)
    end_row = min(MAP_H, int((cam_y + HEIGHT) // TILE_SIZE) + 4)

    # Subtle background floor noise/scuffs on the empty "g" floor.
    for row in range(start_row, end_row):
        for col in range(start_col, end_col):
            if PRISON_LINES[row][col] != "g":
                continue
            x = col * TILE_SIZE - cam_x
            y = row * TILE_SIZE - cam_y
            rng = random.Random(col * 92821 + row * 71317)
            pygame.draw.line(screen, (36, 52, 34), (x, y), (x + TILE_SIZE, y), 1)
            pygame.draw.line(screen, (36, 52, 34), (x, y), (x, y + TILE_SIZE), 1)
            if rng.random() < 0.35:
                ox = x + rng.randint(10, 54)
                oy = y + rng.randint(10, 54)
                pygame.draw.line(screen, (44, 60, 40), (ox, oy), (ox + rng.randint(-10, 10), oy + rng.randint(-10, 10)), 1)
            if rng.random() < 0.08:
                pygame.draw.ellipse(screen, (32, 46, 30), (x + rng.randint(4, 30), y + rng.randint(4, 30), rng.randint(14, 26), rng.randint(8, 14)))

    # Long exterior pipes (decorating exterior walls).
    pipe_runs = [
        (4.0, 5.5, 46.5, 5.5),
        (3.5, 18.0, 20.5, 18.0),
        (6.0, 33.5, 28.0, 33.5),
        (2.0, 12.5, 2.0, 28.5),
        (50.0, 10.5, 50.0, 34.5),
    ]
    for x1, y1, x2, y2 in pipe_runs:
        sx1, sy1 = x1 * TILE_SIZE - cam_x, y1 * TILE_SIZE - cam_y
        sx2, sy2 = x2 * TILE_SIZE - cam_x, y2 * TILE_SIZE - cam_y
        if _screen_visible(min(sx1, sx2), min(sy1, sy2), abs(sx2 - sx1) + 8, abs(sy2 - sy1) + 8):
            _draw_bg_wall_pipes(sx1, sy1, sx2, sy2)

    # Hand-placed background props in dead/outside space.
    props = [
        ("crates", 3.5, 8.5), ("barrels", 10.5, 4.5, 3),
        ("cart", 26.5, 4.5), ("crates", 30.5, 6.5),
        ("booth", 37.5, 27.0), ("crates", 46.5, 8.5),
        ("barrels", 8.5, 38.5, 4), ("bench", 7.5, 18.5),
        ("bench", 21.5, 4.5), ("cart", 4.8, 30.5),
        ("crates", 38.5, 40.5), ("barrels", 35.5, 41.5, 2),
    ]
    for kind, tx, ty, *rest in props:
        sx = tx * TILE_SIZE - cam_x
        sy = ty * TILE_SIZE - cam_y
        if not _screen_visible(sx - 80, sy - 80, 160, 160):
            continue
        if kind == "crates":
            _draw_bg_crate_stack(sx, sy, 0.9)
        elif kind == "barrels":
            _draw_bg_barrels(sx, sy, rest[0])
        elif kind == "booth":
            _draw_bg_gen_booth(sx, sy)
        elif kind == "bench":
            _draw_bg_bench(sx, sy)
        elif kind == "cart":
            _draw_bg_archive_cab(sx, sy)

    # A couple of warm lamps / floor utility lights.
    for tx, ty in ((10.5, 10.5), (24.5, 4.5), (44.5, 16.5), (6.5, 34.5)):
        sx = int(tx * TILE_SIZE - cam_x)
        sy = int(ty * TILE_SIZE - cam_y)
        if not _screen_visible(sx - 80, sy - 80, 160, 160):
            continue
        glow = pygame.Rect(sx - 40, sy - 8, 80, 16)
        pygame.draw.rect(screen, (56, 48, 34), glow, border_radius=8)
        pygame.draw.rect(screen, (170, 150, 100), glow, 1, border_radius=8)
        for off in (-20, 0, 20):
            pygame.draw.line(screen, (210, 190, 140), (glow.centerx + off, glow.y + 4), (glow.centerx + off, glow.bottom - 4), 1)


def _draw_ch_corridor_and_details(cam_x, cam_y):
    """Decorative detail pass for the prison-block map.

    The tile pass alone makes the floor-level corridor look like a flat sheet. This
    pass adds authored, non-collidable flourishes without changing room
    architecture or corridor readability or pathways.
    """
    _draw_cell_block_corridor_details(cam_x, cam_y)
    _draw_guard_room_details(cam_x, cam_y)
    _draw_armory_details(cam_x, cam_y)
    _draw_laundry_details(cam_x, cam_y)
    _draw_boiler_wing_yard_swamp_edge(cam_x, cam_y)
    _draw_yard_lines_details(cam_x, cam_y)


def _draw_cell_block_corridor_details(cam_x, cam_y):
    """Dress the row-to-row corridor that connects the four cells."""
    x0 = 4 * TILE_SIZE - cam_x
    y0 = 15 * TILE_SIZE - cam_y
    w = 38 * TILE_SIZE
    h = 2 * TILE_SIZE
    if x0 > WIDTH + 80 or x0 + w < -80 or y0 > HEIGHT + 80 or y0 + h < -80:
        return

    old_clip = screen.get_clip()
    screen.set_clip(pygame.Rect(int(x0), int(y0), int(w), int(h)))
    try:
        # Soft shading between floor spots near the plates of this block.
        mark = pygame.Surface((int(w), int(h)), pygame.SRCALPHA)
        for yy in range(int(h)):
            a = int(24 + 40 * (yy / h))
            pygame.draw.line(mark, (0, 0, 0, a), (0, yy), (int(w), yy))
        screen.blit(mark, (int(x0), int(y0)))
```

```
# Generated by Django 4.2.11 on 2024-04-15 19:47

from django.db import migrations, models


class Migration(migrations.Migration):

    dependencies = [
        ("core", "0003_alter_user_options_alter_user_managers_and_more"),
    ]

    operations = [
        migrations.AddField(
            model_name="user",
            name="bio",
            field=models.TextField(blank=True, default=""),
            preserve_default=False,
        ),
        migrations.AddField(
            model_name="user",
            name="website",
            field=models.URLField(blank=True, default=""),
            preserve_default=False,
        ),
    ]
```

```
cy = int(9.8 * TILE_SIZE - cam_y)
            # [lines 2-4 of the draw routine illegible]
            for i in range(4):
                pygame.draw.rect(screen, (60, 65, 80), (sx - 20, sy - 10 + i * 24, 40, 16), border_radius=3)
                pygame.draw.circle(screen, (100, 90, 60), (sx + 16, sy - 10 + i * 24), 3)

    # Archive access door marker, east side.
    ax = 30 * TILE_SIZE + TILE_SIZE // 2 - cam_x
    ay = 45 * TILE_SIZE + TILE_SIZE // 2 - cam_y
    if -80 < ax < WIDTH + 80 and -80 < ay < HEIGHT + 80:
        sign = font_tutorial.render("ARCHIVES ->", True, (150, 160, 170))
        screen.blit(sign, (int(ax - sign.get_width() - 10), int(ay - 8)))
        pygame.draw.rect(screen, (40, 60, 80), (int(ax - 16), int(ay - 16), 32, 32), 2, border_radius=4)

def _draw_codex_map_points(cam_x, cam_y):
    """Draw readability-only posts for the Codex-wing map."""
    _draw_loot_room_frame(cam_x, cam_y, 5, 16, 8, 11, gaps_right=[(16, 8)], label="ARCHIVE", label_col=(140, 160, 180))
    _draw_loot_room_frame(cam_x, cam_y, 15, 27, 5, 8, gaps_bottom=[(16, 11)], label="SECURITY", label_col=(160, 150, 140))
    _draw_loot_room_frame(cam_x, cam_y, 15, 30, 33, 36, gaps_bottom=[(23, 36)], label="BATTERY BACKUP", label_col=(190, 160, 120))
    _draw_loot_room_frame(cam_x, cam_y, 34, 43, 33, 37, gaps_left=[(33, 35)], gaps_right=[(36, 43)], label="SIGNAL RELAY", label_col=(140, 210, 180))
    _draw_loot_room_frame(cam_x, cam_y, 33, 45, 39, 44, gaps_top=[(39, 45)], label="SECTOR SERVER", label_col=(170, 165, 180))

    # Security booth details.
    bx = 31 * TILE_SIZE - cam_x
    by = 37 * TILE_SIZE - cam_y
    if _screen_visible_rect(bx, by, 3*TILE_SIZE, 4*TILE_SIZE):
        glass = pygame.Rect(int(19.4*TILE_SIZE - cam_x), int(38.8*TILE_SIZE - cam_y), 86, 50)
        pygame.draw.rect(screen, (40, 55, 65), glass, border_radius=3)
        pygame.draw.rect(screen, (90, 130, 150), glass, 2, border_radius=3)
        for off in range(-40, 80, 16):
            pygame.draw.line(screen, (140, 160, 170, 60), (glass.x + off, glass.y), (glass.x + off + 18, glass.bottom), 1)
        desk = pygame.Rect(int(33.4*TILE_SIZE - cam_x), int(38.4*TILE_SIZE - cam_y), 104, 36)
        pygame.draw.ellipse(screen, (8,8,8,75), (desk.x-8, desk.y+30, desk.w+16, 18))
        pygame.draw.rect(screen, (95, 65, 50), desk, border_radius=4)
        pygame.draw.rect(screen, (160, 50, 70), desk, 2, border_radius=4)
        mon = pygame.Rect(desk.x + 14, desk.y - 24, 36, 20)
        pygame.draw.rect(screen, (40, 50, 60), mon, border_radius=3)
        pygame.draw.rect(screen, (90, 55, 80), mon, 2, border_radius=3)
        pygame.draw.circle(screen, (80, 150, 140), (mon.right - 8, mon.y + 6), 3)
        for i in range(3):
            pygame.draw.rect(screen, (100, 90, 70), (desk.x + 70 + i*10, desk.y + 10, 8, 18))
        chair = pygame.Rect(int(33.6*TILE_SIZE - cam_x), int(39.6*TILE_SIZE - cam_y), 30, 30)
        pygame.draw.ellipse(screen, (8,8,8,60), (chair.x-4, chair.y+20, 38, 18))
        pygame.draw.rect(screen, (35, 38, 50), chair, border_radius=6)
        pygame.draw.rect(screen, (75, 78, 95), chair, 2, border_radius=6)

    # Cable tray connecting archive -> relay/vault/infinity.
    sx0 = 18 * TILE_SIZE - cam_x
    sy = 35 * TILE_SIZE - cam_y + 18
    sx1 = 43 * TILE_SIZE - cam_x
    if _screen_visible_rect(sx0, sy-80, sx1-sx0, 160):
        pygame.draw.rect(screen, (38, 43, 48), (int(sx0), int(sy), int(sx1-sx0), 10), border_radius=3)
        pygame.draw.rect(screen, (58, 68, 80), (int(sx0), int(sy), int(sx1-sx0), 10), 1, border_radius=3)
        for i, col in enumerate([(160,60,60), (60,150,160), (70,150,70), (190,160,70)]):
            y = int(sy + 2 + i*2)
            pygame.draw.line(screen, col, (int(sx0+6), y), (int(sx1-6), y), 1)
        for bx, branch_y in [(24.5, 36.0), (30.5, 33.0), (36.0, 36.0)]:
            bxx = int(bx*TILE_SIZE - cam_x)
            byy = int(branch_y*TILE_SIZE - cam_y)
            pygame.draw.line(screen, (55, 56, 56), (bxx, int(sy+5)), (bxx, byy), 3)
            pygame.draw.line(screen, (90, 105, 100), (bxx+4, int(sy+5)), (bxx+4, byy), 1)

    sign = font_tutorial_small.render("VOID LOG ACCESS ->", True, (70, 140, 160))
    sign.set_alpha(150)
    screen.blit(sign, (int(26.6*TILE_SIZE - cam_x), int(34.4*TILE_SIZE - cam_y)))
    spool_x = int(28.0*TILE_SIZE - cam_x)
    spool_y = int(34.7*TILE_SIZE - cam_y)
    if -80 < spool_x < WIDTH+80 and -80 < spool_y < HEIGHT+80:
        pygame.draw.ellipse(screen, (8,8,8,70), (spool_x-20, spool_y+8, 44, 14))
        pygame.draw.circle(screen, (72,85,88), (spool_x, spool_y), 16)
        pygame.draw.circle(screen, (35,42,45), (spool_x, spool_y), 6)
        pygame.draw.line(screen, (140,88,60), (spool_x+10, spool_y+7), (spool_x+26, spool_y+18), 3)
    for i in range(3):
        ox = int((31.4+i*0.65)*TILE_SIZE - cam_x)
        oy = int(34.8*TILE_SIZE - cam_y)
        pygame.draw.polygon(screen, (160, 190, 80), [(ox,oy), (ox+10,oy), (ox+5,oy+10), (ox-5,oy+10)])

    # Relay room work bay + bench.
    rx = 38 * TILE_SIZE - cam_x
    ry = 38 * TILE_SIZE - cam_y
    if _screen_visible_rect(rx, ry, 7*TILE_SIZE, 5*TILE_SIZE):
        workbay = pygame.Rect(int(38.4*TILE_SIZE - cam_x), int(39.2*TILE_SIZE - cam_y), 140, 54)
        pygame.draw.rect(screen, (38, 40, 45), workbay, border_radius=6)
        pygame.draw.rect(screen, (75,100,90), workbay, 2, border_radius=6)
        pts = [(workbay.x+20, workbay.y+20), (workbay.x+40, workbay.y+34), (workbay.x+62, workbay.y+18), (workbay.x+84, workbay.y+30), (workbay.x+106, workbay.y+16), (workbay.x+124, workbay.y+28)]
        pygame.draw.lines(screen, (90,170,120), False, pts, 2)
        for pt in pts:
            pygame.draw.circle(screen, (190,200,160), pt, 3)
        mast_x = int(39.8*TILE_SIZE - cam_x)
        mast_y = int(38.4*TILE_SIZE - cam_y)
        pygame.draw.line(screen, (70,80,75), (mast_x, mast_y), (mast_x, mast_y+48), 3)
        for off in (10,20,30):
            pygame.draw.line(screen, (100,130,100), (mast_x-8, mast_y+off), (mast_x+8, mast_y+off), 1)

    # Server-cage east overlay.
    sx = 44*TILE_SIZE - cam_x
    sy2 = 43*TILE_SIZE - cam_y
    sw = 10*TILE_SIZE
    sh = 5*TILE_SIZE
    if _screen_visible_rect(sx, sy2, sw, sh):
        cage = pygame.Surface((int(sw), int(sh)), pygame.SRCALPHA)
        for off in range(0, int(sw), 18):
            pygame.draw.line(cage, (70,100,90,45), (off,0), (off+int(sh), int(sh)), 1)
            pygame.draw.line(cage, (70,100,90,45), (off+int(sh), 0), (off, int(sh)), 1)
        screen.blit(cage, (int(sx), int(sy2)))
        pygame.draw.rect(screen, (80,110,100), (int(sx+8), int(sy2+8), int(sw-16), int(sh-16)), 2)

def _relay_pos():
    return (RELAY_TILE_X * TILE_SIZE + TILE_SIZE // 2,
            RELAY_TILE_Y * TILE_SIZE + TILE_SIZE // 2)

def _codex_breaker_pos():
    # East wall of the battery backup room.
    return (20 * TILE_SIZE + TILE_SIZE // 2,
            40 * TILE_SIZE + TILE_SIZE // 2)

def _codex_path_pos():
    # Security booth desk terminal near the archive entrance.
    return (34 * TILE_SIZE + TILE_SIZE // 2,
            38 * TILE_SIZE + TILE_SIZE // 2)

def _restore_codex_power():
    if getattr(STATE, "hs_codex_power_restored", False):
        return
    STATE.hs_codex_power_restored = True
    bx, by = _codex_breaker_pos()
    STATE.floating_texts.append(
        (bx, by - 40, "BREAKERS ONLINE", (255, 220, 140), 180))
    try:
        audio.play("sfx_unlock", volume=0.45)
        audio.play("tutorial_glitch", volume=0.25)
    except Exception:
        pass
    _radio("Redline: Power grid restored. Now find a relay authorization key.",
           color=(255, 200, 140), duration=380, mark="codex_power_restored")

def _acquire_codex_path():
    if getattr(STATE, "hs_codex_path_acquired", False):
        return
    STATE.hs_codex_path_acquired = True
    px, py = _codex_path_pos()
    STATE.floating_texts.append(
        (px, py - 40, "RELAY AUTHORIZATION OBTAINED", (160, 230, 255), 190))
    try:
        audio.play("pickup", volume=0.45)
        audio.play("type_click", volume=0.35)
    except Exception:
        pass
    _radio("REDLINE: Authorization is good. Get back to the relay.",
           color=(180, 220, 255), duration=380, mark="codex_path_acquired")

def _start_codex_relay_tuning():
    if getattr(STATE, "hs_relay_tuned", False):
        return
    if getattr(STATE, "hs_relay_active", False):
        return
    if len(getattr(STATE, "enemies", [])) > 0:
        rx, ry = _relay_pos()
        STATE.floating_texts.append((rx, ry - 40, "CLEAR HOSTILES FIRST", (240, 140, 140), 150))
        _set_objective(f"Clear hostiles: {len(STATE.enemies)} guards left.")
        return
    if not getattr(STATE, "hs_codex_power_restored", False):
        rx, ry = _relay_pos()
        STATE.floating_texts.append(
            (rx, ry - 40, "NO POWER - CHECK BATTERY BREAKER", (240, 180, 140), 160))
        _set_objective("Restore auxiliary power in battery backup.")
        return
    if not getattr(STATE, "hs_codex_path_acquired", False):
        rx, ry = _relay_pos()
        STATE.floating_texts.append(
            (rx, ry - 40, "AUTHORIZATION REQUIRED", (200, 180, 240), 150))
        _set_objective("Copy relay authorization from the Security booth.")
        return
    STATE.hs_relay_active = True
    STATE.hs_relay_timer = 0
    _set_objective("Tuning relay...")
    try:
        audio.play("tutorial_glitch", volume=0.3)
        audio.play("radio_static", volume=0.35)
    except Exception:
        pass

def _update_codex_relay():
    if getattr(STATE, "tutorial_stage", None) != STAGE_CODEX:
        return
    if not getattr(STATE, "hs_relay_active", False):
        return

    duration = int(getattr(STATE, "hs_relay_duration", 240) or 240)
    STATE.hs_relay_timer = getattr(STATE, "hs_relay_timer", 0) + 1
    t = STATE.hs_relay_timer

    # Longer, staged tuning sequence: noisy scan -> handshake -> sync -> patch -> lock.
    # Audio ticks are sparse and brighter as the overlay feels like a machine
    # stabilizing, not a UFO.
    if t in (20, 70, 120, 160, 200, 230):
        try:
            audio.play("type_click", volume=0.15)
            audio.play("radio_static", volume=0.08)
        except Exception:
            pass

    if t >= duration:
        STATE.hs_relay_active = False
        STATE.hs_relay_tuned = True
        STATE.hs_relay_timer = duration
        for door in getattr(STATE, "doors", []):
            if getattr(door, "hs_archive_door", False):
                door.locked = False
                door.target_open = True
                door.motion_timer = 100
        _set_objective("Signal: Continue. Reach Archives.")
        try:
            audio.play("tutorial_glitch", volume=0.35)
            audio.play("radio_static", volume=0.25)
        except Exception:
            pass
        _radio("Redline: Archives interface. A face you, Archives access is open.",
               color=(120, 230, 200), duration=420, mark="codex_relay_tuned")

def _draw_codex_relay_prompt(cam_x, cam_y):
    if getattr(STATE, "tutorial_stage", None) != STAGE_CODEX:
        return
    if getattr(STATE, "hs_relay_active", False):
        return
    rx, ry = _relay_pos()
    sx, sy = rx - cam_x, ry - cam_y
    if not (-80 < sx < WIDTH + 80 and -80 < sy < HEIGHT + 80):
        return
    if getattr(STATE, "hs_relay_tuned", False):
        prompt = "RELAY ONLINE"
        col = (150, 240, 190)
    elif not getattr(STATE, "hs_codex_power_restored", False):
        prompt = "RELAY OFFLINE // NO POWER"
        col = (240, 170, 95)
    elif not getattr(STATE, "hs_codex_path_acquired", False):
        prompt = "RELAY LOCKED // AUTH REQUIRED"
        col = (190, 160, 75)
    elif STATE.player and math.hypot(STATE.player.rect.centerx - rx, STATE.player.rect.centery - ry) < 90:
        prompt = "[E] INITIALIZE RELAY LINK"
        col = (160, 230, 190)
    else:
        prompt = "RELAY READY"
        col = (90, 200, 180)
    a = int(155 + 100 * math.sin(pygame.time.get_ticks() * 0.01))
    label = font_tutorial_small.render(prompt, True, col)
    label.set_alpha(a)
    screen.blit(label, (int(sx - label.get_width() // 2), int(sy - 64)))
    # Soft marker ring.
    ring = pygame.Surface((52, 52), pygame.SRCALPHA)
    pygame.draw.circle(ring, (*col, 40), (26, 26), 24)
    pygame.draw.circle(ring, (*col, 120), (26, 26), 24, 2)
    screen.blit(ring, (int(sx - 26), int(sy - 26)))

def _draw_codex_dynamic_lighting(cam_x, cam_y):
    """Draw active lighting and breaker/relay feedback for the Codex wing."""
    timer = getattr(STATE, "hs_codex_powered_timer", 0)
    powered = getattr(STATE, "hs_codex_power_restored", False)

    if not powered:
        # Emergency power: only the relay/booth signal lamps with minimal darkness.
        # Flashlight dropoff: lower radius vignette on the dark areas.
        pulse = 0.5 + 0.5 * math.sin(timer * 0.05)
        dark = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
        dark.fill((0, 0, 10, int(60 + 20 * pulse)))
        screen.blit(dark, (0, 0))
        if timer % 180 < 12:
            dropout = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
            dropout.fill((0, 0, 0, 50))
            screen.blit(dropout, (0, 0))
    else:
        # Restored green status glow overlay on the relay room.
        rx, ry = _relay_pos()
        sx, sy = int(rx - cam_x), int(ry - cam_y)
        glow = pygame.Surface((180, 180), pygame.SRCALPHA)
        pygame.draw.circle(glow, (60, 180, 140, 30), (90, 90), 88)
        pygame.draw.circle(glow, (80, 200, 160, 50), (90, 90), 50)
        screen.blit(glow, (sx - 90, sy - 90), special_flags=pygame.BLEND_RGBA_ADD)

def _draw_codex_objective_prompts(cam_x, cam_y):
    """Sparse world prompts for the power/authorization objectives only."""
    objectives = [
        (_codex_breaker_pos(),
         "POWER RESTORED" if getattr(STATE, "hs_codex_power_restored", False)
         else "[E] RESTORE AUX POWER",
         getattr(STATE, "hs_codex_power_restored", False)),
        (_codex_path_pos(),
         "AUTH COPIED" if getattr(STATE, "hs_codex_path_acquired", False)
         else "[E] COPY RELAY AUTH",
         getattr(STATE, "hs_codex_path_acquired", False)),
    ]
    for (wx, wy), text, complete in objectives:
        sx, sy = wx - cam_x, wy - cam_y
        if not (-80 < sx < WIDTH + 80 and -80 < sy < HEIGHT + 80):
            continue
        near = bool(STATE.player and math.hypot(
            STATE.player.rect.centerx - wx,
            STATE.player.rect.centery - wy) < 78)
        col = (130, 220, 160) if complete else ((255, 230, 140) if near else (200, 160, 100))
        alpha = int(170 + 70 * math.sin(pygame.time.get_ticks() * 0.01)) if near else 150
        label = font_tutorial_small.render(text, True, col)
        label.set_alpha(max(80, min(255, alpha)))
        screen.blit(label, (int(sx - label.get_width() // 2), int(sy - 58)))
        pygame.draw.circle(screen, col, (int(sx), int(sy)), 10, 2 if not near else 3)

    # Waypoint archive-link display outside the east portal.
    ax = 46 * TILE_SIZE + TILE_SIZE // 2
    ay = 34 * TILE_SIZE + TILE_SIZE // 2
    sx, sy = ax - cam_x, ay - cam_y
    if -80 < sx < WIDTH + 80 and -80 < sy < HEIGHT + 80:
        unlocked = getattr(STATE, "hs_relay_tuned", False)
        text = "ARCHIVES LINK READY" if unlocked else "ARCHIVES LINK OFFLINE"
        col = (130, 230, 190) if unlocked else (120, 70, 70)
        label = font_tutorial_small.render(text, True, col)
        screen.blit(label, (int(sx - label.get_width() // 2), int(sy - 48)))

def _draw_codex_relay_overlay():
    if not getattr(STATE, "hs_relay_active", False):
        return
    t = getattr(STATE, "hs_relay_timer", 0)
    duration = int(getattr(STATE, "hs_relay_duration", 240) or 240)
    progress = max(0.0, min(1.0, t / float(duration)))

    panel_w, panel_h = 640, 300
    panel_x = WIDTH // 2 - panel_w // 2
    panel_y = HEIGHT // 2 - panel_h // 2

    dim = pygame.Surface((WIDTH, HEIGHT), pygame.SRCALPHA)
    dim.fill((0, 0, 0, 140))
    screen.blit(dim, (0, 0))

    panel = pygame.Surface((panel_w, panel_h), pygame.SRCALPHA)
    pygame.draw.rect(panel, (12, 18, 18, 235), panel.get_rect(), border_radius=8)
    pygame.draw.rect(panel, (70, 150, 130, 180), panel.get_rect(), 2, border_radius=8)
    pygame.draw.line(panel, (45, 90, 80, 200), (16, 36), (panel_w - 16, 36), 1)

    title = font_tutorial.render("RELAY SYNC FORMAT", True, (150, 230, 200))
    panel.blit(title, (18, 10))
    sub = font_tutorial_small.render("CODEX-SECTOR RELAY // ARCHIVE ROUTE", True, (90, 150, 130))
    panel.blit(sub, (panel_w - sub.get_width() - 18, 14))

    # Main waveform scope.
    scope = pygame.Rect(18, 48, 380, 150)
    pygame.draw.rect(panel, (4, 12, 12), scope, border_radius=4)
    pygame.draw.rect(panel, (40, 80, 70), scope, 1, border_radius=4)
    # Scope grid.
    for gx in range(scope.x + 20, scope.right, 30):
        pygame.draw.line(panel, (15, 35, 30), (gx, scope.y + 2), (gx, scope.bottom - 2), 1)
    for gy in range(scope.y + 15, scope.bottom, 20):
        pygame.draw.line(panel, (15, 35, 30), (scope.x + 2, gy), (scope.right - 2, gy), 1)

    # Target carrier line and unstable waveform. As progress rises, amplitude and
    # phase jitter collapse toward a clean signal.
    pygame.draw.line(panel, (30, 70, 60), (scope.x + 4, scope.centery), (scope.right - 4, scope.centery), 1)
    pts = []
    phase_jitter = math.sin(t * 0.091) * (1.0 - progress) * 3.0
    amp = 34 * (1.0 - progress) + 7
    for i in range(scope.w - 8):
        x = scope.x + 4 + i
        harmonic = math.sin(i * 0.060 + t * 0.08 + phase_jitter)
        noise = math.sin(i * 0.31 + t * 0.31) * (1.0 - progress) * 10
        y = scope.centery + harmonic * amp + noise
        pts.append((x, y))
    if len(pts) > 1:
        pygame.draw.lines(panel, (110, 240, 170), False, pts, 2)
    # Secondary ghost/echo fades early on.
    if progress < 0.65:
        pts2 = []
        for i in range(scope.w - 8):
            x = scope.x + 4 + i
            y = scope.centery + math.sin(i * 0.051 - t * 0.06) * (amp * 0.55) + math.sin(i * 0.21) * 6 * (1.0 - progress)
            pts2.append((x, y))
        if len(pts2) > 1:
            pygame.draw.lines(panel, (50, 100, 120, 90), False, pts2, 1)

    # Right side: signal/buffer meter.
    meter = pygame.Rect(414, 48, 208, 150)
    pygame.draw.rect(panel, (6, 12, 12), meter, border_radius=4)
    pygame.draw.rect(panel, (40, 80, 70), meter, 1, border_radius=4)
    meter_title = font_tutorial_small.render("SIGNAL LOCK", True, (120, 190, 160))
    panel.blit(meter_title, (meter.centerx - meter_title.get_width() // 2, meter.y + 6))
    for i in range(8):
        bh = 8 + i * 11
        bx = meter.x + 16 + i * 23
        by = meter.bottom - 18 - bh
        locked = progress >= (i + 1) / 8.0
        col = (90, 200, 140) if locked else (30, 45, 40)
        pygame.draw.rect(panel, col, (bx, by, 16, bh), border_radius=2)
        pygame.draw.rect(panel, (70, 100, 90), (bx, by, 16, bh), 1, border_radius=2)

    # Step boxes - stages sequence from the left/flow view.
    steps = [
        ("SCAN", 0.00),
        ("HANDSHAKE", 0.22),
        ("SYNC", 0.44),
        ("PATCH", 0.66),
        ("LOCK", 0.88),
    ]
    base_y = 208
    for idx, (name, threshold) in enumerate(steps):
        bx = 18 + idx * 120
        by = base_y
        complete = progress >= threshold
        active = complete and (idx == len(steps) - 1 or progress < steps[idx + 1][1])
        bg = (20, 44, 38) if not complete else (30, 60, 50)
        border = (80, 180, 140) if complete else ((180, 160, 90) if active else (50, 70, 60))
        pygame.draw.rect(panel, bg, (bx, by, 108, 30), border_radius=4)
        pygame.draw.rect(panel, border, (bx, by, 108, 30), 1, border_radius=4)
        txt = font_tutorial_small.render(name, True, (200, 240, 220) if complete else (120, 140, 130))
        panel.blit(txt, (bx + 54 - txt.get_width() // 2, by + 8))

    # Progress row.
    pygame.draw.rect(panel, (25, 45, 40), (18, 250, panel_w - 36, 10), border_radius=4)
    pygame.draw.rect(panel, (100, 220, 170), (18, 250, int((panel_w - 36) * progress), 10), border_radius=4)
    pct = font_tutorial_small.render(f"SYNC {int(progress * 100)}%", True, (180, 240, 210))
    panel.blit(pct, (panel_w - pct.get_width() - 18, 268))

    # Warning/hint text.
    hint = "HOLD POSITION // RELAY HANDSHAKE IN PROGRESS"
    if progress > 0.85:
        hint = "CARRIER STABLE // ROUTE ARCHIVES UNLOCKING"
    hsurf = font_tutorial_small.render(hint, True, (200, 180, 120))
    panel.blit(hsurf, (18, 268))

    screen.blit(panel, (panel_x, panel_y))

def _draw_enemy_type_overlay(surface, enemy, cam_x, cam_y):
    """Overlays a unique visual differentiator.

    Draw these as one-glance layers on top of the stock enemy.draw() silhouette,
    not as separate sprites. The goal for code-game types: distinct value.
    """
    et = getattr(enemy, "hs_type", "guard")
    if et == "guard" or enemy.hp <= 0:
        return
    x = enemy.rect.centerx - cam_x
    y = enemy.rect.centery - cam_y
    a = enemy.angle
    dx, dy = math.cos(a), math.sin(a)
    px, py = -dy, dx

    def _rot_poly(cx, cy, length, half_w, col, outline=(20, 20, 20), offset=0):
        ax = cx + dx * length * 0.5 + px * offset
        ay = cy + dy * length * 0.5 + py * offset
        bx = cx - dx * length * 0.5 + px * offset
        by = cy - dy * length * 0.5 + py * offset
        outer = [(ax + px*(half_w+1), ay + py*(half_w+1)), (ax - px*(half_w+1), ay - py*(half_w+1)),
                 (bx - px*(half_w+1), by - py*(half_w+1)), (bx + px*(half_w+1), by + py*(half_w+1))]
        pygame.draw.polygon(surface, outline, outer)
        inner = [(ax + px*half_w, ay + py*half_w), (ax - px*half_w, ay - py*half_w),
                 (bx - px*half_w, by - py*half_w), (bx + px*half_w, by + py*half_w)]
        pygame.draw.polygon(surface, col, inner)
        return inner

    if et == "brawler":
        # Armored vest + compact weapon, not holding one on some platform.
        vest = [(x - 10, y - 14), (x + 4, y - 16), (x + 14, y - 10), (x + 10, y + 8), (x - 10, y + 12)]
        pygame.draw.polygon(surface, (54, 60, 44), vest)
        pygame.draw.polygon(surface, (100, 88, 60), vest, 2)
        # Red shoulder flash, visible but not loud.
        pygame.draw.line(surface, (180, 50, 40), (x - 10, y - 12), (x - 4, y - 15), 3)
        pygame.draw.line(surface, (180, 50, 40), (x + 4, y - 15), (x + 10, y - 12), 3)
        # Compact SMG barrel overlay aligned with enemy aim.
        _rot_poly(x + dx*14, y + dy*14, 18, 2.4, (30, 30, 34))
        pygame.draw.circle(surface, (30, 32, 36), (int(x + dx*24), int(y + dy*24)), 3)
        # Helmet/goggles on head.
        hx, hy = x + dx*4, y + dy*4
        pygame.draw.line(surface, (255, 90, 60), (hx - px*5, hy - py*5), (hx + px*5, hy + py*5), 2)

    elif et == "shield":
        # Riot shield in the silhouette's features. No floating label.
        cx = x + dx * 18
        cy = y + dy * 18
        sw, sh = 9, 16
        shield = [
            (cx + px*sw - dx*3, cy + py*sw - dy*3),
            (cx - px*sw - dx*3, cy - py*sw - dy*3),
            (cx - px*sw + dx*3, cy - py*sw + dy*3),
            (cx + px*sw + dx*3, cy + py*sw + dy*3),
        ]
        pygame.draw.polygon(surface, (80, 95, 110), shield)
        inner = [
            (cx + px*(sw-2) - dx*1.5, cy + py*(sw-2) - dy*1.5),
            (cx - px*(sw-2) - dx*1.5, cy - py*(sw-2) - dy*1.5),
            (cx - px*(sw-2) + dx*1.5, cy - py*(sw-2) + dy*1.5),
            (cx + px*(sw-2) + dx*1.5, cy + py*(sw-2) + dy*1.5),
        ]
        pygame.draw.polygon(surface, (150, 180, 200), inner)
        pygame.draw.polygon(surface, (210, 230, 240), shield, 2)
        # Viewport slit.
        pygame.draw.line(surface, (20, 30, 40),
                         (cx - px*4 - dx*2, cy - py*4 - dy*2),
                         (cx + px*4 - dx*2, cy + py*4 - dy*2), 2)
        # Shield anchor stripe.
        pygame.draw.line(surface, (255, 60, 60), (cx - dx*4, cy - dy*4), (cx + dx*4, cy + dy*4), 2)

    elif et == "marksman":
        # Backpack/radio pack + headset, but still same human silhouette.
        pack_cx = x - dx*8
        pack_cy = y - dy*8
        _rot_poly(pack_cx, pack_cy, 10, 7, (50, 60, 40), offset=0)
        pygame.draw.line(surface, (70, 110, 100), (int(pack_cx+5), int(pack_cy-8)), (int(pack_cx+5), int(pack_cy-22)), 2)
        # Antenna/headset cable.
        pygame.draw.line(surface, (180, 170, 150), (int(pack_cx), int(pack_cy)), (int(pack_cx - px*10), int(pack_cy - py*10)), 1)
        # Headset lens-scope band.
        hx, hy = x + dx*4, y + dy*4
        pygame.draw.arc(surface, (160, 170, 180), (int(hx-8), int(hy-8), 16, 16), a + 0.8, a + 2.2, 2)
        pygame.draw.circle(surface, (80, 200, 180), (int(hx + px*5), int(hy + py*5)), 2)
        # Radio handset/cable.
        bar = pygame.Rect(int(x + px*10 - 3), int(y + py*10 - 4), 6, 8)
        pygame.draw.rect(surface, (35, 38, 40), bar, border_radius=2)
        pygame.draw.rect(surface, (100, 110, 100), bar, 1, border_radius=2)

    elif et == "bodyguard":
        # Long rifle and scope, integrated with base drip-hide.
        tipx = (x + dx*30, y + dy*30)
        butt = (x - dx*8, y - dy*8)
        pygame.draw.line(surface, (8, 8, 8), butt, tipx, 6)
        pygame.draw.line(surface, (38, 40, 44), butt, tipx, 4)
        pygame.draw.line(surface, (90, 95, 100), (int(butt[0]+dx*6), int(butt[1]+dy*6)), (int(tipx[0]-dx*2), int(tipx[1]-dy*2)), 1)
        # Scope and lenses.
        scope = (x + dx*10 - px*3, y + dy*10 - py*3)
        pygame.draw.circle(surface, (20, 20, 24), (int(scope[0]), int(scope[1])), 4)
        pygame.draw.circle(surface, (80, 180, 200), (int(scope[0]), int(scope[1])), 2)
        pygame.draw.circle(surface, (5, 5, 5), (int(scope[0]), int(scope[1])), 1)
        # Camouflage stripe.
        hx, hy = x + dx*4, y + dy*4
        pygame.draw.arc(surface, (70, 90, 60), (int(hx-9), int(hy-9), 18, 18), a + 0.6, a + 2.4, 3)
        pygame.draw.line(surface, (100, 70, 40), (hx - px*6, hy - py*6), (hx + px*6, hy + py*6), 2)

def _draw_archive_wing_interior(cam_x, cam_y):
    """Set piece for the redesigned Lab/Vault Archives chain."""
    # -- Shoulder lobby landing --
    lx = 40 * TILE_SIZE - cam_x
    ly = 30 * TILE_SIZE - cam_y
    lw, lh = 12 * TILE_SIZE, 8 * TILE_SIZE
    if _screen_visible_rect(lx, ly, lw, lh):
        # Stacked shoulder desks.
        for sx in (42.4, 45.2):
            desk = pygame.Rect(int(sx * TILE_SIZE - cam_x - 20),
                               int(32 * TILE_SIZE - cam_y - 10), 52, 30)
            pygame.draw.rect(screen, (28, 40, 40), desk.inflate(6, 6), border_radius=6)
            pygame.draw.rect(screen, (55, 80, 75), desk, border_radius=4)
            pygame.draw.rect(screen, (130, 160, 140), desk, 2, border_radius=4)
            pygame.draw.line(screen, (20, 30, 30),
                             (desk.centerx, desk.top + 4),
                             (desk.centerx, desk.bottom - 4), 1)
        tag = font_tutorial_small.render("ARCHIVE SHOULDER // NO ACCESS", True, (140, 150, 140))
        tag.set_alpha(150)
        screen.blit(tag, (int(44 * TILE_SIZE - cam_x - tag.get_width() // 2),
                          int(31 * TILE_SIZE - cam_y)))
        sticker = font_tutorial_small.render("ARCHIVE TIMER", True, (140, 100, 100))
        sticker.set_alpha(150)
        screen.blit(sticker, (int(46 * TILE_SIZE - cam_x - sticker.get_width() // 2),
                              int(36 * TILE_SIZE - cam_y)))

    # -- Archive (northern) gallery --
    ax = 38 * TILE_SIZE - cam_x
    ay = 18 * TILE_SIZE - cam_y
    aw, ah = 6 * TILE_SIZE, 12 * TILE_SIZE
    if _screen_visible_rect(ax, ay, aw, ah):
        # Tall library rows down the archive center.
        for gy in range(0, 10, 2):
```

```
Traceback (most recent call last):
  File "/usr/lib/python3.11/multiprocessing/resource_tracker.py", line 209, in main
    cache = {rtype: set() for rtype in _CLEANUP_FUNCS}
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.11/multiprocessing/resource_tracker.py", line 209, in <dictcomp>
    cache = {rtype: set() for rtype in _CLEANUP_FUNCS}
             ^^^^^^^^^^^^
KeyboardInterrupt: 
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/usr/lib/python3.11/multiprocessing/resource_tracker.py", line 212, in main
    for line in f:
  File "<frozen codecs>", line 322, in decode
```

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```
# Generated by configure.ac; do not edit by hand.
# Copyright (C) 1992-2012 Free Software Foundation, Inc.
# This configure script is free software; the Free Software Foundation
# gives unlimited permission to copy, distribute and modify it.
## --------------------- ##
## M4sh Initialization.  ##
## --------------------- ##
# Be more Bourne compatible
DUALCASE=1; export DUALCASE # for MKS sh
if test -n "${ZSH_VERSION+set}" && (emulate sh) >/dev/null 2>&1; then :
  emulate sh
  NULLCMD=:
  # Pre-4.2 versions of Zsh do word splitting on ${1+"$@"}, which
  # is contrary to our usage.  Disable this feature.
  alias -g '${1+"$@"}'='"$@"'
  setopt NO_GLOB_SUBST
else
  case `(set -o) 2>/dev/null` in #(
  *posix*) :
    set -o posix ;; #(
  *) :
     ;;
esac
fi
as_nl='
'
export as_nl
# Printing a long string crashes Solaris 7 /usr/bin/printf.
as_echo='\\\
```

```
                                                
```

```
```

```
```

```
# Generated by Django 4.2.11 on 2024-03-22 11:39

import django.contrib.postgres.fields.ranges
import django.db.models.deletion
import uuid


class Migration(migrations.Migration):

    initial = True

    dependencies = [
        ("contenttypes", "0002_remove_content_type_name"),
    ]

    operations = [
        migrations.CreateModel(
            name="Author",
            fields=[
                (
                    "id",
                    models.UUIDField(
                        default=uuid.uuid4,
                        editable=False,
                        primary_key=True,
                        serialize=False,
                    ),
                ),
                ("name", models.CharField(max_length=255)),
                ("birth_year", models.IntegerField(blank=True, null=True)),
                ("death_year", models.IntegerField(blank=True, null=True)),
                ("bio", models.TextField(blank=True)),
                ("created_at", models.DateTimeField(auto_now_add=True)),
            ],
            options={
                "ordering": ["name"],
                "indexes": [
                    models.Index(fields=["name"], name="author_name_idx"),
                    models.Index(fields=["birth_year"], name="author_birth_year_idx"),
                ],
            },
        ),
        migrations.CreateModel(
            name="Publication",
            fields=[
                (
                    "id",
                    models.UUIDField(
                        default=uuid.uuid4,
                        editable=False,
                        primary_key=True,
                        serialize=False,
                    ),
                ),
                ("title", models.CharField(max_length=500)),
                ("publication_info", models.JSONField(blank=True, default=dict)),
                ("publication_range", django.contrib.postgres.fields.ranges.DateTimeRangeField(blank=True, null=True)),
                ("created_at", models.DateTimeField(auto_now_add=True)),
            ],
            options={
                "ordering": ["-created_at"],
                "indexes": [
                    models.Index(fields=["title"], name="publication_title_idx"),
                    models.Index(fields=["publication_info"], name="publication_info_idx"),
                ],
            },
        ),
        migrations.CreateModel(
            name="Tag",
            fields=[
                (
                    "id",
                    models.UUIDField(
                        default=uuid.uuid4,
                        editable=False,
                        primary_key=True,
                        serialize=False,
                    ),
                ),
                ("name", models.CharField(max_length=100, unique=True)),
                ("slug", models.SlugField(max_length=100, unique=True)),
            ],
            options={
                "ordering": ["name"],
                "indexes": [
                    models.Index(fields=["slug"], name="tag_slug_idx"),
                ],
            },
        ),
        migrations.CreateModel(
            name="Comment",
            fields=[
                (
                    "id",
                    models.UUIDField(
                        default=uuid.uuid4,
                        editable=False,
                        primary_key=True,
                        serialize=False,
                    ),
                ),
                ("object_id", models.UUIDField()),
                ("body", models.TextField()),
                ("created_at", models.DateTimeField(auto_now_add=True)),
                (
                    "content_type",
                    models.ForeignKey(
                        on_delete=django.db.models.deletion.CASCADE,
                        to="contenttypes.contenttype",
                    ),
                ),
            ],
            options={
                "ordering": ["created_at"],
                "indexes": [
                    models.Index(
                        fields=["content_type", "object_id"],
                        name="comment_target_idx",
                    ),
                ],
            },
        ),
    ]
```